BY USING THESE STANDARD PLANS, THE USER AGREES TO RELEASE THE COUNTY OF SAN DIEGO FROM ANY AND ALL CLAIMS, LIABILITIES, SUITS, AND DEMANDS ON ACCOUNT OF ANY INJURY, DAMAGE, OR LOSS TO PERSONS OR PROPERTY, INCLUDING INJURY OR DEATH, OR ECONOMIC LOSSES, ARISING OUT OF THE USE OF THESE CONSTRUCTION DOCUMENTS. THE USE OF THESE PLANS DOES NOT ELIMINATE OR REDUCE THE USER'S RESPONSIBILITY TO VERIFY ANY AND ALL INFORMATION.

SHEET INDEX

PP	PLOT PLAN
BMP-1	BMP PLAN
SW.	STORMWATER FORMS
A1	FLOOR PLAN
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A6	SECTIONS
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CS-1	MIN. CONSTRUCTION SPECIFICATIONS
MR	MANDATORY REQUIREMENTS SUMMARY
T24	ENERGY EFFICIENCY COMPLIANCE FORMS

GENERAL CODES

THIS PROJECT SHALL COMPLY WITH THE FOLLOWING BUILDING CODES AND ASSOCIATED COUNTY OF SAN DIEGO AMENDMENTS:

- 2025 CALIFORNIA RESIDENTIAL CODE
- 2025 CALIFORNIA BUILDING CODE
- 2025 CALIFORNIA GREEN BUILDING STANDARDS CODE
- 2025 CALIFORNIA ELECTRICAL CODE
- 2025 CALIFORNIA MECHANICAL CODE
- 2025 CALIFORNIA PLUMBING CODE
- 2025 CALIFORNIA FIRE CODE
- 2025 CALIFORNIA BUILDING ENERGY EFFICIENCY STANDARDS
- 2025 CALIFORNIA WILDLAND-URBAN INTERFACE CODE

DESIGN BASIS

CONVENTIONAL LIGHT FRAME CONSTRUCTION

- ROOF LIVE LOAD: 20 PSF
- SNOW LOAD: 50 PSF
- ULTIMATE WIND SPEED: 110 MPH
- EXPOSURE CATEGORY: C
- SITE CLASS: D
- RISK CATEGORY: II
- S_{DS}: 1.25
- SEISMIC DESIGN CATEGORY: D_2
- ALLOW SOIL VERTICAL BEARING PRESSURE: 1500 PSF
- ALLOW SOIL LATERAL BEARING PRESSURE: 100 PSF/FT

ALL TABLES BELOW ARE REQUIRED TO INCLUDE ALL IMPERVIOUS / PERVIOUS / LAND DISTURBANCE AREAS FROM **ALL OPEN PERMITS** ON THIS SITE

IMPERVIOUS AREA INFORMATION

SITE ID	ITEM	NEW or REPLACED AREA (SF)	EXISTING AREA (SF)
1	PROPOSED DWELLING (ROOF AREA)	1233	
2			
3			
4			
5			
	TOTAL (SF)		

PERVIOUS AREA INFORMATION

SITE ID	ITEM	AREA (SF)	NOTES
1			
2			
3			
4			
5			
	TOTAL (SF)		

PERVIOUS ELEMENT MANUFACTURER: ___________________________

PERVIOUS ELEMENT SLOPE AND DIRECTION OF SLOPE: ____________

MAINTENANCE PROGRAM: ______________________________________

PERVIOUS ELEMENT CROSS SECTION LOCATED ON SHEET: __________

CONSTRUCTED PERVIOUS SURFACES SHALL NOT BE SEALED.

LAND DISTURBANCE AREA INFORMATION

SITE ID	ITEM	AREA (SF)	NOTES
1	GRADING PERMIT		
2	BUILDING PERMIT		
3	SEPTIC PERMIT		
4	LANDSCAPING PERMIT		
5			
	TOTAL (SF)		

FOR PROJECTS IN A WILDFIRE ZONE, EXTERIOR GLAZING IN DOORS AND WINDOWS NOW REQUIRE **A MINIMUM OF TWO PANES OF TEMPERED GLAZING.**

N

ENGINEERING SCALE: 1" =

VICINITY MAP

OWNER INFORMATION

NAME:

ADDRESS:

PHONE:

EMAIL:

CONTACT INFORMATION

NAME:

ADDRESS:

PHONE:

EMAIL:

PROJECT INFORMATION

PROJECT SCOPE: PROPOSED 1000 SF DETACHED DWELLING UNIT ADU ☐ SFD ☐

APN:

SITE ADDRESS:

PROPERTY CONNECTED TO THE ELECTRICAL GRID: ☐ YES ☐ NO

PROPERTY SERVICED BY: ☐ NATURAL GAS ☐ PROPANE (SHOW THE PROPANE TANK ON PLOT PLAN)

ALL PROPOSED BUILDINGS, STRUCTURES, ADDITIONS, MODIFICATIONS TO BUILDINGS/STRUCTURES MUST COMPLY WITH THE APPROVED LOCATION, AS SHOWN ON THE COUNTY APPROVED PLOT PLAN. AT THE DISCRETION OF THE COUNTY, THE PROPERTY OWNER MAY BE REQUIRED TO PROVIDE PROOF OF CURRENT PLACEMENT OF EACH ON THE PARCEL. THIS MAY INCLUDE A STAMPED AND SIGNED SETBACK CERTIFICATE PREPARED BY A CALIFORNIA LICENSED SURVEYOR OR CIVIL ENGINEER. (COUNTY BUILDING CODE 91.1.107.2)

ENERGY COMPLIANCE & SOLAR NOTES

PROPERLY COMPLETED AND SIGNED CERTIFICATES OF INSTALLATION (CF2R FORMS) SHALL BE PROVIDED TO THE INSPECTOR IN THE FIELD. FOR PROJECTS REQUIRING HERS VERIFICATION, THE CF2R FORMS SHALL BE REGISTERED WITH A CALIFORNIA-APPROVED HERS PROVIDER DATA REGISTRY." CF2R FORMS ARE AVAILABLE AT HTTP://WWW.SDCOUNTY.CA.GOV/PDS/BLDG/ENERGY-STDS.HTML. (CBEES 10-103)

PROPERLY COMPLETED CERTIFICATES OF VERIFICATION (CF3R FORMS) SHALL BE PROVIDED TO THE INSPECTOR IN THE FIELD FOR ITEMS REQUIRING HERS VERIFICATION. CF3R FORMS SHALL BE REGISTERED WITH A CALIFORNIA-APPROVED HERS PROVIDER DATA REGISTRY." CF3R FORMS ARE AVAILABLE AT HTTP://WWW.SDCOUNTY.CA.GOV/PDS/BLDG/ENERGY-STDS.HTML. (CBEES 10-103)

SOLAR PV NOTE:

- IF REQUIRED BY ENERGY COMPLIANCE DOCUMENTATION, PROVIDE SOLAR PV SYSTEM UNDER SEPARATE PERMIT. SYSTEM SIZE SHALL COMPLY WITH APPROVED ENERGY DOCUMENTATION.
- IF SOLAR PV IS **NOT** REQUIRED, REFER TO SOLAR READY NOTES ON SHEET A2.

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

BY USING THESE STANDARD PLANS, THE USER AGREES TO RELEASE THE COUNTY OF SAN DIEGO FROM ANY AND ALL CLAIMS, LIABILITIES, SUITS, AND DEMANDS ON ACCOUNT OF ANY INJURY, DAMAGE, OR LOSS TO PERSONS OR PROPERTY, INCLUDING INJURY OR DEATH, OR ECONOMIC LOSSES, ARISING OUT OF THE USE OF THESE CONSTRUCTION DOCUMENTS. THE USE OF THESE PLANS DOES NOT ELIMINATE OR REDUCE THE USER'S RESPONSIBILITY TO VERIFY ANY AND ALL INFORMATION.

PROJECT TITLE

DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

PLOT PLAN

SHEET NUMBER

PP

EROSION CONTROL NOTES

1. ALL BUILDING PADS TO BE DIKED AND THE DIKES MAINTAINED TO PREVENT WATER FROM FLOWING FROM THE PAD UNTIL THE STREETS AND DRIVEWAYS ARE PAVED AND WATER CAN FLOW FROM THE PADS WITHOUT CAUSING EROSION, OR CONSTRUCT DRAINAGE FACILITIES TO THE SATISFACTION OF THE COUNTY DEPARTMENT OF PUBLIC WORKS THAT WILL ALLOW WATER TO DRAIN FROM THE PAD WITHOUT CAUSING EROSION.

2. TOPS OF ALL SLOPES TO BE DIKED OR TRENCHED TO PREVENT WATER FROM FLOWING OVER THE CREST OF SLOPES.

3. MANUFACTURED SLOPES AND PADS SHALL BE ROUNDED VERTICALLY AND HORIZONTALLY AS APPROPRIATE TO BLEND WITH THE SURROUNDING TOPOGRAPHY.

4. AS SOON AS CUTS OR EMBANKMENTS ARE COMPLETED, BUT NOT LATER THAN OCTOBER 1, ALL CUT AND FILL SLOPES SHALL BE STABILIZED WITH A HYDROMULCH MIXTURE OR AN EQUAL TREATMENT APPROVED BY THE COUNTY DEPARTMENT OF PUBLIC WORKS BETWEEN OCTOBER 1 AND APRIL 15. APPROVED SLOPE PROTECTION MEASURES SHALL PROCEED IMMEDIATELY BEHIND THE EXPOSURE OF CUT SLOPES AND/OR THE CREATION OF EMBANKMENT SLOPES.

5. CATCH BASINS, DESILTING BASINS AND STORM DRAIN SYSTEMS SHALL BE INSTALLED TO THE SATISFACTION OF THE COUNTY DEPARTMENT OF PUBLIC WORKS.

6. SAND BAG CHECK DAMS TO BE PLACED IN A MANNER APPROVED BY THE COUNTY DEPARTMENT OF PUBLIC WORKS IN UNPAVED STREETS WITH GRADIENTS IN EXCESS OF 2% AND ON OR IN OTHER GRADED OR EXCAVATED AREAS AS REQUIRED BY THE COUNTY DEPARTMENT OF PUBLIC WORKS.

7. THE DEVELOPER TO MAINTAIN THE PLANTING AND EROSION CONTROL MEASURES DESCRIBED ABOVE UNTIL RELIEVED OF THE SAME BY THE COUNTY DEPARTMENT OF PUBLIC WORKS. THE DEVELOPER TO REMOVE ALL SOIL INTERCEPTED BY THE SAND BAGS, CATCH BASINS AND DESILTING BASINS AND KEEP THESE FACILITIES CLEAN AND FREE OF SILT AND SAND AS DIRECTED BY THE COUNTY DEPARTMENT OF PUBLIC WORKS. THE DEVELOPER SHALL REPAIR ANY ERODED SLOPES AS DIRECTED BY THE COUNTY DEPARTMENT OF PUBLIC WORKS.

PLANTING & IRRIGATION NOTES

ALL PERMANENT AND TEMPORARY EROSION CONTROL PLANTING AND IRRIGATION SHALL BE INSTALLED AND MAINTAINED AS REQUIRED BY SECTIONS 87.417 AND 87.418 OF THE COUNTY'S GRADING ORDINANCE AND SHALL BE COMPLIANT WITH THE COUNTY'S WATER CONSERVATION IN LANDSCAPE ORDINANCE, THE WATER EFFICIENT LANDSCAPE DESIGN MANUAL, AND THE BMP DESIGN MANUAL:

A. SLOPE EROSION CONTROL:

1. AT A MINIMUM, ALL MANUFACTURED SLOPE AREAS SHALL BE COVERED WITHIN 10 DAYS OF COMPLETION OF GRADING WITH HYDROSEED/MULCH, PUNCHED STRAW MULCH, JUTE NETTING OR OTHER APPROVED GEOTEXTILE MATERIAL CAPABLE OF CONTROLLING SURFACE SOIL EROSION;

   a. EXCEPT WHERE APPROVED OTHERWISE, ALL SLOPES AND ANY OTHER AREAS DISTURBED IN CONJUNCTION WITH GRADING ACTIVITIES SHALL BE MAINTAINED UNTIL VEGETATION IS WELL ESTABLISHED, WITH COVERAGE EQUAL TO AT LEAST 70 PERCENT OF COVERAGE, AS COMPARED TO THE NATIVE BACKGROUND PLANTS IS ACHIEVED. THIS THRESHOLD MUST BE MET BEFORE OCCUPANCY OF THE SITE WILL BE PERMITTED;

   b. A MINIMUM OF 50 PERCENT OF THE TOTAL SLOPE AREAS OF MANUFACTURED SLOPES SHALL BE PLANTED WITH DEEP ROOTING PLANTINGS (I.E., THOSE WITH A TYPICAL ROOT DEPTH OF APPROXIMATELY 5 FEET OR GREATER). FOR SEEDED PLANTINGS, AT LEAST 50 PERCENT OF THE VIABLE SEED COUNT SHALL BE DEEP ROOTING SPECIES;

2. ALL PLANT MATERIALS ON MANUFACTURED SLOPES SHALL BE APPROPRIATE TO THE SITE CONDITIONS, SHALL BE WATER EFFICIENT WHEN ESTABLISHED AND SHALL BE ADEQUATELY SPACED TO CONTROL SOIL EROSION;

   a. ALL SLOPES IN EXCESS OF 15 FEET SHALL BE PLANTED WITH ROOTED CONTAINER STOCK AT AN AVERAGE RATE OF ONE PER 100 SQUARE FEET UNLESS APPROVED OTHERWISE. CONTAINERS SHALL BE A MINIMUM OF ONE GALLON FOR SHRUBS AND FIVE GALLON FOR TREES. ALL CONTAINER STOCK SHALL BE PROVIDED WITH A TEMPORARY IRRIGATION SYSTEM;

   b. SLOPES GREATER THAN 25 PERCENT SHALL NOT BE IRRIGATED WITH AN IRRIGATION SYSTEM WITH A PRECIPITATION RATE EXCEEDING 0.75 INCHES PER HOUR.

B. IRRIGATION

1. ALL IRRIGATION SYSTEMS SHALL BE INSTALLED TO AVOID RUNOFF, SEEPAGE, LOW HEAD DRAINAGE, OVER-SPRAY OR OTHER SIMILAR CONDITIONS ONTO ADJACENT PROPERTY, NON-IRRIGATED AREAS,WALKS, ROADWAYS OR STRUCTURES;

2. APPROVED BACKFLOW PREVENTION UNITS ARE REQUIRED ON ALL POTABLE WATER IRRIGATION SYSTEMS;

3. AFTER PLANT ESTABLISHMENT, THE IRRIGATION SYSTEM IS TO PROVIDE SUFFICIENT WATER TO SUSTAIN PLANTS IN A HEALTHY, GROWING CONDITION;

4. LANDSCAPE WATER METERS, SHALL BE INSTALLED FOR ALL NON-RESIDENTIAL IRRIGATED LANDSCAPES OF 1,000 SQ. FT. OR MORE, AND FOR ALL RESIDENTIAL IRRIGATED LANDSCAPES OF 5,000 SQ. FT. OR GREATER. A LANDSCAPE WATER METER SHALL BE EITHER A CUSTOMER SERVICE METER DEDICATED TO LANDSCAPE USE PROVIDED BY THE LOCAL WATER PURVEYOR; OR A PRIVATELY OWNED METER OR SUB-METER;

5. FLOW SENSORS THAT DETECT AND REPORT HIGH FLOW CONDITIONS CREATED BY SYSTEM DAMAGE OR MALFUNCTION ARE REQUIRED FOR ALL NON-RESIDENTIAL LANDSCAPES, AND RESIDENTIAL LANDSCAPES OF 5,000 SQ. FT. OR LARGER;

6. MASTER SHUT-OFF VALVES ARE REQUIRED;

7. AUTOMATIC CONTROL SYSTEMS ARE REQUIRED, AND MUST BE ABLE TO ACCOMMODATE ALL ASPECTS OF THE INSTALLED EROSION CONTROL LANDSCAPE, INCLUDING MULTIPLE SCHEDULES, REPEAT CYCLES, EVAPOTRANSPIRATION OR SOIL MOISTURE SENSING AND RAIN SENSING OVERRIDE DEVICES. CONTROL MECHANISMS FOR AN EVAPOTRANSPIRATION (WEATHER BASED) SYSTEM OR SOIL MOISTURE DETECTING SYSTEMS, UTILIZING NON-VOLATILE MEMORY SHALL BE ACCOMMODATED WITHIN THE CONTROLLER ENCLOSURE. BATTERY OPERATED VALVES OR SOLAR POWERED CONTROLLERS MAY BE USED IN AREAS WHERE ELECTRICITY IS NOT YET AVAILABLE, HOWEVER, WHEN AVAILABLE, SYSTEMS SHALL BE CONVERTED;

8. ALL IRRIGATION SYSTEMS SHALL BE ADJUSTED SEASONALLY AND AS WEATHER AND PLANT CONDITIONS WARRANT;

9. SHUTOFF VALVES SHALL BE PROVIDED AT POINTS OF CONNECTION TO DIVIDE THE IRRIGATION SYSTEM INTO CONTROLLABLE UNITS. CHECK VALVES OR ANTI-DRAIN VALVES ARE REQUIRED FOR ALL IRRIGATION SYSTEMS;

10. ALL PIPE AND WIRE UNDER VEHICULAR PAVING SHALL BE INSTALLED IN PVC SCHEDULE 40 SLEEVES. SLEEVES SHALL BE AT LEAST TWICE THE DIAMETER OF THE PIPE OR WIRE BUNDLES TO BE ENCLOSED;

11. PERMANENT ON-GRADE SYSTEMS MAY ONLY BE ALLOWED FOR AREAS WITH HIGHLY EROSIVE OR ROCKY SOILS AND SHALL NOT BE ALLOWED ADJACENT TO PEDESTRIAN TRAFFIC;

12. ALL ON-GRADE LINES SHALL BE SECURED TO SLOPES EVERY TEN FEET OR LESS. THE ENDS OF ALL LATERALS SHALL ALSO BE STAKED;

13. ON-GRADE LATERAL PIPING IS ALLOWED FOR TEMPORARY SYSTEMS;

14. ALL REQUIRED LANDSCAPING SHALL BE PLANTED AND IRRIGATION SYSTEMS INSTALLED PRIOR TO THE ROUGH GRADING INSPECTION. THE COUNTY OFFICIAL SHALL DETERMINE AT FINAL GRADING INSPECTION THAT PLANTING IS ESTABLISHED AND THAT ALL IRRIGATION SYSTEMS ARE OPERATIONAL;

C. SLOPE HYDROSEEDING MIX SHALL CONSIST OF THE FOLLOWING:

1. NON-IRRIGATED SEED MIX SHALL NOT BE ALLOWED AND NO INVASIVE, FIRE PRONE SPECIES SHALL BE INCLUDED;

2. SEED MIX SHALL CONSIST OF NO LESS THAN:

   a. ALL SEED MATERIALS SHALL BE TRANSPORTED TO THE JOBSITE IN UNOPENED CONTAINERS WITH THE CALIFORNIA DEPARTMENT OF FOOD AND AGRICULTURE CERTIFICATION TAG ATTACHED TO, OR PRINTED ON SAID CONTAINERS.
   b. NON-PHYTO-TOXIC WETTING AGENTS MAY BE ADDED TO THE HYDROSEED SLURRY AT THE DISCRETION OF THE CONTRACTOR.

3. TYPE 1 MULCH APPLIED AT THE RATE OF NO LESS THAN 2000 LBS. PER ACRE. TYPE 6 MULCH (STRAW) MAY BE SUBSTITUTED, ALL OR PART, FOR HYDRAULICALLY APPLIED FIBER MATERIAL. WHEN STRAW IS USED IT MUST BE ANCHORED TO THE SLOPE BY MECHANICALLY PUNCHING NO LESS THAN 50% OF THE STRAW INTO THE SOIL.

4. FERTILIZER CONSISTING OF AMMONIUM PHOSPHATE, SULFATE, 16-20-0, WITH 15% SULPHUR APPLIED AT THE RATE OF 500 LBS. PER ACRE.

5. AREAS TO BE HYDROSEEDED SHALL BE PREPARED PRIOR TO HYDROSEEDING BY:

   a. ROUGHENING THE SURFACE TO BE PLANTED BY ANY OR COMBINATION OF:

      a.a. TRACK WALKING SLOPES STEEPER THAN 6:1
      a.b. HARROWING AREAS 6:1 OR FLATTER THAT ARE SUFFICIENTLY FRIABLE.
      a.c. RIPPING AREAS THAT WILL NOT BREAK UP USING ITEMS a.a. OR a.b. ABOVE.

6. CONDITION THE SOILS SO THAT IT IS SUITABLE FOR PLANTING BY:

   a. ADJUSTING THE SURFACE SOIL MOISTURE TO PROVIDE A DAMP BUT NOT SATURATED SEED BED.
   b. THE ADDITION OF SOIL AMENDMENTS, PH ADJUSTMENT, LEACHING OR COVERING SALINE SOILS TO PROVIDED VIABLE CONDITIONS FOR GROWTH.

7. HYDROSEEDED AREAS SHALL BE MAINTAINED TO PROVIDE A VIGOROUS GROWTH UNTIL THE PROJECT IS PERMANENTLY LANDSCAPED OR, FOR AREAS WHERE HYDROSEEDING IS THE PERMANENT LANDSCAPING, UNTIL THE PROJECT IS COMPLETED AND ALL GRADING BONDS RELEASED.

HYDROSEED NOTES:

1. PRIOR TO THE ACCEPTANCE OF THE HYDROSEEDED AREAS BY THE COUNTY , A CERTIFICATION REPORT MUST BE SUBMITTED BY A REGISTERED LANDSCAPE ARCHITECT, STATING THAT THE HYDROSEEDING WAS DONE ACCORDING TO THE PROJECT SPECIFICATIONS AND THAT THE GROWTH IS ADEQUATELY ESTABLISHED TO PREVENT EROSION.

2. ALL DISTURBED SLOPES TO BE HYDROSEEDED UNLESS OTHERWISE NOTED.

PLANT LEGEND:

** PROVIDE A PLANT LEGEND WITH BOTH BOTANICAL AND COMMON NAMES LISTED ALONG WITH SYMBOLS FOR IDENTIFICATION AND LOCATION. PROVIDE CONTAINER SIZES, QUANTITIES OF EACH SPECIES, AND ON CENTER SPACE REQUIREMENTS. NO INVASIVE OR FIRE PRONE SPECIES SHALL BE INTRODUCED TO THE SITE.

** PROVIDE A PLANT LEGEND OF ALL PROPOSED HYDROSEED SPECIES WITH BOTH BOTANICAL AND COMMON NAMES LISTED. INCLUDE POUNDS PER ACRE FOR EACH SEED SPECIES. NO INVASIVE OR FIRE PRONE SPECIES SHALL BE INTRODUCED TO SITE.

NOTE:

VEGETATION STABILIZATION (PLANTING OR HYDROSEEDING) MAY BE INSTALLED BETWEEN MAY 1ST AND AUGUST 15TH. SLOPE IRRIGATION MUST BE IN PLACE AND TO BE OPERABLE FOR SLOPES >3'. VEGETATION MUST BE WATERED AND ESTABLISHED PRIOR TO OCTOBER 1ST. THE OWNER SHALL SUPPLEMENT VEGETATION WITH BARK OR MULCH BY OCTOBER 1ST IF VEGETATION ESTABLISHMENT DOES NOT OCCUR BY THAT DATE. EROSION CONTROL MEASURES MUST ALSO BE USED WHILE LANDSCAPING IS BEING ESTABLISHED. ESTABLISHED VEGETATION SHALL HAVE A SUBSURFACE MAT OF INTERTWINED MATURE ROOTS WITH A UNIFORM VEGETATION COVERAGE OF 70% OF THE NATURAL VEGETATIVE COVERAGE OR MORE ON ALL DISTURBED AREAS.

ALL SLOPES OVER THREE FEET MUST HAVE ESTABLISHED VEGETATIVE COVER PRIOR TO FINAL PERMIT APPROVAL

BMP LEGEND

PDS 659	BROW DITCH	⇒ ⇒
PDS 659	BERM	→ B →
DIRECTION OF LOT DRAINAGE		→ →

MATERIALS & WASTE MANAGEMENT BMPs:

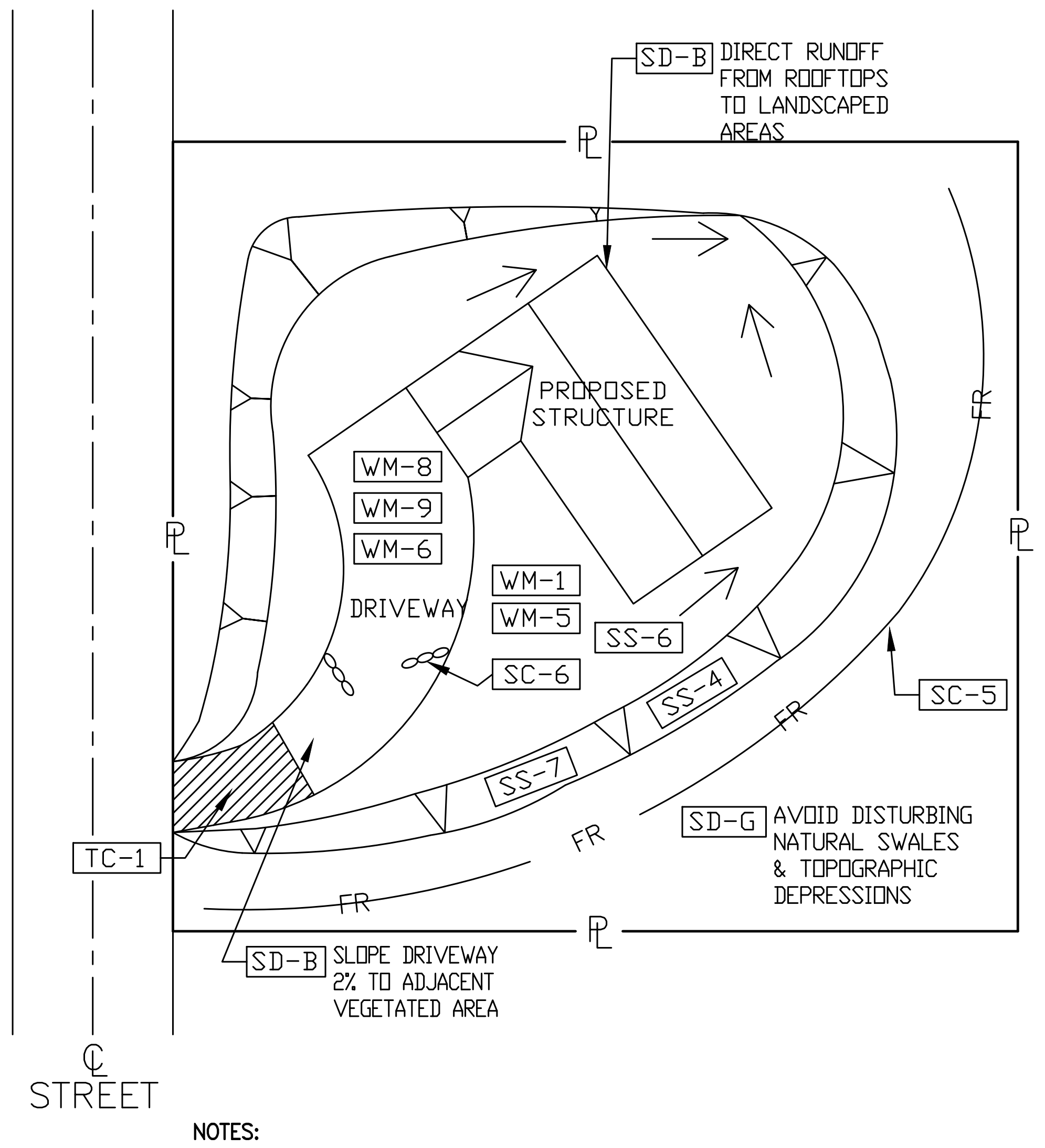
WM-1	MATERIAL DELIVERY & STORAGE
WM-4	SPILL PREVENTION AND CONTROL
WM-8	CONCRETE WASTE MANAGEMENT
WM-5	SOLID WASTE MANAGEMENT
WM-9	SANITARY WASTE MANAGEMENT
WM-6	HAZARDOUS WASTE MANAGEMENT

TEMPORARY RUNOFF CONTROL BMPs:

SS-2	PRESERVATION OF EXISTING VEGETATION	⌒PEV ⌒PEV ⌒
SS-3	BONDED OR STABILIZED FIBER MATRIX (WINTER)	⌒M ⌒M ⌒
SS-4	HYDROSEEDING (SUMMER)	⌒TSP ⌒TSP ⌒
SS-6 / SS-8	STRAW OR WOOD MULCH	⌒S/W ⌒S/W ⌒
SS-7	PHYSICAL STABILIZATION (WINTER)	⌒EBM ⌒EBM ⌒
SS-10	ENERGY DISSIPATOR	▨
SC-1	SILT FENCE	—■—■—
SC-2 / PDS 659	SEDIMENT / DESILTING BASIN	⬭
SC-5	FIBER ROLLS	—FR —FR —
SC-6 / SC-8	GRAVEL OR SAND BAGS	○○○○○
SC-7	STREET SWEEPING AND VACUUMING	
SC-10	STORM DRAIN INLET PROTECTION	▦
NS-2	DEWATERING FILTRATION	—(DW)—(DW)—
TC-1	STABILIZED CONSTRUCTION ENTRANCE	▨
TC-2	CONSTRUCTION ROAD STABILIZATION	▩
TC-3	ENTRANCE / EXIT TIRE WASH	▨

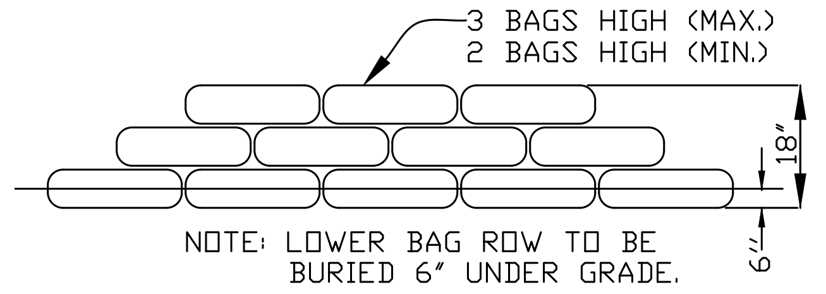
BASELINE BMPs FOR EXISTING AND PROPOSED SITE FEATURES

SD-B	DIRECT RUNOFF TO PERVIOUS AREAS
SD-C	INSTALL GREEN ROOF
SD-E	INSTALL RAIN BARRELS
SD-G	CONSERVE NATURAL FEATURES
SD-H	PROVIDE BUFFERS AROUND WATER BODIES
SD-I	CONSTRUCT SURFACES FROM PERMEABLE MATERIALS
SD-K	SUSTAINABLE LANDSCAPING

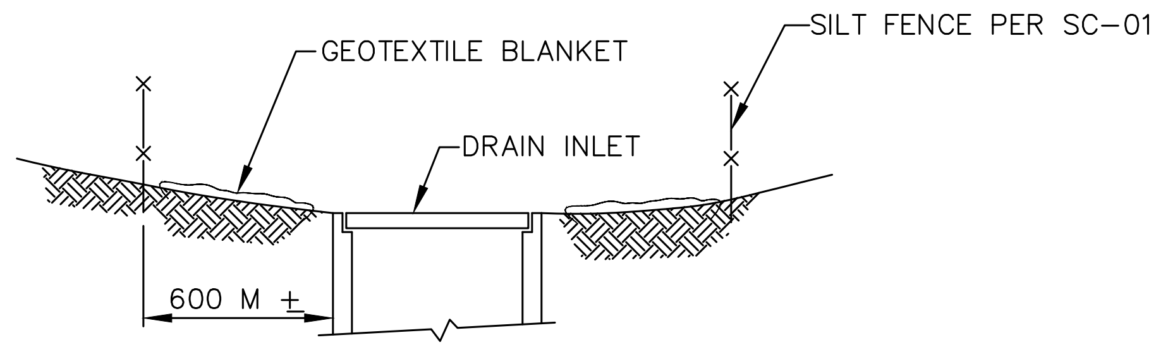

SD-B DIRECT RUNOFF FROM ROOFTOPS TO LANDSCAPED AREAS
PL
PROPOSED STRUCTURE
WM-8
WM-9
WM-6
DRIVEWAY
WM-1
WM-5
SS-6
SC-6
SS-4
SS-7
SC-5
FR
FR
FR
FR
TC-1
SD-G AVOID DISTURBING NATURAL SWALES & TOPOGRAPHIC DEPRESSIONS
SD-B SLOPE DRIVEWAY 2% TO ADJACENT VEGETATED AREA
℄
STREET

NOTES:
- TO BE USED IN CONJUNCTION WITH STORM WATER QUALITY MANAGEMENT PLAN (SWQMP) FOR STANDARD PROJECTS (FORM: STANDARD SWQMP)
- THIS PLAN REPRESENTS THE SITE BMP PLAN. ACTUAL IMPLEMENTATION OF SITE BMPS IS AT THE DISCRETION OF THE BUILDING INSPECTOR.

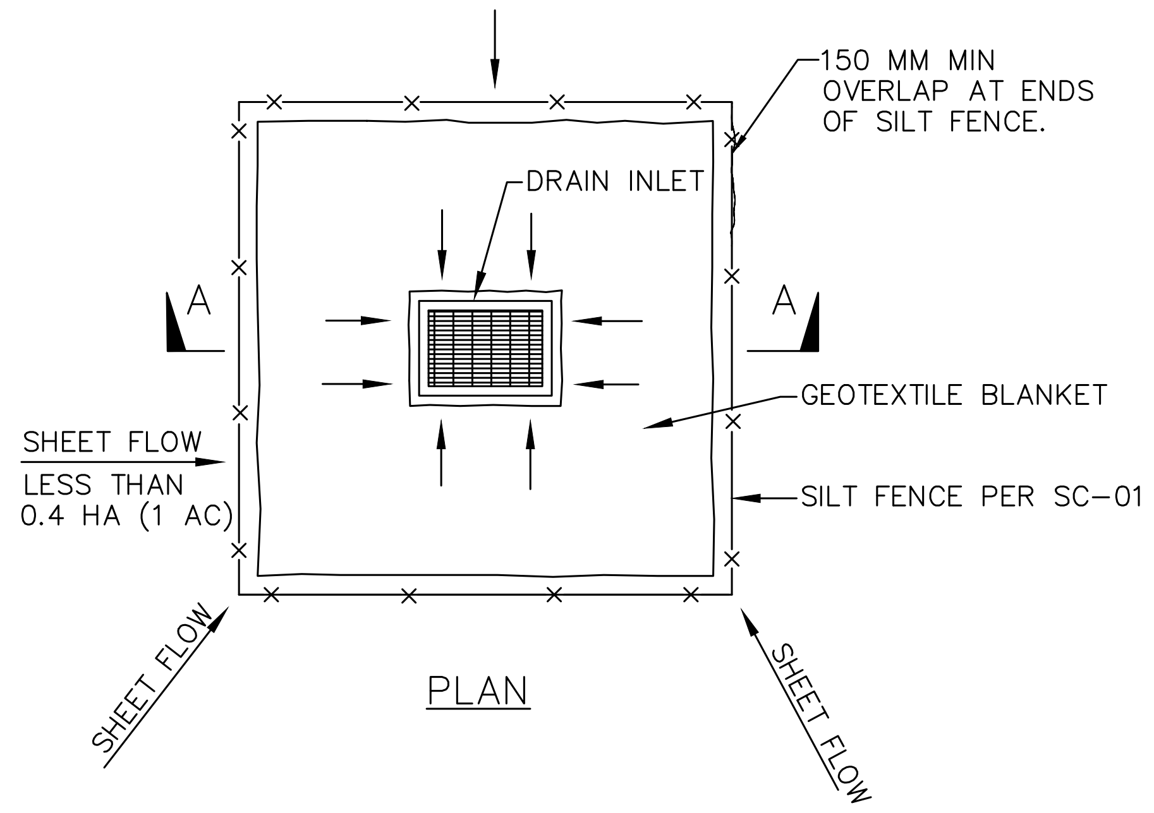
STANDARD BMP PLAN
NO SCALE


3 BAGS HIGH (MAX.)
2 BAGS HIGH (MIN.)
18"
6"
NOTE: LOWER BAG ROW TO BE BURIED 6" UNDER GRADE.

GRAVEL BAG CHECK DAM
NO SCALE


GEOTEXTILE BLANKET
SILT FENCE PER SC-01
DRAIN INLET
600 M ±

SECTION A-A

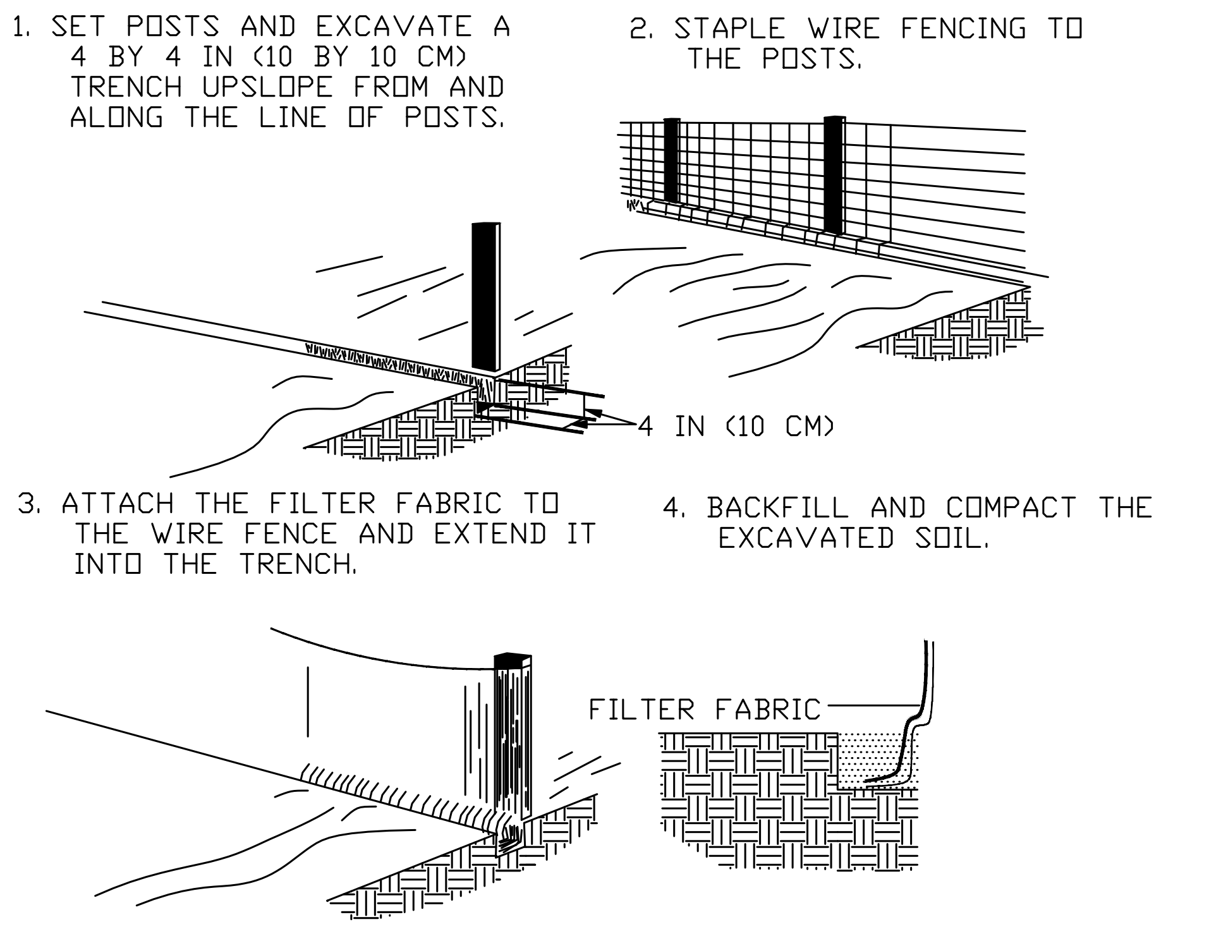

150 MM MIN OVERLAP AT ENDS OF SILT FENCE.
DRAIN INLET
A
A
GEOTEXTILE BLANKET
SHEET FLOW
LESS THAN 0.4 HA (1 AC)
SILT FENCE PER SC-01
SHEET FLOW
SHEET FLOW

PLAN

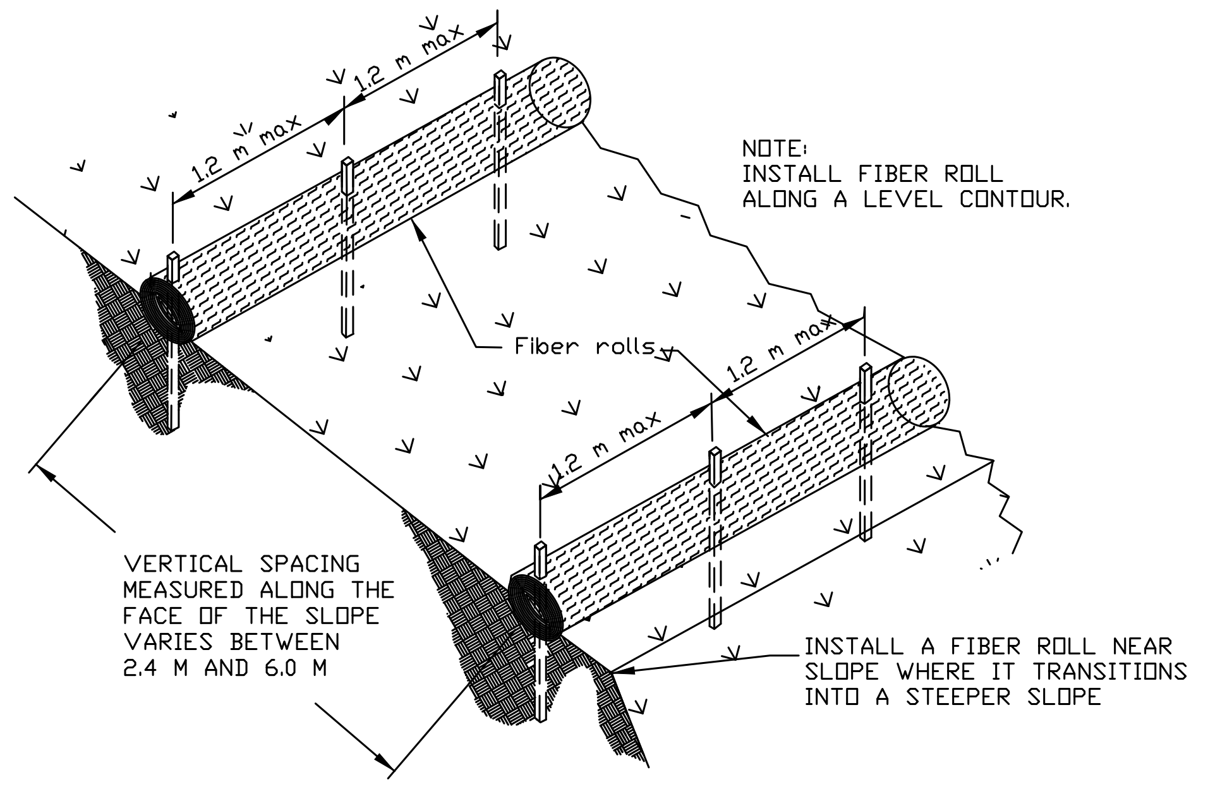
NOTES:
1. FOR USE IN AREAS WHERE GRADING HAS BEEN COMPLETED AND FINAL SOIL STABILIZATION AND SEEDING ARE PENDING.
2. NOT APPLICABLE IN PAVED AREAS.
3. NOT APPLICABLE WITH CONCENTRATED FLOWS.

DRAIN INLET PROTECTION TYPE 1
NO SCALE

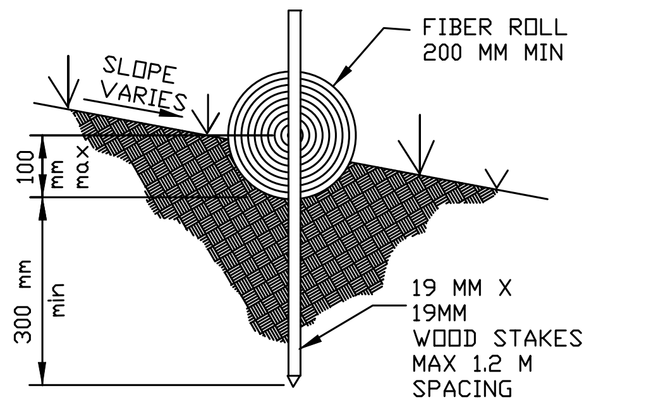
TYPICAL BMP DETAILS


1. SET POSTS AND EXCAVATE A 4 BY 4 IN (10 BY 10 CM) TRENCH UPSLOPE FROM AND ALONG THE LINE OF POSTS.
2. STAPLE WIRE FENCING TO THE POSTS.
4 IN (10 CM)
3. ATTACH THE FILTER FABRIC TO THE WIRE FENCE AND EXTEND IT INTO THE TRENCH.
4. BACKFILL AND COMPACT THE EXCAVATED SOIL.
FILTER FABRIC

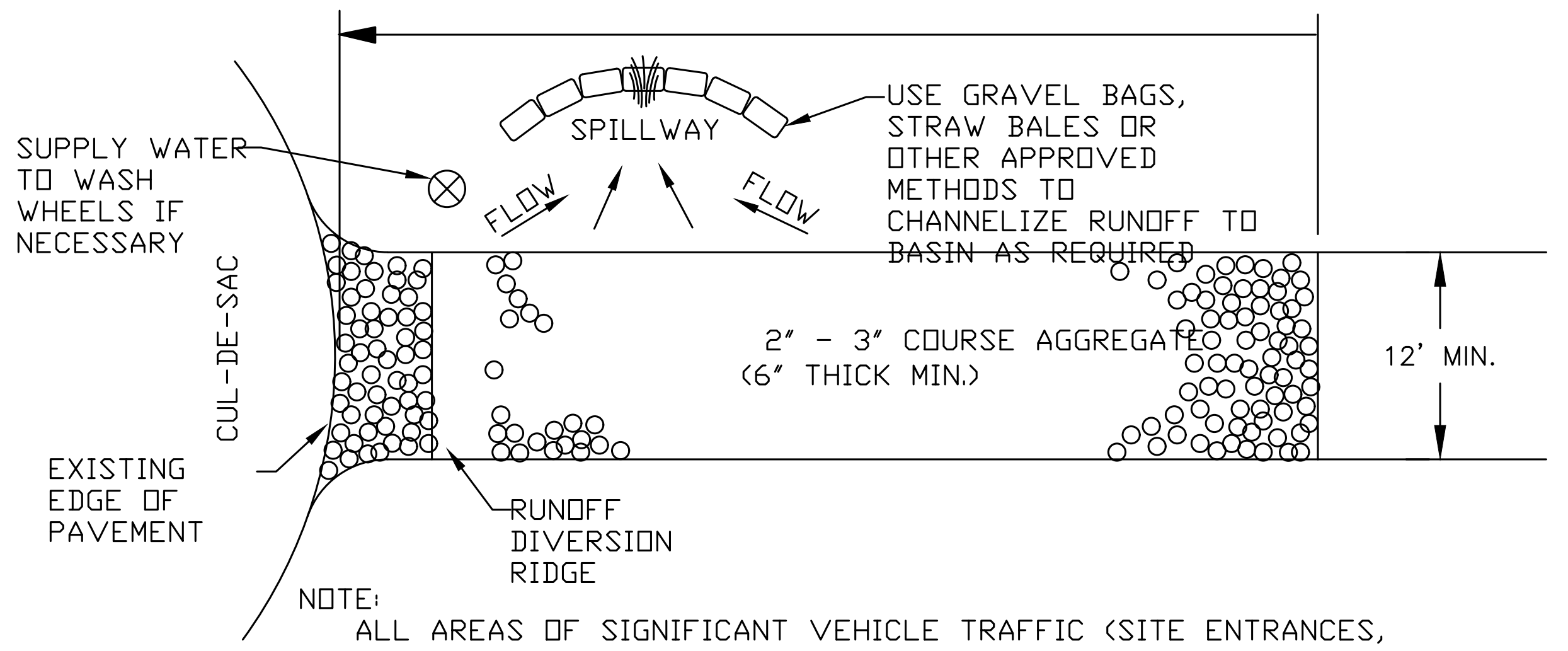
SILT FENCE
NO SCALE


1.2 m max
1.2 m max
1.2 m max
1.2 m max
NOTE: INSTALL FIBER ROLL ALONG A LEVEL CONTOUR.
Fiber rolls
VERTICAL SPACING MEASURED ALONG THE FACE OF THE SLOPE VARIES BETWEEN 2.4 M AND 6.0 M
INSTALL A FIBER ROLL NEAR SLOPE WHERE IT TRANSITIONS INTO A STEEPER SLOPE

TYPICAL FIBER ROLL INSTALLATION
NO SCALE


SLOPE VARIES
FIBER ROLL 200 MM MIN
100 mm max
300 mm min
19 MM X 19MM WOOD STAKES MAX 1.2 M SPACING

ENTRENCHMENT DETAIL
NO SCALE


SUPPLY WATER TO WASH WHEELS IF NECESSARY
SPILLWAY
FLOW
FLOW
USE GRAVEL BAGS, STRAW BALES OR OTHER APPROVED METHODS TO CHANNELIZE RUNOFF TO BASIN AS REQUIRED
CUL-DE-SAC
2" - 3" COURSE AGGREGATE (6" THICK MIN.)
12' MIN.
EXISTING EDGE OF PAVEMENT
RUNOFF DIVERSION RIDGE

NOTE:
ALL AREAS OF SIGNIFICANT VEHICLE TRAFFIC (SITE ENTRANCES, ACCESS ROADS, PARKING LOTS, ETC.) SHOULD BE STABILIZED IMMEDIATELY AFTER GRADING TO PREVENT EROSION AND CONTROL DUST. SITE ENTRANCES AND EXITS ARE ESPECIALLY IMPORTANT. USE GRAVEL APPROACHES TO LIMIT TRACKING OF SEDIMENT OFFSITE.

STABILIZED CONSTRUCTION ENTRANCE
NO SCALE

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

STANDARD BMP PLAN

SHEET NUMBER

BMP

County of San Diego
Stormwater Quality Management Plan (SWQMP)
Attachment 1: Stormwater Intake Form for All Permit Applications

This form establishes Stormwater Quality Management Plan (SWQMP) requirements for Development Projects per Sections 67.809 and 67.811 of the County of San Diego Watershed Protection Ordinance (WPO). See ***Stormwater Intake Form Instructions*** for additional guidance and explanation of terms.

Part 1. Project Information	
Project Name:	
Record ID (Permit) No(s):	
Assessor's Parcel No(s):	
Street Address (or Intersection):	
City, State, Zip:	

Part 2. Applicant / Project Proponent Information	
Name:	
Company:	
Street Address:	
City, State, Zip:	
Phone Number	
Email:	

Part 3. Required Information for All Development Projects

Ⓐ **1. Existing (pre-development) impervious surfaces (ft²)**	**2. Created or replaced impervious surfaces (ft²)**	**3. Total disturbed area (acres or ft²)**

Ⓑ ☐ Check here and provide a WDID# if this project is subject to the California Construction General Permit (Order No. 2009-0009-DWQ)[1]

WDID # (if issued)

For County Use Only	Reviewed By:	Review Date:
☐ Standard SWQMP	☐ PDP SWQMP	☐ Green Streets PDP Exempt SWQMP

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[1] Available at: https://www.waterboards.ca.gov/water_issues/programs/stormwater/construction.html

Part 4. Priority Classification & SWQMP Form Selection

Ⓐ ***If your project is the following … (select one)***	Ⓑ ***You must complete …***
☐ **Standard Project** ☐ a. Project is East of the Pacific/Salton Sea Divide ☐ b. None of the PDP criteria below applies	→ **Standard *SWQMP Form***
☐ **Priority Development Project (PDP)** ☐ 1. Project is part of an existing PDP, OR ☐ 2. Project does any of the following: ☐ a. Creates or replaces a total of 10,000 ft² or more of impervious surface ☐ b. Creates or replaces a combined total of 5,000 ft² or more of impervious surface within one or more of the following uses: (1) parking lots; (2) streets, roads, highways, freeways, and/or driveways; (3) restaurants; and (4) hillsides ☐ c. Creates or replaces a combined total of 5,000 ft² or more of impervious surface within one or more of the following uses: (1) automotive repair shops; and (2) retail gasoline outlets ☐ d. Discharges directly to an Environmentally Sensitive Area (ESA) AND creates or replaces 2,500 ft² or more of impervious surface ☐ e. Disturbs one or more acres of land (43,560 ft²) and is expected to generate pollutants post-construction ☐ f. Is a redevelopment project that creates or replaces 5,000 ft² or more of impervious surface on a site already having at least 10,000 ft² of impervious surface Refer to Section 1.4 of the County BMP Design Manual for additional guidance.	→ **PDP *SWQMP Form/ PDP SWQMP Amendment required with scope change***
☐ **Green Streets PDP Exemption**[2] Select this option only if the entire project qualifies for Green Streets PDP Exemption. Refer to Section 1.4.3 of the County BMP Design Manual for additional guidance.	→ **Green *Streets PDP Exempt SWQMP Form***

Part 5. Applicant Signature

I have reviewed the information in this form, and it is true and correct to the best of my knowledge.

Applicant / Project Proponent Signature: Date:

- ***Upon completion*** *submit this form to the County.*
- ***If requested****, attach supporting documentation to justify selections made or exemptions claimed.*
- ***If this is a PDP that is part of a larger existing PDP****, you will be required to attach a copy of the existing SWQMP to the newer SWQMP submittal.*

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[2] ***Green Streets PDP Exemption Projects*** are those claiming exemption from PDP classification per WPO Section 67.811(b)(2) because they consist exclusively of *either* 1) development of new sidewalks, bike lanes, and/or trails; *or* 2) improvements to existing roads, sidewalks, bike lanes, and/or trails.

County of San Diego
Stormwater Quality Management Plan (SWQMP)
For Standard Projects
Use for all Standard Projects (see Storm Water Intake Form, Part 4)

Standard Project

Project Information			
Project Name			
Project Address			
Assessor's Parcel # (APN)			
Permit # / Record ID			

Project Applicant / Project Proponent			
Name			
Address			
Phone		**Email:**	

SWQMP Preparer			
Name			
Company (if applicable)			
Address			
Phone		**Email:**	
PE Number (if applicable)			

Preparer's Certification

I understand that the County of San Diego has adopted minimum requirements for managing urban runoff, including storm water, from land development activities, as described in the County of San Diego BMP Design Manual. The BMP Design Manual is a design manual for compliance with local County of San Diego Watershed Protection Ordinance (Sections 67.801 et seq.) and regional MS4 Permit (California Regional Water Quality Control Board San Diego Region Order No. R9-2013-0001, as amended by Order No. R9-2015-0001 and Order No. R9-2015-0100) requirements for storm water management.

This SWQMP is intended to comply with applicable requirements of the BMP Design Manual. I certify that it has been completed to the best of my ability and accurately reflects the project being proposed and the applicable BMPs proposed to minimize the potentially negative impacts of this project's land development activities on water quality. I understand and acknowledge that the plan check review of this SWQMP by County staff is confined to a review and does not relieve me as the person in charge of overseeing the selection and design of storm water BMPs for this project, of my responsibilities for project design.

Signature	**Date**

COUNTY ACCEPTED
SWQMP Approved By: *Approval Date:*

**** NOTE* Approval does not constitute compliance with regulatory requirements.***

General Directions

Note: These directions may be omitted from the final SWQMP submittal.

Attachment 1: Stormwater Intake Form (Required)

Include a copy of your completed ***Storm Water Intake Form*** as **Attachment 1** to this submittal.

Table 1: Baseline BMPs for Existing and Proposed Site Features

Complete **Table 1** to document existing and proposed site features and the BMPs to be implemented for them. All BMPs must be implemented ***where applicable and feasible***. Applicability is generally assumed if a feature exists or is proposed.

BMPs must be implemented for **Table 1** features ***where feasible***. Leaving the box for a BMP unchecked means it will not be implemented (either partially or fully) either because it is inapplicable or infeasible. Explanations must be provided in **Table 4**. **Table 1** provides specific instructions on when explanations are required.

Table 2: Baseline BMPs for Pollutant-generating Sources

Table 2 is not required for Small Residential Projects. Applicants should check the box at the top of the table to confirm it does not apply.

> *Small Residential Projects* are those requiring *either:* a Building Permit, Minor Residential Grading Permit, or Site Plan Permit for a single family home; *or* a Tentative Parcel Map Permit for up to 4 single family homes and a remainder parcel.

All other projects must complete **Table 2** to identify applicable requirements for documenting pollutant-generating sources/ features and source control BMPs.

BMPs must be implemented for **Table 2** features ***where feasible***. Leaving the box for a BMP unchecked means it will not be implemented (either partially or fully) either because it is inapplicable or infeasible. Explanations must be provided in **Table 4**. **Table 2** provides specific instructions on when explanations are required.

Table 3: Temporary Construction Phase BMPs

Complete **Table 3** to document which BMPs will be implemented for each construction activity on the site. For each applicable activity, all BMPs listed must be fully implemented ***where applicable and feasible***. Provide justifications for not fully implementing BMPs in **Table 4**, Part 2.

Table 4: Required Explanations

Use **Table 4** to provide justifications for Table 1, 2, and 3 BMPs as shown below.

• Table 1 and 2 Site Features & Sources • Table 3 Construction Activities	Any feature or activity for which NO BMPs are selected (you must explain why all BMPs are infeasible)	Individual BMPs that will either not be implemented, or will only be partially implemented

Table 5: Construction Plan Set Requirements

Complete **Table 5** to verify that the construction plans incorporate all applicable site features, activities, and BMPs identified in this SWQMP (**Tables 1, 2, and 3)**.

Table 1 – Baseline BMPs for Existing and Proposed Site Features

A. BMPs for Existing Natural Site Features (See Fact Sheet BL-1)

1. Check the boxes below for each existing feature on the site. *2. Select the BMPs to be implemented for each identified feature. Explain why any BMP not selected is infeasible in Table 4.*

	Conserve natural features (SD-G)	**Provide buffers around waterbodies (SD-H)**
☐ Natural waterbodies	☐	☐
☐ Natural storage reservoirs & drainage corridors	☐	---
☐ Natural areas, soils, & vegetation (incl. trees)	☐	---

1. Check the boxes below for each proposed feature. *2. Select the BMPs to be implemented for each proposed feature. If neither BMP **SD-B** nor **SD-I** is selected for a feature, explain why both BMPs are infeasible in Table 4.*

	a. Direct runoff to pervious areas (SD-B)	**b. Construct surfaces from permeable materials (SD-I)**	**c. Minimize the size of impervious areas**
☐ Streets and roads	☐	☐	☐ *Check this box to confirm that all impervious areas on the site will be minimized where feasible.* *If this box is not checked, identify the surfaces that cannot be minimized in Table 4, and explain why it is infeasible to do so.*
☐ Sidewalks & walkways	☐	☐	
☐ Parking areas & lots	☐	☐	
☐ Driveways	☐	☐	
☐ Patios, decks, & courtyards	☐	☐	
☐ Hardcourt recreation areas	☐	☐	
☐ Other:	☐	☐	

C. ☐ **BMPs for Rooftop Areas:** *Check this box if rooftop areas are proposed and select at least one BMP below.*

If no BMPs are selected, explain why they are infeasible in Table 4.

1. Direct runoff to pervious areas (SD-B)	**2. Install green roofs (SD-C)**	**3. Install rain barrels (SD-E)**
☐	☐	☐

D. ☐ **BMPs for Landscaped Areas:** *Check this box if landscaping is proposed and select at least one BMP below.*

If no BMPs are selected, explain why they are infeasible in Table 4.

1. Sustainable Landscaping (SD-K)
☐

Note: All features and BMPs must be shown on applicable construction plans. See applicable Fact Sheets in Appendix C of the BMP Design Manual for additional information.

Note: Use Table 4 to explain BMP infeasibility or inapplicability, or to describe features or BMPs not listed in this table. Additional explanation may be required by the County.

Table 2 – Baseline BMPs for Pollutant-generating Sources

☐ If this is a **Small Residential Project**, check this box and skip the rest of this table.

A. Management of Stormwater Discharges

1. Identify all proposed outdoor work areas below (☐ *Check here if none are proposed*)

2. Which BMPs will be used to prevent materials from contacting rainfall or runoff? (See Fact Sheet BL-5) *(Select all feasible BMPs for each work area[1])*

3. Where will runoff from the work area be routed? (See Fact Sheet BL-6) *(Select one or more option for each work area)*

	Overhead covering (rooftops, etc.) (SC-A)	**Separation of flows from adjacent areas (berms, etc.) (SC-B)**	**Wind protection (screens, etc.) (SC-C)**	**Sanitary sewer[2] (SC-D)**	**Containment system (SC-E)**	**Stormwater S-BMP or SSD-BMP[3]**	**Other[4]**
☐ Trash & Refuse Storage	☐	☐	☐	☐	☐	☐	☐
☐ Materials & Equipment Storage	☐	☐	☐	☐	☐	☐	☐
☐ Loading & Unloading	☐	☐	---	☐	☐	☐	☐
☐ Fueling	☐	☐	---	☐	☐	☐	☐
☐ Maintenance & Repair	☐	☐	---	☐	☐	☐	☐
☐ Vehicle & Equipment Cleaning	☐	☐	---	☐	☐	☐	☐
☐ Other:	☐	☐	---	☐	☐	☐	☐

B. Prevention of Non-stormwater Discharges (See Fact Sheet BL-7)

Select one option for each feature below:

• **Storm drain inlets and catch basins …**	☐ are not proposed	☐ will be labeled with stenciling or signage to discourage dumping **(SC-F)**
• **Educational BMP Signage …**	☐ are not proposed	☐ will be labeled with educational signage for BMP **(SC-G)**
• **Interior work surfaces, floor drains, & sumps …**	☐ are not proposed	☐ will not discharge directly or indirectly to the MS4 or receiving waters
• **Drain lines (e.g., air conditioning, boiler, etc.) …**	☐ are not proposed	☐ will not discharge directly or indirectly to the MS4 or receiving waters
• **Fire sprinkler test water …**	☐ are not proposed	☐ will not discharge directly or indirectly to the MS4 or receiving waters

Note: All outdoor features and BMPs in this table must be shown on applicable construction plans. See applicable Fact Sheets in Appendix C of the BMP Design Manual for additional information.

Note: Use Table 4 to explain BMP infeasibility or inapplicability, or to describe features or BMPs not listed in this table. Additional explanation may be required by the County.

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[1] Each BMP is required where feasible. If none are selected for any feature, explain why they are infeasible in Table 4.

[2] Separate wastewater agency approvals may be required.

[3] Structural Treatment Control BMPs (S-BMPs) and Significant Site Design BMPs (SSD-BMPs) are permanent BMPs for Priority Development Projects (PDPs). Most Standard Projects do not have pre-existing or proposed S-BMPs and SSD-BMPs.

[4] Describe other proposed options for managing stormwater discharges in Table 4.

Table 3 –Minimum Construction Stormwater BMPs

Minimum Required BMPs by Activity Type Select all applicable activities and at least one BMP for each	**References** **Caltrans[5]**	**County of San Diego**
☐ **Erosion Control for Disturbed Slopes** (choose at least 1 per season)		
☐ Vegetation Stabilization Planting[6] (Summer)	SS-2, SS-4	
☐ Hydraulic Stabilization Hydroseeding[6] (Summer)	SS-4	
☐ Bonded Fiber Matrix or Stabilized Fiber Matrix[7] (Winter)	SS-3	
☐ Physical Stabilization Erosion Control Blanket[7] (Winter)	SS-7	
☐ **Erosion control for disturbed flat areas (slope < 5%)**		
☐ County Standard Lot Perimeter Protection Detail	SC-2	PDS 659[8]
☐ Use of Item A erosion control measures on flat areas	SS-3, SS-4, SS-7	
☐ County Standard Desilting Basin (must treat all site runoff)	SC-2	PDS 660[9]
☐ Mulch, straw, wood chips, soil application	SS-6, SS-8	
☐ **Energy dissipation (required to control velocity for concentrated runoff or dewatering discharge)**		
☐ Energy Dissipater Outlet Protection	SS-10	RSD D-40[10]
☐ **Sediment control for all disturbed areas**		
☐ Silt Fence	SC-1	
☐ Fiber Rolls (Straw Wattles)	SC-5	
☐ Gravel & Sand Bags	SC-6, SC-8	
☐ Dewatering Filtration	NS-2	
☐ Storm Drain Inlet Protection	SC-10	
☐ Engineered Desilting Basin (sized for 10-year flow)	SC-2	
☐ **Preventing offsite tracking of sediment**		
☐ Stabilized Construction Entrance	TC-1	
☐ Construction Road Stabilization	TC-2	
☐ Entrance/Exit Tire Wash	TC-3	
☐ Entrance/Exit Inspection & Cleaning Facility	TC-1	
☐ Street Sweeping and Vacuuming	SC-7	
☐ **Materials Management**		
☐ Material Delivery & Storage	WM-1	
☐ Spill Prevention and Control	WM-4	
☐ **Waste Management[11]**		
☐ Waste Management Concrete Waste Management	WM-8	
☐ Solid Waste Management	WM-5	
☐ Sanitary Waste Management	WM-9	
☐ Hazardous Waste Management	WM-6	

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[5] See Caltrans 2017 Construction Site Best Management Practices (BMP) Manual, available at: https://dot.ca.gov/programs/construction/storm-water-and-water-pollution-control/manuals-and-handbooks

[6] Planting or Hydroseeding may be installed between May 1st and August 15th. Slope irrigation must be in place and operable for slopes >3 feet. Vegetation must be watered and established prior to October 1st. A contingency physical BMP must be implemented by August 15th if vegetation is not established by that date. If landscaping is proposed, erosion control measures must also be used while landscaping is being established. Established vegetation must have a subsurface mat of intertwined mature roots with a uniform vegetative coverage of 70 percent of the natural vegetative coverage or more on all disturbed areas.

[7] All slopes over three feet must have established vegetative cover prior to final permit approval.

[8] County PDS 659. Standard Lot Perimeter Protection Design System (Bldg. Division)

[9] County PDS 660. County Standard Desilting Basin for Disturbed Areas of 1 Acre or Less Bldg. Division

[10] Regional Standard Drawing D-40 – Rip Rap Energy Dissipater (also acceptable for velocity reduction)

[11] Applicants are responsible to apply appropriate BMPs for specific wastes (e.g., BMP WM-8 for concrete).

Table 4 – Required Explanations

Part 1: Baseline BMPs (from Tables 1 and 2)

- **Required Justifications**: Provide explanations of BMP inapplicability and/or infeasibility as indicated per Tables 1 and 2.
- **If Requested**: Justify why individual BMPs will not be implemented or will only be partially implemented.
- **Additional Explanation**: Describe any proposed features and/or BMPs not listed in Tables 1 or 2.

BMP-Feature Combination		**Explanation**
Feature		
BMP		
Feature		
BMP		
Feature		
BMP		

Part 2: Justifications for Temporary Construction Phase BMPs (from Table 3)

- **Required Justifications**: Justify all construction activity types for which NO BMPs were selected.
- **If Requested**: Justify why specific individual BMPs were not selected.
- **Additional Explanation**: Describe any proposed features and/or BMPs not listed in Table 3.

BMP-Activity Combination		**Explanation**
Activity Type		
BMP		
Activity Type		
BMP		

Table 5 – Construction Plan Set Requirements[12]

Use this checklist to confirm that the required information is included on the construction plans.
☐ Include all source control and site design measures described in the SWQMP.
☐ Include all construction BMPs described in the SWQMP.

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[12] For Building Permit Applications, refer to Form PDS 272, https://www.sandiegocounty.gov/content/dam/sdc/pds/docs/pds272.pdf

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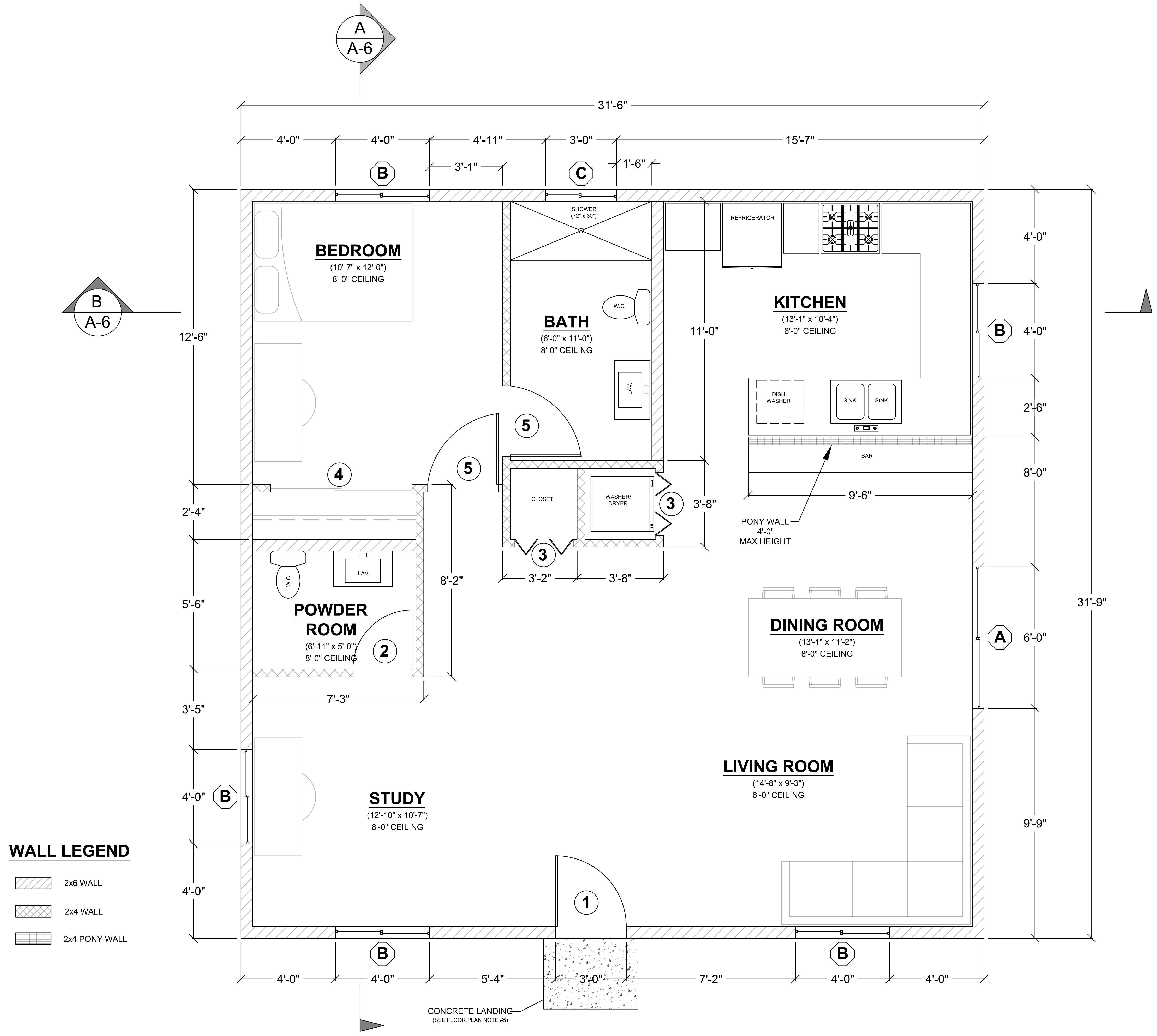
COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

STORMWATER INTAKE AND STORMWATER STANDARD (SWQMP) FORMS

SHEET NUMBER

SW1

(REV. 01/01/2026)



FLOOR PLAN
3/8" = 1'-0"

WINDOW SCHEDULE

MARK	DIMENSION	TYPE	TEMPERED	NOTES
A	6'-0" x 4'-0"	SLIDING		
B	4'-0" x 4'-0"	SLIDING		
C	3'-0" x 2'-0"	SLIDING	Y	

EXTERIOR WINDOWS, WINDOW WALLS AND GLAZED DOORS, WINDOWS WITHIN EXTERIOR DOORS, AND SKYLIGHTS PER ANY OF THE FOLLOWING (COUNTY CONSOLIDATED WUI CODE 96.2.504.8):

A. MULTILAYERED GLAZED PANELS WITH AT LEAST TWO PANES OF TEMPERED GLASS COMPLYING WITH SECTION 2406 OF THE CALIFORNIA BUILDING CODE. SKYLIGHTS/SOLAR TUBES SHALL BE TEMPERED GLASS PER LISTING REPORT OR MODIFIED SUBJECT TO FIELD INSPECTION.
B. GLASS BLOCK
C. GLAZING WITH A FIRE-PROTECTION RATING OF NOT LESS THAN 20 MINUTES WHEN TESTED ACCORDING TO NFPA 257 OR UL 9, AND SHALL BE EXEMPT FROM THE HOSE STREAM TEST.

DOOR SCHEDULE

MARK	DIMENSION	TYPE	TEMPERED	NOTES
①	3'-0" x 6'-8"	SWINGING		1-3/8" SOLID CORE
②	2'-6" x 6'-8"	SWINGING		
③		BI-FOLD		LAUNDRY ROOM
④	6'-0" x 6'-8"	SLIDING		6FT CLOSET
⑤	3'-0" x 6'-8"	SWINGING		

EXTERIOR DOORS SHALL COMPLY WITH ONE OF THE FOLLOWING (COUNTY CONSOLIDATED WUI CODE 96.2.504.9):

A. NONCOMBUSTIBLE CONSTRUCTION
B. SOLID-CORE WOOD NOT LESS THAN 1-¾ INCHES THICK
C. THE EXTERIOR DOOR SHALL BE CONSTRUCTED OF SOLID CORE WOOD THAT COMPLIES WITH THE FOLLOWING REQUIREMENTS:
   - STILES AND RAILS SHALL NOT BE LESS THAN 1-⅜ INCHES THICK
   - RAISED PANELS SHALL NOT BE LESS THAN 1-¼ INCHES THICK, EXCEPT FOR THE EXTERIOR PERIMETER OF THE RAISED PANEL THAT SHALL BE PERMITTED TO TAPER TO A TONGUE NOT LESS THAN ⅜ INCH THICK
D. HAVE A FIRE PROTECTION RATING OF NOT LESS THAN 20 MINUTES WHEN TESTED ACCORDING TO NFPA 252, UL 10B OR UL 10C.

FLOOR PLAN NOTES

1. EXTERIOR WALLS WITHIN 3 FEET OF PROPERTY LINE (WITH SPRINKLERS) OR 5 FEET OF PROPERTY LINE (WITHOUT SPRINKLERS) REQUIRE 1-HOUR FIRE RATING FOR EXPOSURE TO BOTH SIDES.
2. PROJECTIONS:
   - PROHIBITED WITHIN 2 FEET OF PROPERTY LINE
   - 1-HOUR FIRE RATING ON THE UNDERSIDE WITHIN 3FT OF PROPERTY LINE (WITH SPRINKLERS)
   - 1-HOUR FIRE RATING ON THE UNDERSIDE WITHIN 5FT OF PROPERTY LINE (WITHOUT SPRINKLERS)
3. OPENINGS:
   - PROHIBITED WITHIN 3FT OF PROPERTY LINE
   - MAXIMUM 25% OF WALL AREA WITHIN 5 FEET OF PROPERTY LINE (WITHOUT SPRINKLERS)
4. PENETRATIONS:
   - 1-HOUR FIRE-RATED PENETRATIONS OF WALLS WITHIN 3FT OF PROPERTY LINE (WITH SPRINKLERS)
   - 1-HOUR FIRE-RATED PENETRATIONS OF WALLS WITHIN 5FT OF PROPERTY LINE (WITHOUT SPRINKLERS)
5. CONCRETE LANDING WITH MIN 36" DEPTH AND A MAXIMUM OF 1-½" LOWER THAN TOP OF DOOR THRESHOLD

OPTIONAL ROLL-IN SHOWER PLAN NOTES

1. SHOWER COMPARTMENT SEAT
   - MUST BE FOLDING TYPE, NOT TO EXCEED MORE THAN 6 INCHES FROM MOUNTING WALL WHEN FOLDED
   - LOCATED WITHIN 27 INCHES OF SHOWER CONTROLS
   - MOUNTED MINIMUM 17 INCHES AND MAXIMUM 19 INCHES ABOVE BATHROOM FINISHED FLOOR.
   - SEAT INSTALLED ON SIDE WALL ADJACENT TO CONTROLS AND EXTENDING FROM BACK WALL TO POINT WITHIN 3 INCHES OF SHOWER COMPARTMENT ENTRY
   - STRUCTURAL ADEQUACY OF MOUNTING HARDWARE AND FASTENERS TO ACCOMMODATE 250 POUND POINT LOAD APPLIED AT ANY POINT ON THE GRAB BAR, FASTENER, MOUNTING DEVICE, OR SUPPORTING STRUCTURE
2. SHOWER GRAB BARS
   - MOUNTED MINIMUM 33 INCHES AND MAXIMUM 36 INCHES ABOVE SHOWER FLOOR
   - NOT EXTENDING OVER SHOWER SEAT
   - IF CROSS SECTION IS CIRCULAR, MINIMUM 1-¼" AND MAXIMUM 2" OUTSIDE DIAMETER
   - IF CROSS SECTION IS NON-CIRCULAR, MINIMUM 4" AND MAXIMUM 4.8" PERIMETER AND MAXIMUM 2-¼" CROSS SECTION DIMENSION
   - GRAB BARS MOUNTED ADJACENT TO A WALL, 1-½" ABSOLUTE SPACE BETWEEN WALL AND GRAB BAR
   - MINIMUM 1-½" SPACE BETWEEN GRAB BAR AND PROJECTING OBJECTS BELOW AND AT ENDS
   - MINIMUM 12 INCH SPACE BETWEEN GRAB BAR AND PROJECTING OBJECTS ABOVE
   - SURFACE MATERIAL OF ANY WALLS OR OBJECTS ADJACENT TO GRAB BARS MUST BE FREE OF SHARP OR ABRASIVE ELEMENTS AND HAVE ROUNDED EDGES.
   - STRUCTURAL ADEQUACY OF MOUNTING HARDWARE AND FASTENERS TO ACCOMMODATE 250 POUND POINT LOAD APPLIED AT ANY POINT ON THE GRAB BAR, FASTENER, MOUNTING DEVICE, OR SUPPORTING STRUCTURE
   - WALL REINFORCEMENT TO BE PROVIDED AT LOCATION OF GRAB BARS (E.G. BLOCKING)
3. OPERABLE PARTS OF SHOWER CONTROLS AND FAUCETS:
   - INSTALLED ON BACK WALL OF SHOWER COMPARTMENT ADJACENT TO SEAT WALL
   - LOCATED MINIMUM 19 INCHES AND MAXIMUM 27 INCHES FROM SEAT WALL
   - LOCATED ABOVE GRAB BAR BUT NO HIGHER THAN 48 INCHES ABOVE SHOWER FLOOR
   - CENTERLINE AT MINIMUM 39 INCHES AND MAXIMUM 41 INCHES ABOVE SHOWER FLOOR
   - SINGLE-LEVER DESIGN
   - OPERABLE WITH MAXIMUM 5 POUNDS OF FORCE
   - OPERABLE WITH ONE HAND AND WITHOUT TIGHT GRASPING, PINCHING, OR TWISTING OF WRIST
4. SPRAYER UNIT AND ASSOCIATED OPERABLE PARTS SHALL BE PROVIDED PER THE FOLLOWING:
   - OPERABLE PARTS, INCLUDING HANDLE, TO BE INSTALLED ON BACK WALL OF SHOWER COMPARTMENT MINIMUM 19 INCHES AND MAXIMUM 27 INCHES FROM SEAT WALL
   - OPERABLE PARTS LOCATED ABOVE GRAB BAR BUT NO HIGHER THAN 48 INCHES ABOVE SHOWER FLOOR, MEASURED TO TOP OF MOUNTING BRACKET
   - MINIMUM 59 INCH LONG HOSE
   - CAPABLE FOR USE AS FIXED SHOWER HEAD AND HAND HELD SHOWER
   - ON/OFF CONTROL WITH NON-POSITIVE SHUT OFF
   - ADJUSTABLE -HEIGHT SHOWER HEADS ON VERTICAL BAR SHALL NOT OBSTRUCT USE OF BATHTUB GRAB BARS
5. WHERE SOAP DISHES ARE PROVIDED, MAXIMUM 40 INCHES ABOVE SHOWER FLOOR AND WITHIN REACH LIMITS FROM THE SHOWER SEAT
6. MAXIMUM 2.1% SLOPE IN ALL DIRECTIONS OF ROLL-IN SHOWER FLOORS
7. MAXIMUM ½" HIGH THRESHOLDS WITH MAXIMUM 50% BEVELED SLOPE AT ROLL-IN SHOWERS
8. WHERE DRAINS ARE PROVIDED AT ROLL-IN SHOWERS, MAXIMUM ¼" GRATE OPENINGS FLUSH WITH SHOWER FLOOR SURFACE

AGING-IN-PLACE AND FALL PREVENTION DESIGN

1. REINFORCEMENT FOR GRAB BARS: AT LEAST ONE BATHROOM ON THE ENTRY LEVEL SHALL BE PROVIDED WITH REINFORCEMENT INSTALLED IN ACCORDANCE WITH SECTION R327.1.1. REINFORCEMENT SHALL BE MINIMUM 2X8 SOLID LUMBER, LOCATED BETWEEN 32" AND 39-¼" ABOVE THE FINISHED FLOOR FLUSH WITH WALL FRAMING ON BOTH SIDE WALLS OF THE FIXTURE.
2. ELECTRICAL OUTLETS, SWITCH, AND CONTROL HEIGHTS SHALL BE LOCATED NO MORE THAN 48" MEASURED FROM THE TOP OF THE OUTLET BOX AND NOT LESS THAN 15" MEASURED FROM THE BOTTOM OF THE OUTLET BOX ABOVE THE FINISHED FLOOR (SECTION R327.1.2). SHOW DIMENSION ON ELEVATION.
3. DOORBELL BUTTONS SHALL NOT EXCEED 48" ABOVE EXTERIOR FLOOR OR LANDING. (SECTION R327.1.4). SHOW DIMENSION ON ELEVATION.
4. INTERIOR DOORS: EFFECTIVE JULY 1, 2024, AT LEAST ONE BATHROOM AND ONE BEDROOM ON THE ENTRY LEVEL SHALL PROVIDE A DOORWAY WITH A NET CLEAR OPENING OF NOT LESS THAN 32 INCHES, MEASURED WITH THE DOOR POSITIONED AT AN ANGLE OF 90 DEGREES FROM THE CLOSE POSITION; OR, IN THE CASE OF A TWO- OR THREE-STORY SINGLE FAMILY DWELLING, ON THE SECOND OR THIRD FLOOR OF THE DWELLING IF A BATHROOM OR BEDROOM IS NOT LOCATED ON THE ENTRY LEVEL.

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

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PROJECT TITLE

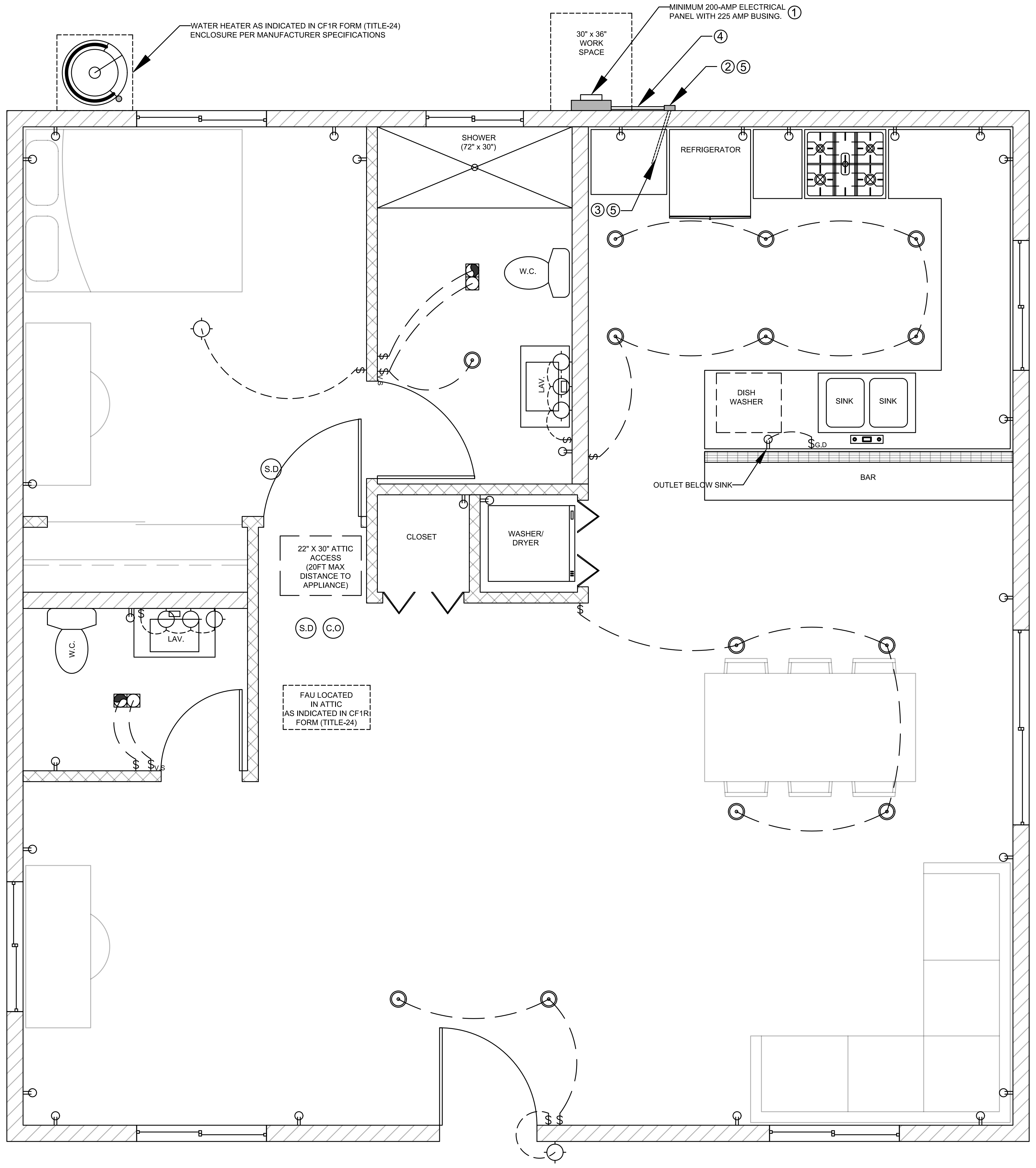
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

FLOOR PLAN

SHEET NUMBER

A1


WATER HEATER AS INDICATED IN CF1R FORM (TITLE-24)
ENCLOSURE PER MANUFACTURER SPECIFICATIONS
30" x 36" WORK SPACE
MINIMUM 200-AMP ELECTRICAL PANEL WITH 225 AMP BUSING. ①
④
②⑤
③⑤
SHOWER (72" x 30")
REFRIGERATOR
W.C.
LAV.
DISH WASHER
SINK
SINK
$G.D
OUTLET BELOW SINK
BAR
S.D
CLOSET
WASHER/ DRYER
22" X 30" ATTIC ACCESS (20FT MAX DISTANCE TO APPLIANCE)
S.D
C.O
W.C.
LAV.
FAU LOCATED IN ATTIC AS INDICATED IN CF1R FORM (TITLE-24)
$
$V.S

ELECTRICAL, PLUMBING, AND MECHANICAL TO BE VERIFIED IN FIELD

ELECTRICAL LEGEND

⊖	DUPLEX OUTLET	◎	HIGH EFFICACY RECESSED LIGHT
$	WALL SWITCH	⌽	GARBAGE DISPOSAL
$G.D	GARBAGE DISPOSAL SWITCH		
$V.S	VACANCY SENSOR		
⊥	4" DIA DRYER VENT		
S.D	SMOKE DETECTOR	✣	FAN & LIGHT COMBO
C.O	CARBON MONOXIDE ALARM		
▨	FAN AND LIGHT COMBINATION		
⌖	HIGH EFFICACY LIGHT FIXTURE		

ELECTRICAL PLAN
3/8" = 1'-0"

UTILITY PLAN NOTES

1. SMOKE DETECTORS TO BE INTERCONNECTED PER CRC R314.4 AND HARD–WIRED WITH BATTERY BACK–UP PER CRC R314.6
2. CARBON MONOXIDE ALARMS TO BE INTERCONNECTED PER CRC R315.7 AND HARD–WIRED WITH BATTERY BACK–UP PER CRC R315.5
3. 4" ø DRYER VENT WITH MAXIMUM 14 FOOT COMBINED HORIZONTAL AND VERTICAL LENGTH WITH TWO 90 DEGREE ELBOWS.
4. A MECHANICAL EXHAUST VENTILATION SYSTEM, SUPPLY VENTILATION SYSTEM, OR COMBINATION THEREOF SHALL BE INSTALLED FOR EACH DWELLING UNIT TO PROVIDE WHOLE–BUILDING VENTILATION WITH OUTDOOR AIR IN COMPLIANCE WITH ASHRAE STANDARD 62.2 AS ADOPTED BY THE CALIFORNIA ENERGY COMMISSION.
5. AN INTERMITTENTLY OR CONTINUOUSLY OPERATING LOCAL MECHANICAL EXHAUST VENTILATION SYSTEM SHALL BE INSTALLED IN EACH BATHROOM WITH A BATHTUB, SHOWER, OR SIMILAR MOISTURE SOURCE AND IN EACH KITCHEN IN COMPLIANCE WITH ASHRAE STANDARD 62.2 AS ADOPTED BY THE CALIFORNIA ENERGY COMMISSION. VENTILATION RATES TO BE PER SECTION 150.0 (O) G.III.B AND TABLES 150.0–E, 150–F, 150.0–G, AND 150.0–H
6. WATER HEATER OR FURNACE SHALL BE A DIRECT–VENT APPLIANCE
7. LISTED GASKETED SELF–CLOSING DOOR REQUIRED FOR GAS FAU

SOLAR READY KEY NOTES ○

WHEN PV SYSTEM NOT REQUIRED PER ENERGY COMPLIANCE DOCUMENTATION OR OTHERWISE EXEMPTED:

1. THE MAIN ELECTRICAL SERVICE PANEL SHALL NOT BE OF A TYPE WITH A CENTER–FED MAIN CIRCUIT BREAKER AND SHALL INCLUDE RESERVED SPACE ALLOWING FOR INSTALLATION OF DOUBLE–POLE CIRCUIT BREAKERS FOR A FUTURE SOLAR PHOTOVOLTAIC SYSTEM. SUCH RESERVED SPACE SHALL BE POSITIONED AT THE OPPOSITE (LOAD) END FROM THE INPUT FEEDER OR MAIN CIRCUIT BREAKER LOCATION. THE RESERVED SPACE SHALL BE PERMANENTLY AND VISIBLY MARKED AS "FOR FUTURE SOLAR PHOTOVOLTAIC"
2. APPROVED MINIMUM 4–INCH SQUARE ELECTRICAL JUNCTION BOX LOCATED WITHIN 72 INCHES HORIZONTALLY AND 12 INCHES VERTICALLY OF MAIN ELECTRICAL SERVICE PANEL
3. MINIMUM 1 INCH DIAMETER LISTED ELECTRICAL METALLIC RACEWAY ORIGINATING AT READILY ACCESSIBLE ATTIC LOCATION WITH PROXIMITY TO SOLAR ZONE AREA AND TERMINATING AT THE REQUIRED ELECTRICAL JUNCTION BOX
4. MINIMUM 1 INCH DIAMETER LISTED ELECTRICAL METALLIC RACEWAY ORIGINATING AT THE REQUIRED ELECTRICAL JUNCTION BOX AND TERMINATING AT THE MAIN ELECTRICAL SERVICE PANEL
5. ELECTRICAL JUNCTION BOX AND SEGMENT OF METALLIC RACEWAY IN THE ATTIC SHALL BE PERMANENTLY AND VISIBLY MARKED AS "FOR FUTURE SOLAR PHOTOVOLTAIC"

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PROJECT TITLE

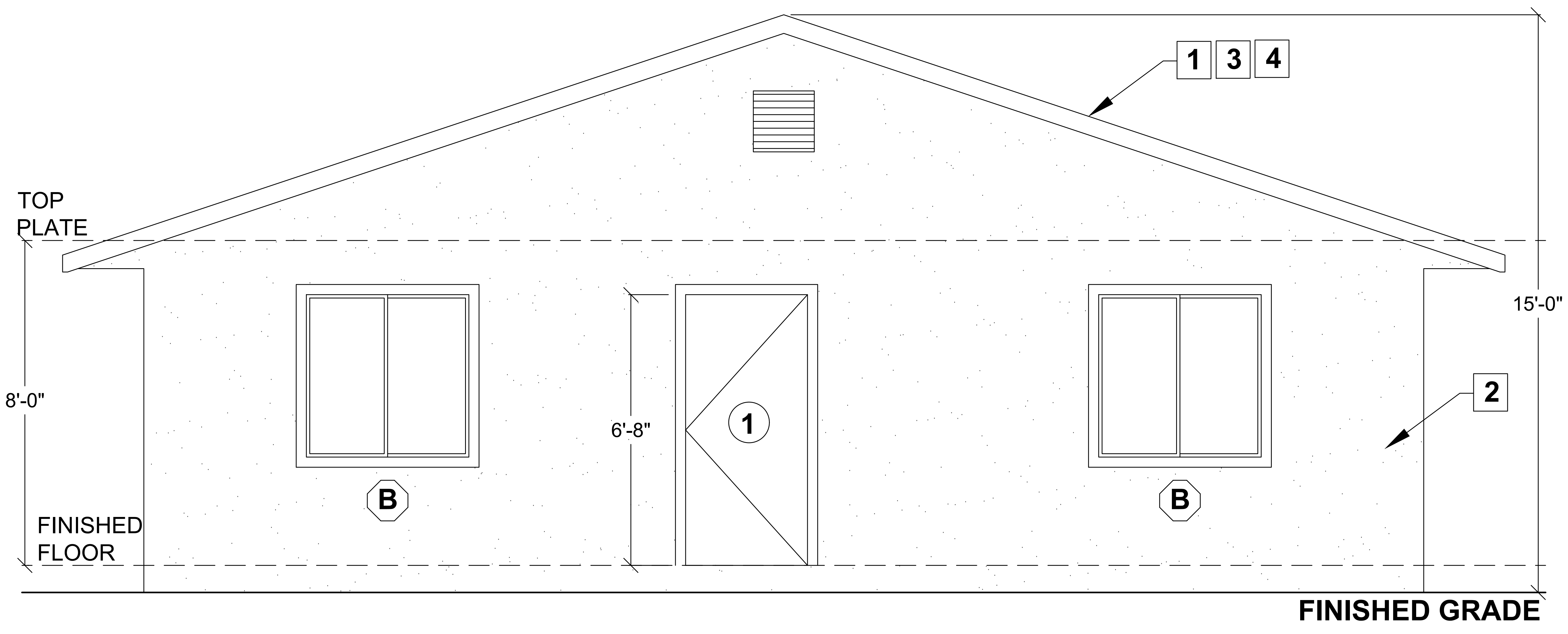
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

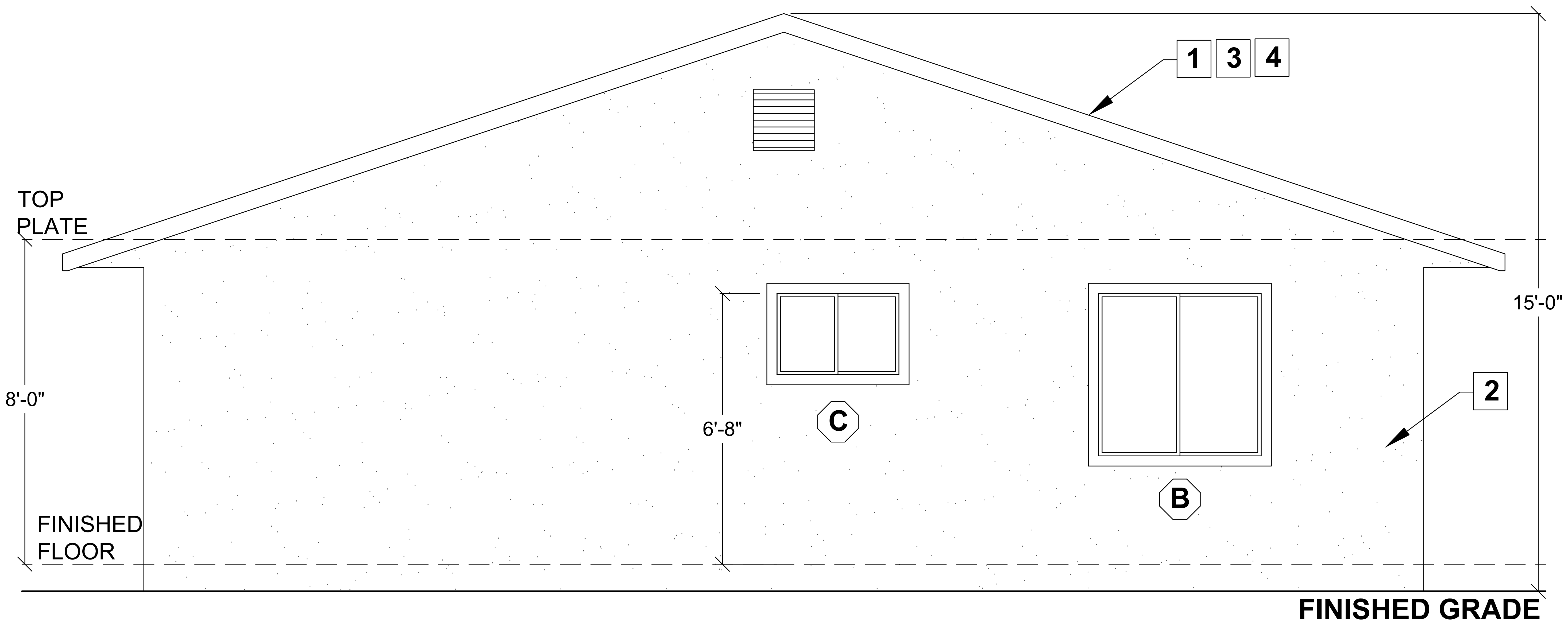
ELECTRICAL
PLAN

SHEET NUMBER

A2



FRONT



BACK

ELEVATIONS

1/2" = 1'-0"

WILDFIRE ZONE PLAN NOTES

1. WHERE THERE IS A VOID UNDER THE ROOF COVERING (SPACE BETWEEN THE ROOF COVERING AND COMBUSTIBLE ROOF DECKING) IT SHALL COMPLY WITH THE FOLLOWING (CWUIC 504.2.1):
   a. BIRD STOPS SHALL BE USED AT THE EAVES WHERE THE PROFILE FITS, TO PREVENT THE ENTRY OF DEBRIS AT THE EAVE.
   b. HIP AND RIDGE CAPS SHALL BE MUDDED IN TO PREVENT INTRUSION OF FIRE OR EMBERS.
   c. PLUS ONE OF THE FOLLOWING:
      1. A 72 LB. (32.7 KG) CAP SHEET COMPLYING WITH ASTM D3909 STANDARD SPECIFICATION FOR "ASPHALT ROLLED ROOFING (GLASS FELT) SURFACED WITH MINERAL GRANULES," SHALL BE INSTALLED OVER THE ROOF DECK.
      2. A LISTED AND LABELED ROOF UNDERLAYMENT THAT HAS BEEN TESTED AS PART OF THE ROOF ASSEMBLY IN ACCORDANCE WITH ASTM E108 OR UL 790 FOR CLASS A FIRE CLASSIFICATION SHALL BE INSTALLED IN ACCORDANCE WITH THE UNDERLAYMENT MANUFACTURER'S INSTALLATION INSTRUCTIONS AND LISTING.
2. EXPOSED VALLEY FLASHINGS SHALL BE CONSTRUCTED WITH NOT LESS THAN 0.019–INCH (NO. 26 GALVANIZED SHEET GAGE) CORROSION–RESISTANT METAL INSTALLED OVER A MINIMUM 36–INCH–WIDE UNDERLAYMENT CONSISTING OF ONE LAYER OF NO. 72 ASTM CAP SHEET RUNNING THE FULL LENGTH OF THE VALLEY. (CWUIC 504.2.2)
3. GUTTERS AND DOWNSPOUTS SHALL BE CONSTRUCTED OF NONCOMBUSTIBLE MATERIAL. GUTTERS SHALL BE PROVIDED WITH AN APPROVED MEANS TO PREVENT THE ACCUMULATION OF LEAVES AND DEBRIS IN THE GUTTER AND SHALL BE NON–COMBUSTIBLE. (COUNTY CONSOLIDATED WUI CODE 96.2.504.4)
4. OPERABLE SKYLIGHTS SHALL BE PROTECTED BY A NONCOMBUSTIBLE MESH SCREEN AND THE DIMENSIONS OF THE OPENINGS IN THE SCREEN SHALL NOT EXCEED 1/8 INCH. (CWUIC 504.8.2)
5. ALL VENTS (ROOF, FOUNDATION, COMBUSTION–AIR, ETC.) SHALL RESIST THE INTRUSION OF FLAMES AND EMBERS. (CWUIC 504.10) GABLE–END AND DORMER VENTS SHALL BE LOCATED NOT LESS THAN 10 FEET FROM LOT LINES. UNDERFLOOR VENTILATION OPENINGS SHALL BE LOCATED AS CLOSE TO GRADE AS PRACTICAL. (CWUIC 504.10.3) WHERE AN ADDITION, ALTERATION, OR ROOF MODIFICATION TAKES PLACE, NON–COMPLIANT WUI VENTS SHALL BE UPGRADED AND/OR RETROFITTED TO RESIST THE INTRUSION OF FLAMES AND EMBERS. (COUNTY CONSOLIDATED WUI CODE 96.2.504.10.4)
6. VENTILATION OPENINGS FOR ENCLOSED ATTICS, EAVE SOFFIT SPACES, ENCLOSED RAFTER SPACES FORMED WHERE CEILINGS ARE APPLIED DIRECTLY TO THE UNDERSIDE OF ROOF RAFTERS, UNDERFLOOR VENTILATION OPENINGS, AND VENT OPENINGS IN EXTERIOR WALLS AND EXTERIOR DOORS SHALL COMPLY WITH ONE OF THE FOLLOWING: (CWUIC 504.10)
   a. VENTILATION OPENINGS SHALL BE FULLY COVERED WITH WILDFIRE FLAME AND EMBER RESISTANT VENTS APPROVED AND LISTED BY THE CALIFORNIA STATE FIRE MARSHAL, OR WUI VENTS, TESTED IN ACCORDANCE WITH ASTM E2886, TO DEMONSTRATE COMPLIANCE WITH ALL THE FOLLOWING REQUIREMENTS:
      1. THERE SHALL BE NO FLAMING IGNITION OF THE COTTON MATERIAL DURING THE EMBER INTRUSION TEST.
      2. THERE SHALL BE NO FLAMING IGNITION DURING THE INTEGRITY TEST PORTION OF THE FLAME INTRUSION TEST.
      3. THE MAXIMUM TEMPERATURE OF THE UNEXPOSED SIDE OF THE VENT SHALL NOT EXCEED 662 DEGREES FAHRENHEIT (350 DEGREES CELSIUS).
   b. VENTS THAT ARE INSTALLED ON A SLOPED ROOF, SUCH AS DORMER VENTS, SHALL COMPLY WITH ALL OF THE FOLLOWING:
      1. VENTS SHALL BE COVERED WITH A MESH WHERE THE DIMENSIONS OF THE MESH THEREIN SHALL BE A MINIMUM OF 1/16 INCH AND SHALL NOT EXCEED 1/8 INCH IN DIAMETER.
      2. THE VENT AND MESH MATERIAL SHALL BE NONCOMBUSTIBLE.
      3. THE VENT AND MESH MATERIAL SHALL BE CORROSION RESISTANT.
7. EXTERIOR WALL COVERINGS COMPLYING WITH THE FOLLOWING (COUNTY CONSOLIDATED WUI CODE 96.2.504.5):

   EXCEPTION: EXTERIOR WALL COVERING ON AN ADDITION MAY MATCH THE EXISTING STRUCTURE PROVIDED THAT THE SQUARE FOOTAGE OF THE ADDITION DOES NOT EXCEED 50% OF THE EXISTING STRUCTURE OR 2,500 SQUARE FEET, WHICHEVER IS LESS, UNLESS THE EXTERIOR WALL COVERING IS WOOD SHINGLE OR SHAKE, IN WHICH CASE THE EXTERIOR WALL COVERING ON THE ADDITION SHALL BE FIRE–RETARDANT, PRESSURE–TREATED WOOD SHINGLES OR SHAKES. ALL OTHER PROVISIONS OF THIS SECTION SHALL APPLY TO THE ADDITION.

   EXCEPTION: REPAIR OR REPLACEMENT OF 50% OR MORE OF AN EXISTING EXTERIOR WALL SHALL REQUIRE THE ENTIRE WALL TO MEET ALL WILDFIRE REQUIREMENTS.
   c. NONCOMBUSTIBLE MATERIAL (STUCCO, CEMENT FIBER BOARD, MASONRY, ETC.)
   d. A MINIMUM OF 6 INCHES OF METAL FLASHING OR NONCOMBUSTIBLE MATERIAL APPLIED VERTICALLY ON THE EXTERIOR OF THE WALL SHALL BE INSTALLED AT THE GROUND, DECKING AND ROOF INTERSECTIONS.
8. DETAIL PATIO COVER, CARPORT, AND TRELLIS CONSTRUCTION, WITH ALL EXPOSED ELEMENTS COMPLYING WITH ANY OF THE FOLLOWING (COUNTY CONSOLIDATED WUI CODE 96.2.504.7):
   a. HEAVY TIMBER CONSTRUCTION
   b. NONCOMBUSTIBLE MATERIAL
   c. MATERIALS APPROVED FOR NOT LESS THAN 1–HOUR FIRE–RESISTANCE–RATED CONSTRUCTION ON THE EXTERIOR SIDE, AS TESTED IN ACCORDANCE WITH ASTM E119 OR UL 263
   d. THE EXTERIOR PORTION OF A 1–HOUR FIRE–RESISTANCE–RATED EXTERIOR ASSEMBLY, AS TESTED IN ACCORDANCE WITH ASTM E119 OR UL 263, APPLIED TO THE UNDERSIDE OF THE CEILING ASSEMBLY, INCLUDING ASSEMBLIES USING THE GYPSUM PANEL AND SHEATHING PRODUCTS LISTED IN THE GYPSUM ASSOCIATION FIRE RESISTANCE DESIGN MANUAL
   e. THE UNDERSIDE OF A FLOOR PROJECTION ASSEMBLY THAT MEETS THE PERFORMANCE CRITERIA IN SECTION 504.7.2 WHEN TESTED IN ACCORDANCE WITH THE TEST PROCEDURES SET FORTH IN ASTM E2957
   f. THE UNDERSIDE OF A FLOOR PROJECTION ASSEMBLY THAT MEETS THE PERFORMANCE CRITERIA IN ACCORDANCE WITH THE TEST PROCEDURES SET FORTH IN SFM STANDARD 12–7A–3
   g. MODIFIED HEAVY TIMBER (MINIMUM 2X EXTERIOR GRADE TONGUE–AND–GROOVE SHEATHING, 4X6 RAFTERS/BEAMS, 6X6 POSTS/COLUMNS, 3X BLOCKING)

WILDFIRE ZONE PLAN NOTES (CONTINUE)

9. DETAIL DECK, BALCONY, AND EXTERIOR STAIR CONSTRUCTION, WITH ALL EXPOSED ELEMENTS COMPLYING WITH THE FOLLOWING (COUNTY CONSOLIDATED WUI CODE 96.2.504.7):

   EXCEPTION: DECK REPAIR OR REPLACEMENT INVOLVING 50% OR MORE OF THE EXISTING DECK OR 1,000 SQUARE FEET OF DECK AREA, WHICHEVER IS LESS, SHALL REQUIRE THE ENTIRE DECK TO MEET ALL REQUIREMENTS OF THIS ITEM
   a. FRAMING (ANY OF THE FOLLOWING):

      EXCEPTION: STRUCTURAL SUPPORTS OF DECKS, BALCONIES, AND SIMILAR PROJECTIONS SKIRTED FROM FLOOR LEVEL TO GROUND LEVEL WITH NONCOMBUSTIBLE MATERIAL
      - HEAVY TIMBER CONSTRUCTION
      - NONCOMBUSTIBLE MATERIAL
      - MATERIALS APPROVED FOR NOT LESS THAN 1–HOUR FIRE–RESISTANCE–RATED CONSTRUCTION ON THE EXTERIOR SIDE, AS TESTED IN ACCORDANCE WITH ASTM E119 OR UL 263
      - THE EXTERIOR PORTION OF A 1–HOUR FIRE–RESISTANCE–RATED EXTERIOR ASSEMBLY, AS TESTED IN ACCORDANCE WITH ASTM E119 OR UL 263, APPLIED TO THE UNDERSIDE OF THE CEILING ASSEMBLY, INCLUDING ASSEMBLIES USING THE GYPSUM PANEL AND SHEATHING PRODUCTS LISTED IN THE GYPSUM ASSOCIATION FIRE RESISTANCE DESIGN MANUAL
      - THE UNDERSIDE OF A FLOOR PROJECTION ASSEMBLY THAT MEETS THE PERFORMANCE CRITERIA IN SECTION 504.7.2 WHEN TESTED IN ACCORDANCE WITH THE TEST PROCEDURES SET FORTH IN ASTM E2957
      - THE UNDERSIDE OF A FLOOR PROJECTION ASSEMBLY THAT MEETS THE PERFORMANCE CRITERIA IN ACCORDANCE WITH THE TEST PROCEDURES SET FORTH IN SFM STANDARD 12–7A–3
      - MODIFIED HEAVY TIMBER (MINIMUM 2X EXTERIOR GRADE TONGUE–AND–GROOVE SHEATHING, 4X BEAMS/JOISTS/STRINGERS/TREADS, 6X6 POSTS/COLUMNS, 3X BLOCKING)
      - A MINIMUM OF A 6–INCH METAL FLASHING, APPLIED VERTICALLY ON THE EXTERIOR OF THE WALL, SHALL BE INSTALLED AT ALL DECK–TO–WALL INTERSECTIONS.
   b. WALKING SURFACE MATERIAL (ANY OF THE FOLLOWING):
      - MATERIAL THAT COMPLIES WITH THE PERFORMANCE REQUIREMENTS OF SECTION 504.7.3.3 WHEN TESTED IN ACCORDANCE WITH ASTM E2632 AND ASTM E2726
      - MATERIAL THAT COMPLIES WITH THE PERFORMANCE REQUIREMENTS OF BOTH SFM STANDARD 12–7A–4 AND SECTION 503.2.4
      - NONCOMBUSTIBLE MATERIAL
      - ANY MATERIAL THAT COMPLIES WITH THE PERFORMANCE REQUIREMENTS OF SFM STANDARD 12–7A–4A AND ANY ATTACHED EXTERIOR WALL COVERING IS NONCOMBUSTIBLE OR IGNITION–RESISTANT BUILDING MATERIALS.
      - ANY MATERIAL THAT COMPLIES WITH THE PERFORMANCE REQUIREMENTS OF SECTION 504.7.3.4 WHEN TESTED IN ACCORDANCE WITH ASTM E2632 AND ANY ATTACHED EXTERIOR WALL COVERING IS NONCOMBUSTIBLE OR IGNITION–RESISTANT BUILDING MATERIALS.
10. GARAGE DOORS (CWUIC 504.9):
   a. AUTOMATIC GARAGE DOOR OPENERS FOR VEHICLE DOORS SERVING A RESIDENTIAL BUILDING SHALL BE EQUIPPED WITH A BATTERY BACKUP FUNCTION.
   b. EXTERIOR GARAGE DOORS SHALL RESIST THE INTRUSION OF EMBERS BY PREVENTING GAPS BETWEEN DOORS AND DOOR OPENINGS, AT THE BOTTOM, SIDES AND TOPS OF DOORS, FROM EXCEEDING 1/8 INCH. GAPS BETWEEN DOORS AND DOOR OPENINGS SHALL BE CONTROLLED BY ONE OR MORE OF THE FOLLOWING METHODS:
      - WEATHER–STRIPPING PRODUCTS WITH TENSILE STRENGTH AND FLAMMABILITY RATING PER CWUIC 504.9.2.
      - DOOR OVERLAPS ONTO JAMBS AND HEADERS.
      - GARAGE DOOR JAMBS AND HEADERS COVERED WITH METAL FLASHING.

ELEVATION KEY NOTES ☐

1. ROOF: CLASS 'A' FIRE RATING

   ROOF MATERIAL: ____

   MANUFACTURER / MODEL: ________________________________

   UNDERLAYMENT: ____

   LISTING REPORT #: ______________________________________

2. EXTERIOR WALL FINISH: __________________ (SEE WILDFIRE NOTE 7)

3. ROOF PITCH: ____

4. RADIANT BARRIER IS REQUIRED YES ☐ NO ☐

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PROJECT TITLE

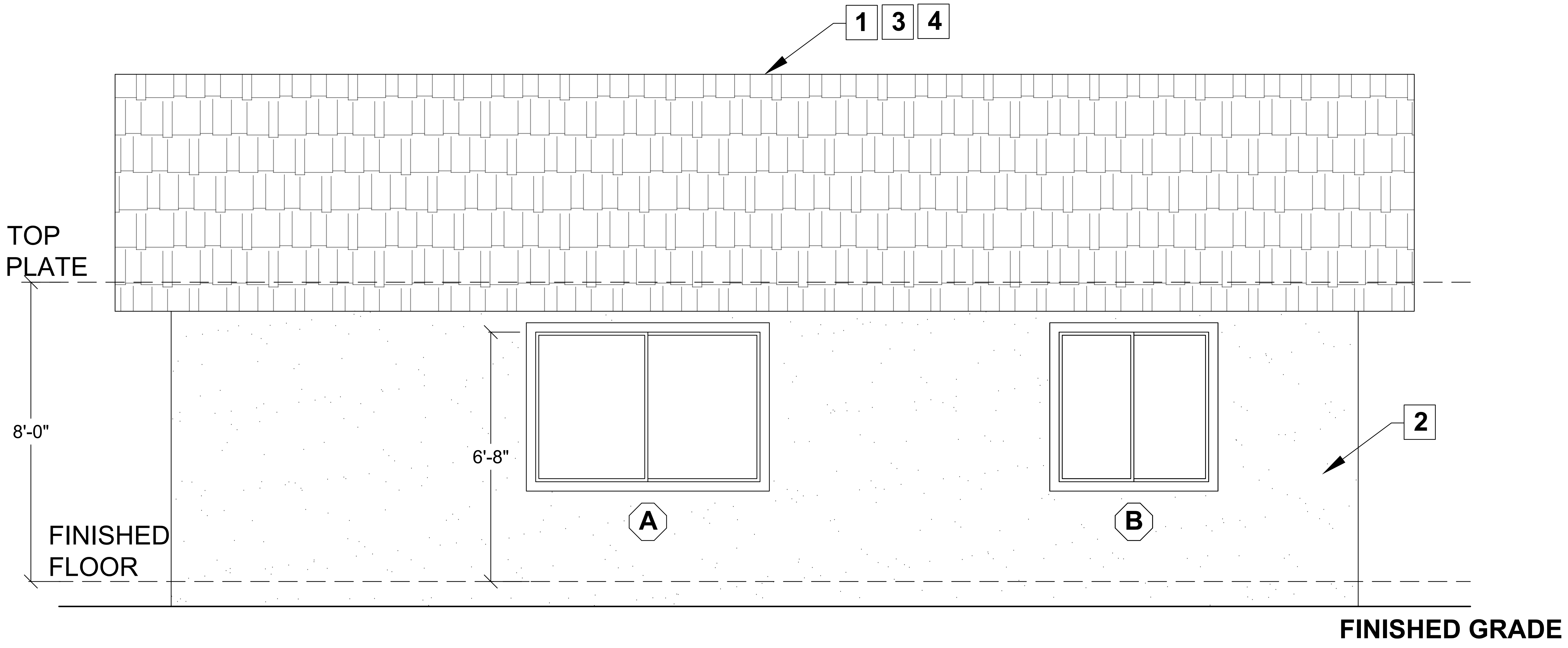
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

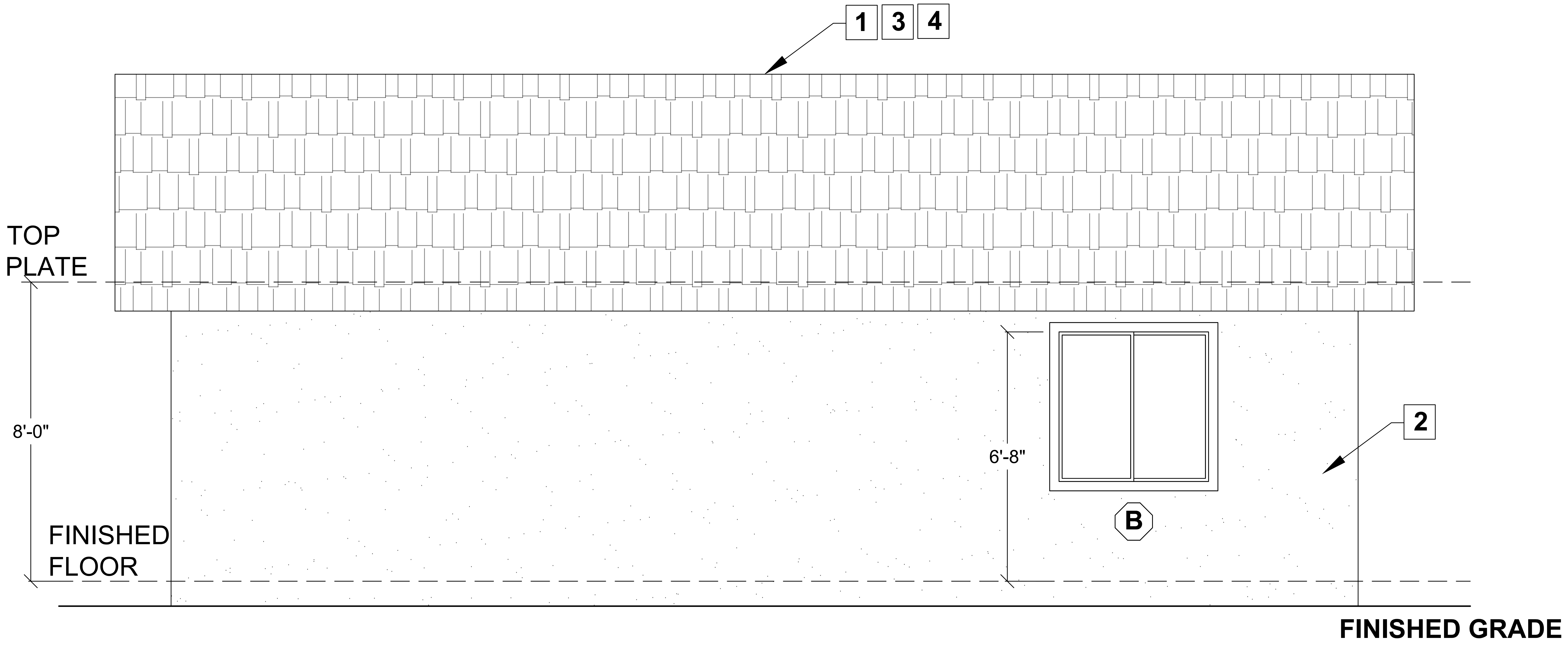
ELEVATIONS

SHEET NUMBER

A3


1
3
4
TOP PLATE
8'-0"
6'-8"
2
A
B
FINISHED FLOOR
FINISHED GRADE

RIGHT


1
3
4
TOP PLATE
8'-0"
6'-8"
2
B
FINISHED FLOOR
FINISHED GRADE

LEFT

ELEVATIONS
1/2" = 1'-0"

ELEVATION KEY NOTES

SEE SHEET A3 FOR KEYNOTES

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
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PROJECT TITLE

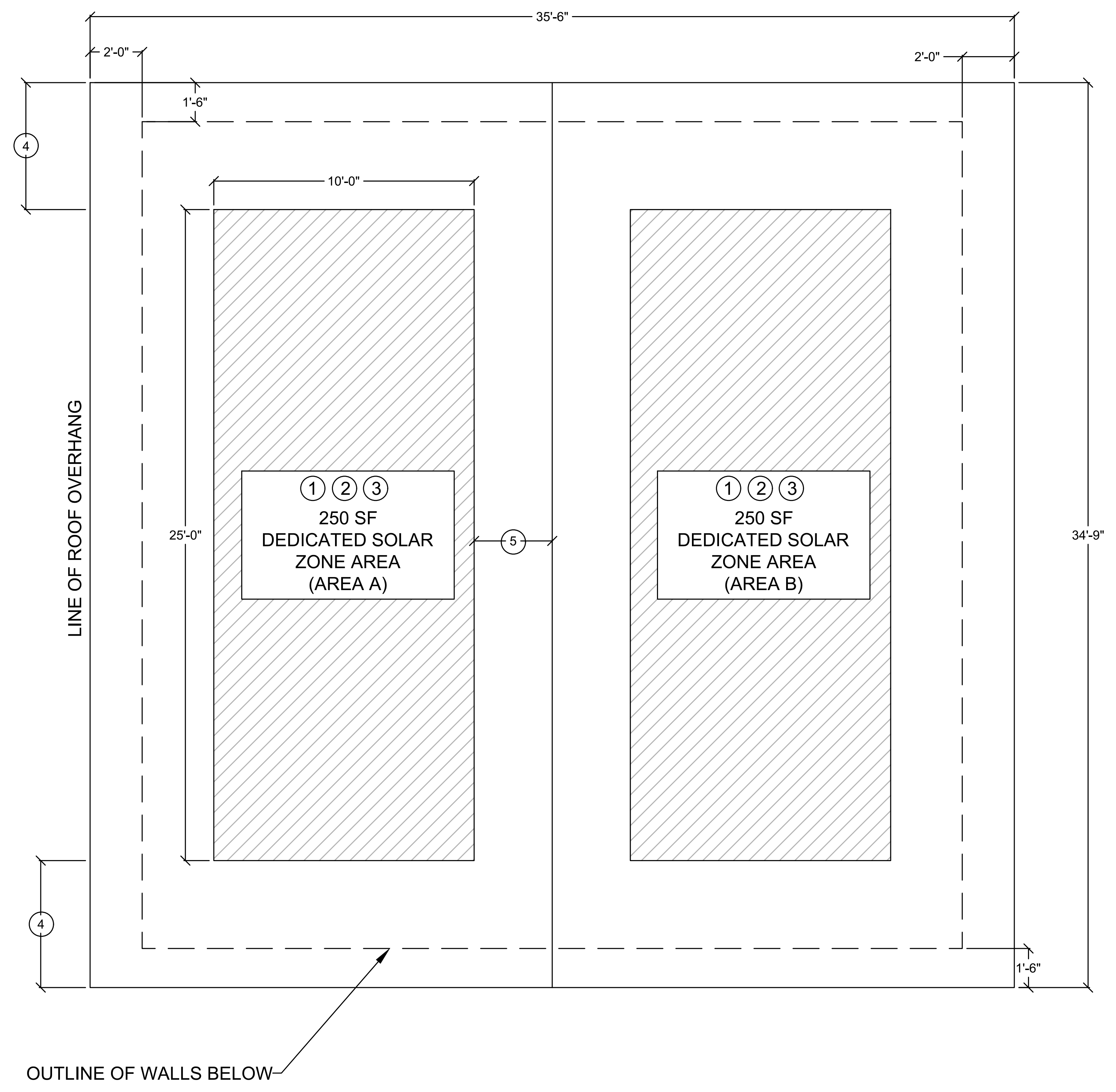
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

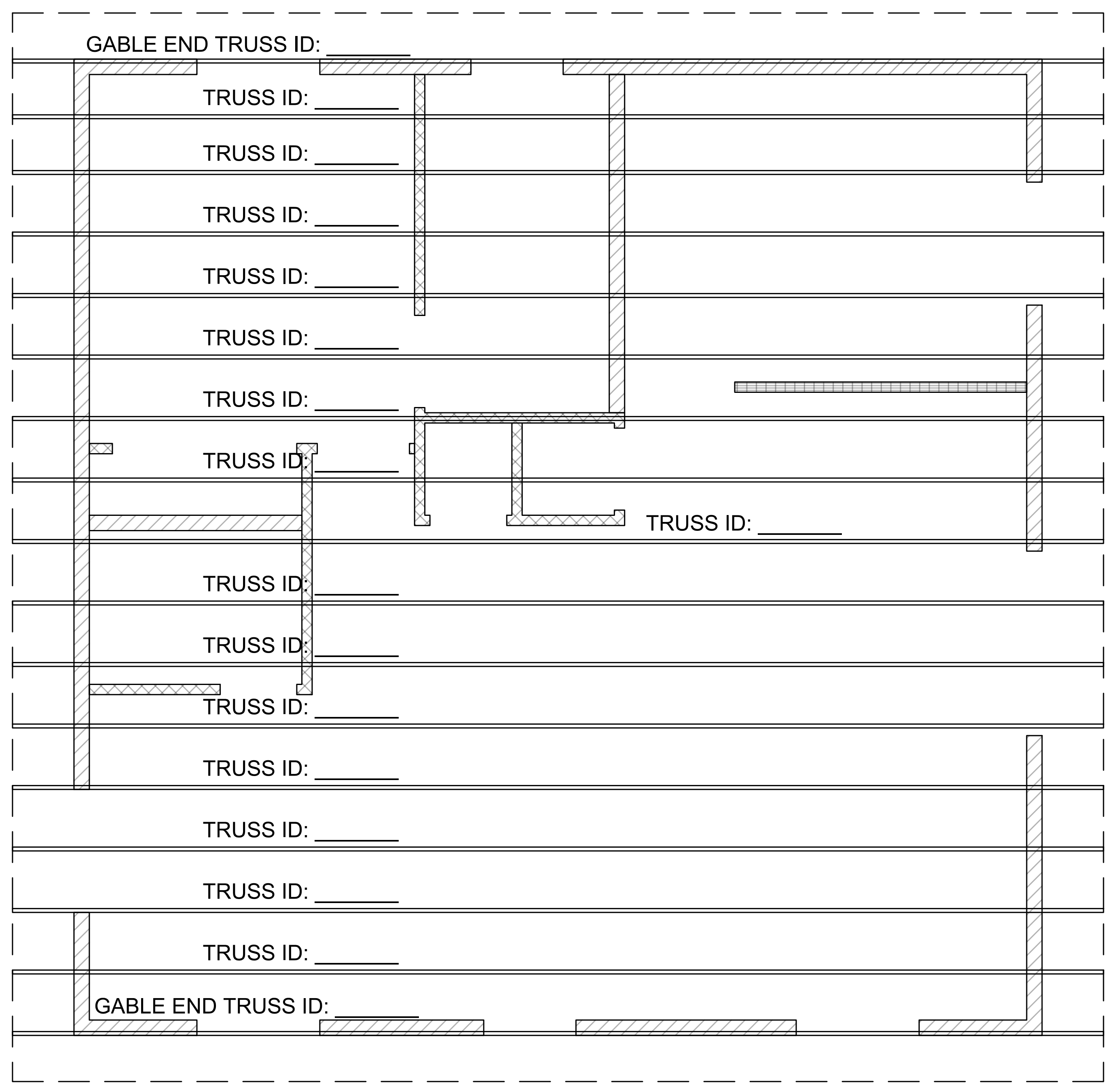
SHEET TITLE

ELEVATIONS

SHEET NUMBER

A4


35'-6"
2'-0"
2'-0"
1'-6"
10'-0"
4
25'-0"
① ② ③
250 SF
DEDICATED SOLAR
ZONE AREA
(AREA A)
5
① ② ③
250 SF
DEDICATED SOLAR
ZONE AREA
(AREA B)
34'-9"
LINE OF ROOF OVERHANG
4
1'-6"
OUTLINE OF WALLS BELOW


GABLE END TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
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TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
TRUSS ID: ______
GABLE END TRUSS ID: ______

ROOF PLAN / TRUSS LAYOUT
1/4" = 1'-0"

SOLAR READY KEY NOTES ○

WHEN PV SYSTEM NOT REQUIRED PER ENERGY COMPLIANCE DOCUMENTATION OR OTHERWISE EXEMPTED:

1. MIN 250 S.F. SOLAR ZONE AREA
2. DEDICATED SOLAR ZONE AREA LOCATED BETWEEN 110 AND 270 DEGREES OF TRUE NORTH – USE AREA A OR B AS NEEDED.
3. NO OBSTRUCTIONS – INCLUDING VENTS, CHIMNEYS, SKYLIGHTS, ARCHITECTURAL FEATURES, ROOF–MOUNTED EQUIPMENT – LOCATED WITHIN SOLAR ZONE.
4. 3' MIN FIRE FIGHTER ACCESS
5. 3' SMOKE VENTILATION SETBACK AT RIDGES

ATTIC VENTILATION

ATTIC VENTILATION REQUIRED
(SEE WILDFIRE NOTES 5 & 6 ON SHEET A3)

NET FREE CROSS VENTILATION AREA = $^1/_{300}$

VENT AREA REQUIRED = 1000 ft^2/ 300 = 3.33 ft^2 x 144 = 480 in^2 < ______ in^2
TOTAL VENT AREA PROVIDED

1. GABLE VENT (MIN ONE VENT AT EACH GABLE END)

   MANUFACTURER:______________________________________

   MODEL: ______________________________________

   QTY: ________________ NFVA: ________________ in^2

   VENT AREA PROVIDED = QTY x NFVA = ________________ in^2

2. EAVE VENT

   MANUFACTURER:______________________________________

   MODEL: ______________________________________

   QTY: ________________ NFVA: ________________ in^2

   VENT AREA PROVIDED = QTY x NFVA = ________________ in^2

3. ROOF VENT

   MANUFACTURER:______________________________________

   MODEL: ______________________________________

   QTY: ________________ NFVA: ________________ in^2

   VENT AREA PROVIDED = QTY x NFVA = ________________ in^2

4. INSTALL BETWEEN 40% AND 50% OF THE REQUIRED NET FREE VENT AREA A MAXIMUM OF 3 FEET BELOW THE RIDGE OR THE HIGHEST POINT OF THE SPACE (MEASURED VERTICALLY), AND INSTALL THE BALANCE OF THE REQUIRED VENTILATION IN THE BOTTOM ONE–THIRD OF THE ATTIC SPACE.

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PROJECT TITLE

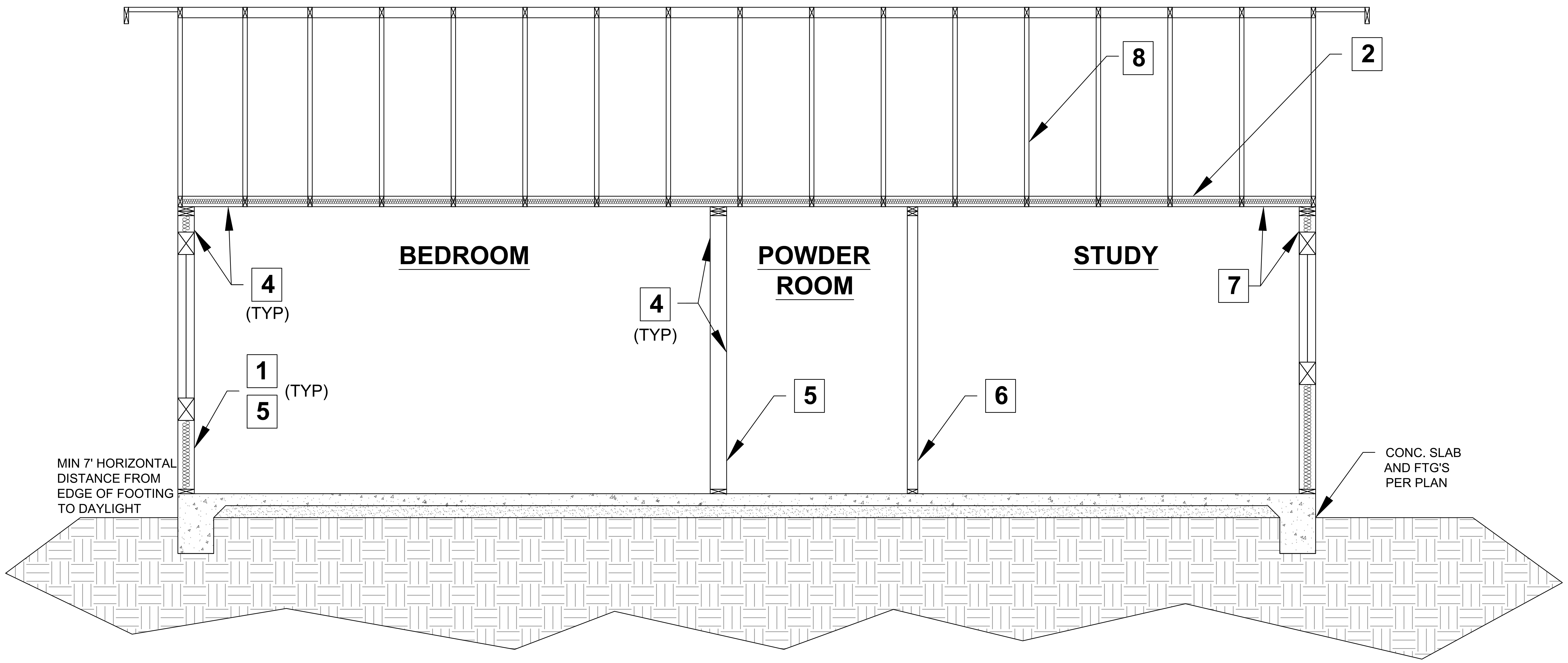
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

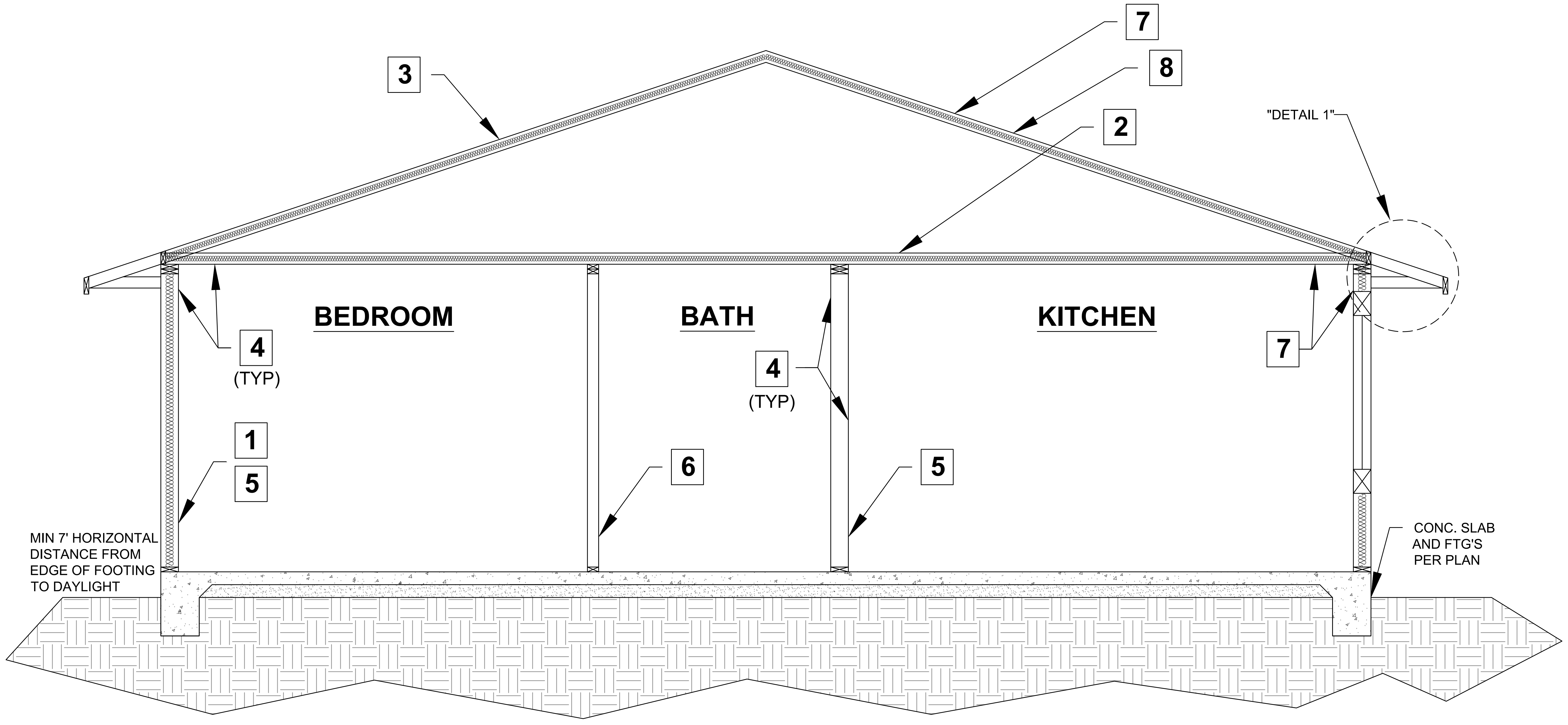
ROOF PLAN
TRUSS LAYOUT

SHEET NUMBER

A5


8
2
BEDROOM
POWDER ROOM
STUDY
4
(TYP)
4
(TYP)
7
1
5
(TYP)
5
6
MIN 7' HORIZONTAL DISTANCE FROM EDGE OF FOOTING TO DAYLIGHT
CONC. SLAB AND FTG'S PER PLAN

SECTION A-A


7
3
8
2
"DETAIL 1"
BEDROOM
BATH
KITCHEN
4
(TYP)
4
(TYP)
7
1
5
6
5
MIN 7' HORIZONTAL DISTANCE FROM EDGE OF FOOTING TO DAYLIGHT
CONC. SLAB AND FTG'S PER PLAN

SECTION B-B

SECTIONS

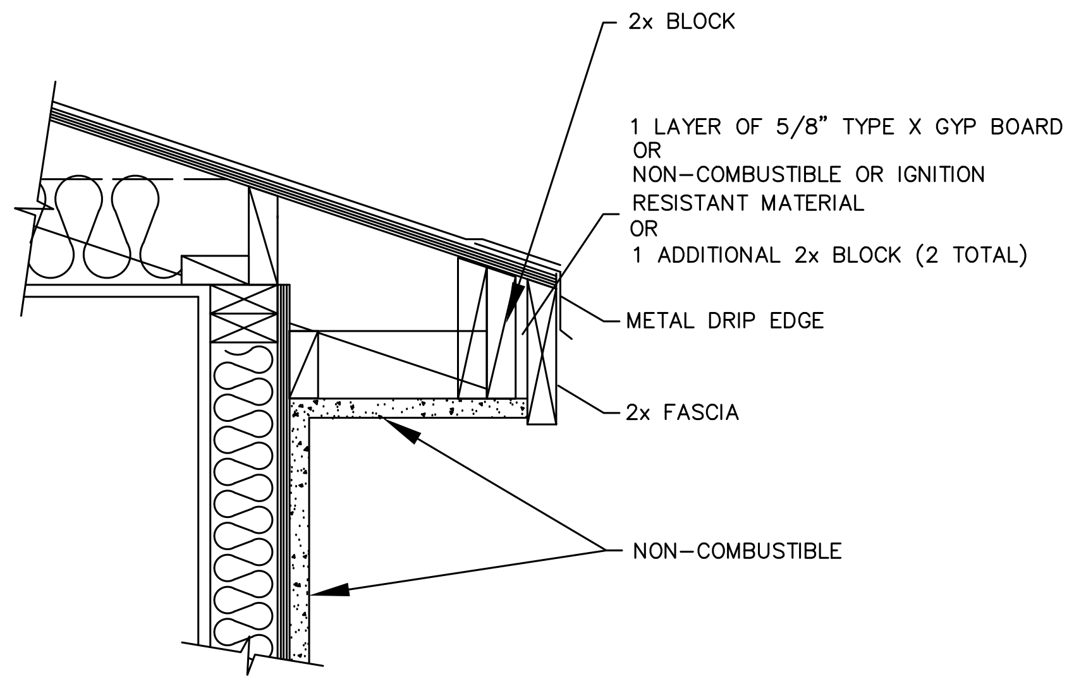
1/2" = 1'-0"

SECTION KEY NOTES

1. WALL INSULATION: ___________________
2. CEILING INSULATION: _________________
3. ROOF (TOP CHORD) INSULATION: __________________
4. INTERIOR FINISH: 1/2" GYPSUM BOARD
5. EXTERIOR WALL/PLUMBING WALL: 2X6 STUD WALL
6. INTERIOR WALL: 2X4 STUD WALL
7. CLIMATE ZONE 14 PROJECT (Y or N) if yes, see below:

   A CLASS I OR II VAPOR RETARDER SHALL BE INSTALLED ON THE CONDITIONED SPACE SIDE OF ALL INSULATION IN ALL EXTERIOR WALLS AND VENTED ATTICS
8. MANUFACTURED TRUSSES

① IGNITION-RESISTANT EAVE


2x BLOCK
1 LAYER OF 5/8" TYPE X GYP BOARD
OR
NON-COMBUSTIBLE OR IGNITION
RESISTANT MATERIAL
OR
1 ADDITIONAL 2x BLOCK (2 TOTAL)
METAL DRIP EDGE
2x FASCIA
NON-COMBUSTIBLE

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

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PROJECT TITLE

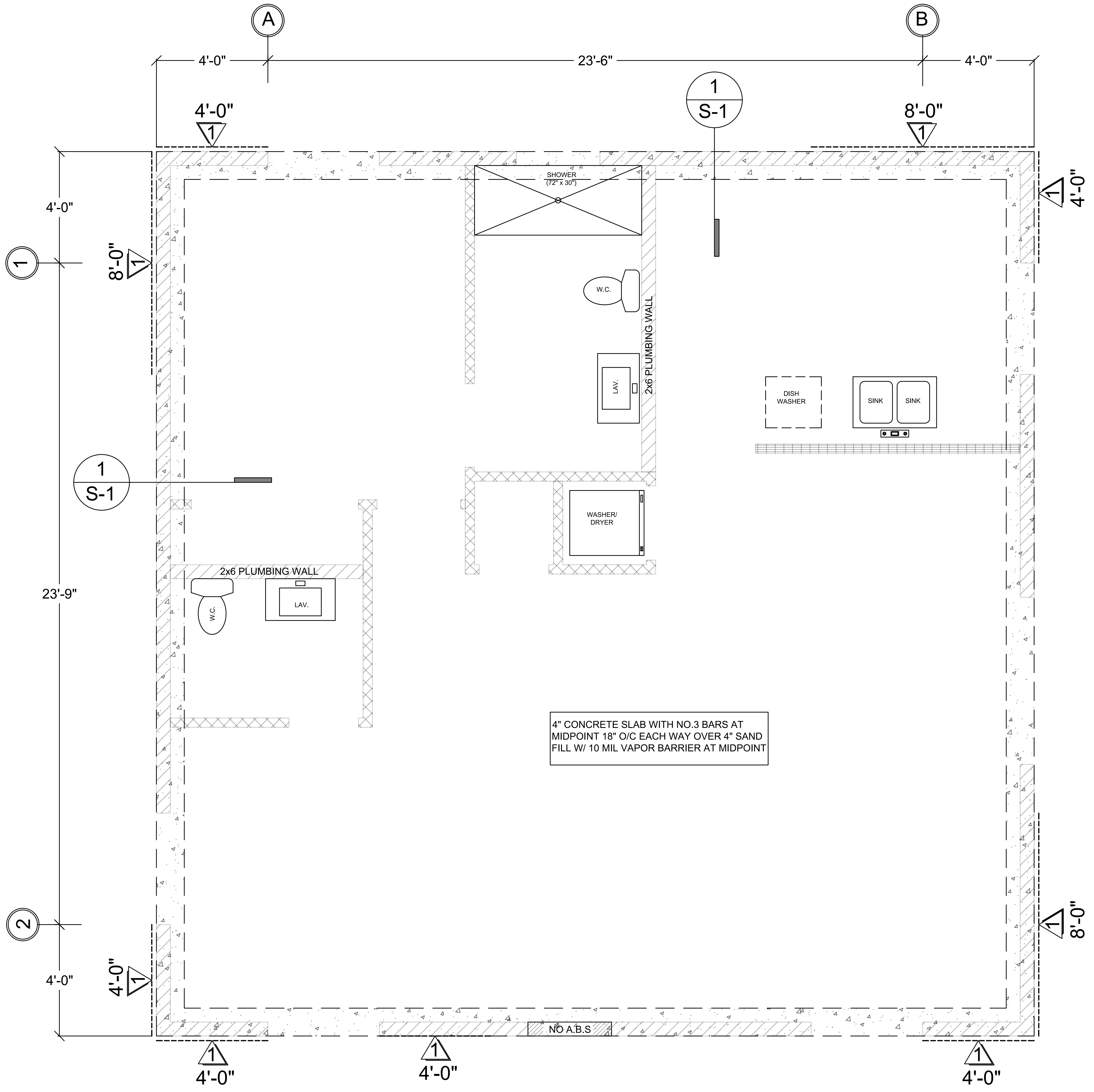
DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

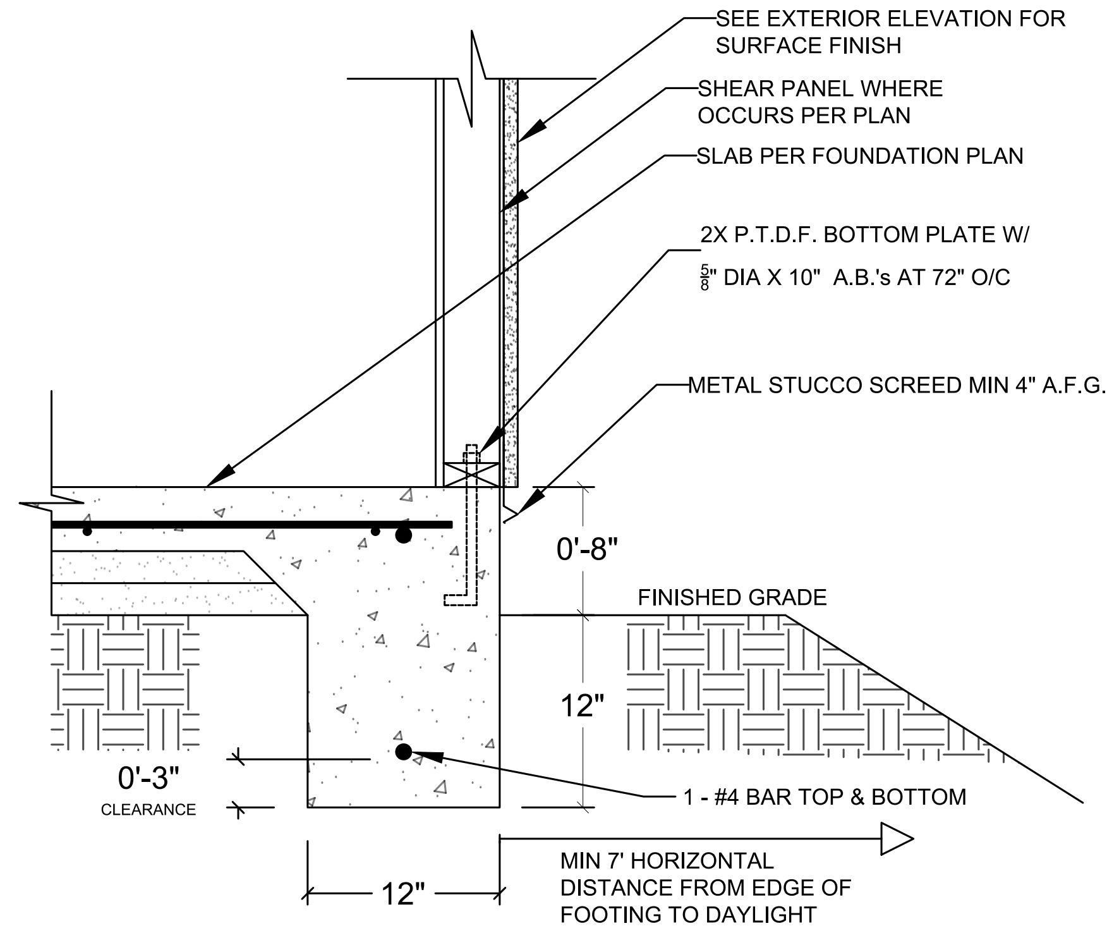
SHEET TITLE

SECTIONS

SHEET NUMBER

A6


A
B
4'-0"
23'-6"
4'-0"
1
S-1
4'-0"
1
8'-0"
1
4'-0"
8'-0"
1
1
4'-0"
23'-9"
4'-0"
2
4'-0"
1
SHOWER
(72" x 30")
W.C.
LAV.
2x6 PLUMBING WALL
DISH WASHER
SINK
SINK
WASHER/ DRYER
1
S-1
2x6 PLUMBING WALL
W.C.
LAV.
4" CONCRETE SLAB WITH NO.3 BARS AT MIDPOINT 18" O/C EACH WAY OVER 4" SAND FILL W/ 10 MIL VAPOR BARRIER AT MIDPOINT
8'-0"
1
NO A.B.S
1
4'-0"
1
4'-0"
1
4'-0"


SEE EXTERIOR ELEVATION FOR SURFACE FINISH
SHEAR PANEL WHERE OCCURS PER PLAN
SLAB PER FOUNDATION PLAN
2X P.T.D.F. BOTTOM PLATE W/ 5/8" DIA X 10" A.B.'s AT 72" O/C
METAL STUCCO SCREED MIN 4" A.F.G.
0'-8"
FINISHED GRADE
12"
0'-3"
CLEARANCE
1 - #4 BAR TOP & BOTTOM
12"
MIN 7' HORIZONTAL DISTANCE FROM EDGE OF FOOTING TO DAYLIGHT

1 EXTERIOR FOOTING

FOUNDATION PLAN NOTES

1. ALL ANCHORS BOLTS SHALL BE 5/8" DIAMETER AND HAVE A MINIMUM EMBEDMENT OF 7 INCHES INTO CONCRETE (UNO) AND NOT SPACED MORE THAN 6 FEET APART
2. 3"X3"X0.229" PLATE WASHERS SHALL BE USED ON EACH SILL PLATE ANCHOR BOLT
3. FOR STANDARD CUT WASHERS PLACED BETWEEN PLATE WASHER AND NUT, HOLE IN PLATE WASHER MAY BE DIAGONALLY SLOTTED WITH MAXIMUM 3/16" LARGER WIDTH THAN BOLT DIAMETER AND MAXIMUM 1–3/4" SLOT LENGTH
4. PROVIDE A MINIMUM OF TWO ANCHOR BOLTS PER SILL PLATE WITH ONE BOLT LOCATED MAXIMUM 12" AND MINIMUM 7 BOLT DIAMETERS FROM EACH END OF EACH SECTION.
5. BOLTS LOCATED IN THE MIDDLE THIRD OF THE SILL PLATE WIDTH
6. FASTENERS FOR PRESSURE–PRESERVATIVE TREATED AND FIRE RETARDANT TREATED WOOD SHALL BE HOT–DIPPED ZINC COATED GALVANIZED, STAINLESS STEEL OR COPPER
7. NO LPG PIPING ASSEMBLIES ALLOWED IN OR BENEATH SLABS WITHIN THE STRUCTURE

WOOD STRUCTURAL PANEL SHEATHING

MARK	MINIMUM NAIL		MINIMUM WOOD STRUCTURAL PANEL SPAN RATING	MINIMUM NOMINAL PANEL THICKNESS (in)	MAXIMUM WALL STUD SPACING (in)	PANEL NAIL SPACING	
	SIZE	PENETRATION (in)				EDGES (inches o/c)	FIELD (inches o/c)
1	6D COMMON	1.5	24:0	3/8"	16	6	12
	8D COMMON	1.75	24:16	7/16"	16	6	12

WOOD STRUCTURAL PANELS SHALL CONFORM TO DOC PS 1, DOC PS 2 OR ANSI/APA PRP 210, CSA O437 OR CSA O325.
PANELS SHALL BE IDENTIFIED BY A GRADE MARK OR CERTIFICATE OF INSPECTION ISSUED BY AN APPROVED AGENCY

VERTICAL JOINTS OF PANEL SHEATHING SHALL OCCUR OVER AND BE FASTENED TO COMMON STUDS.
HORIZONTAL JOINTS IN BRACED WALL PANELS SHALL OCCUR OVER AND BE FASTENED TO COMMON BLOCKING OF A MINIMUM 1 1/2 INCH THICKNESS.

FOUNDATION PLAN
3/8" = 1'-0"

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PROJECT TITLE

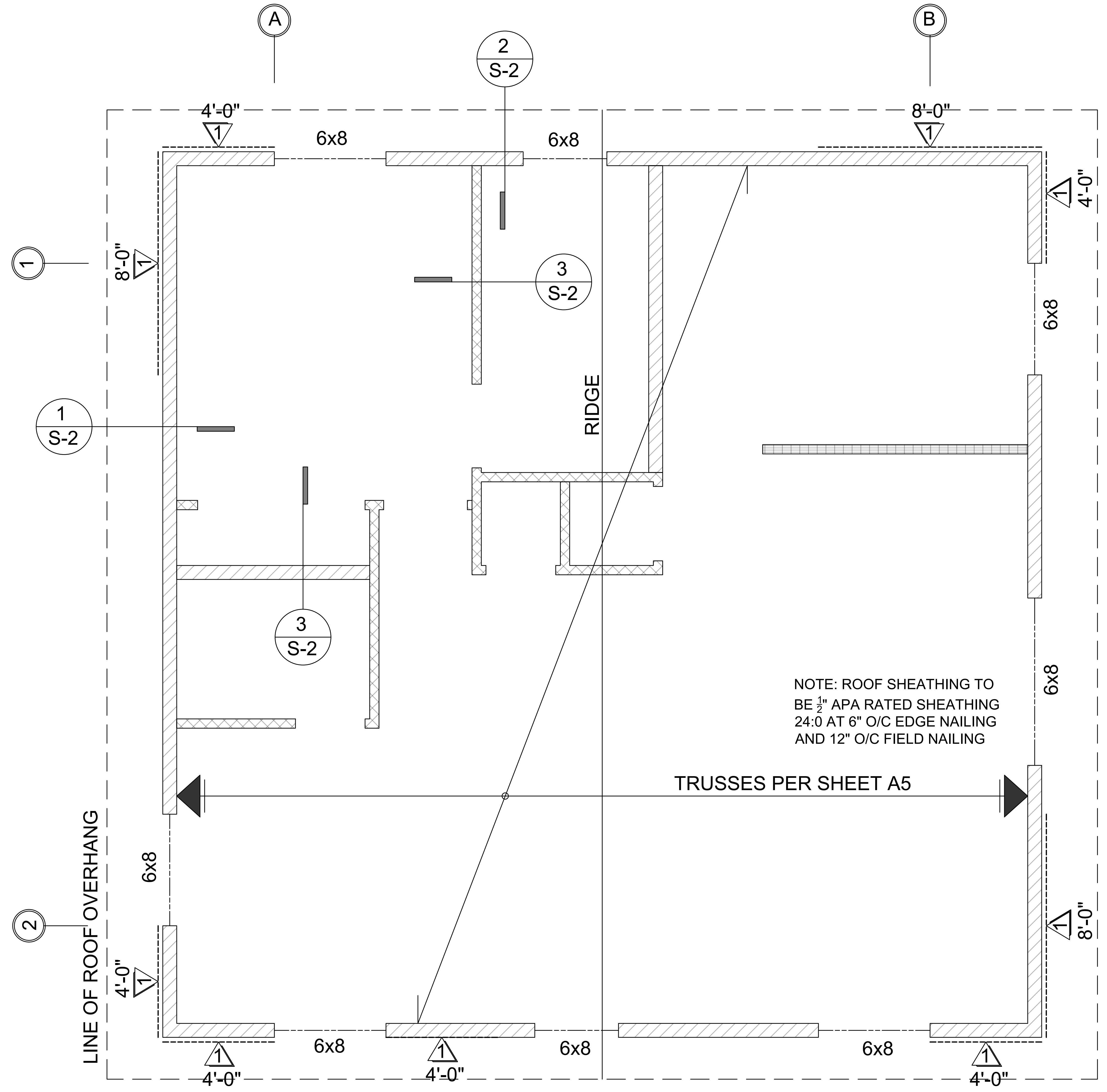
DETACHED DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

FOUNDATION PLAN & DETAILS

SHEET NUMBER

S1


A
B
2
S-2
3
S-2
1
S-2
3
S-2
1
2
4'-0"
8'-0"
6x8
6x8
4'-0"
8'-0"
6x8
6x8
RIDGE
TRUSSES PER SHEET A5
6x8
4'-0"
8'-0"
LINE OF ROOF OVERHANG
6x8
6x8
6x8
4'-0"
4'-0"
4'-0"
1

NOTE: ROOF SHEATHING TO BE $\frac{1}{2}$" APA RATED SHEATHING 24:0 AT 6" O/C EDGE NAILING AND 12" O/C FIELD NAILING

WOOD STRUCTURAL PANEL SHEATHING

MARK	MINIMUM NAIL		MINIMUM WOOD STRUCTURAL PANEL SPAN RATING	MINIMUM NOMINAL PANEL THICKNESS (in)	MAXIMUM WALL STUD SPACING (in)	PANEL NAIL SPACING	
	SIZE	PENETRATION (in)				EDGES (inches o/c)	FIELD (inches o/c)
1	6D COMMON	1.5	24:0	$\frac{3}{8}$"	16	6	12
	8D COMMON	1.75	24:16	$\frac{7}{16}$"	16	6	12

WOOD STRUCTURAL PANELS SHALL CONFORM TO DOC PS 1, DOC PS 2 OR ANSI/APA PRP 210, CSA O437 OR CSA O325.
PANELS SHALL BE IDENTIFIED BY A GRADE MARK OR CERTIFICATE OF INSPECTION ISSUED BY AN APPROVED AGENCY

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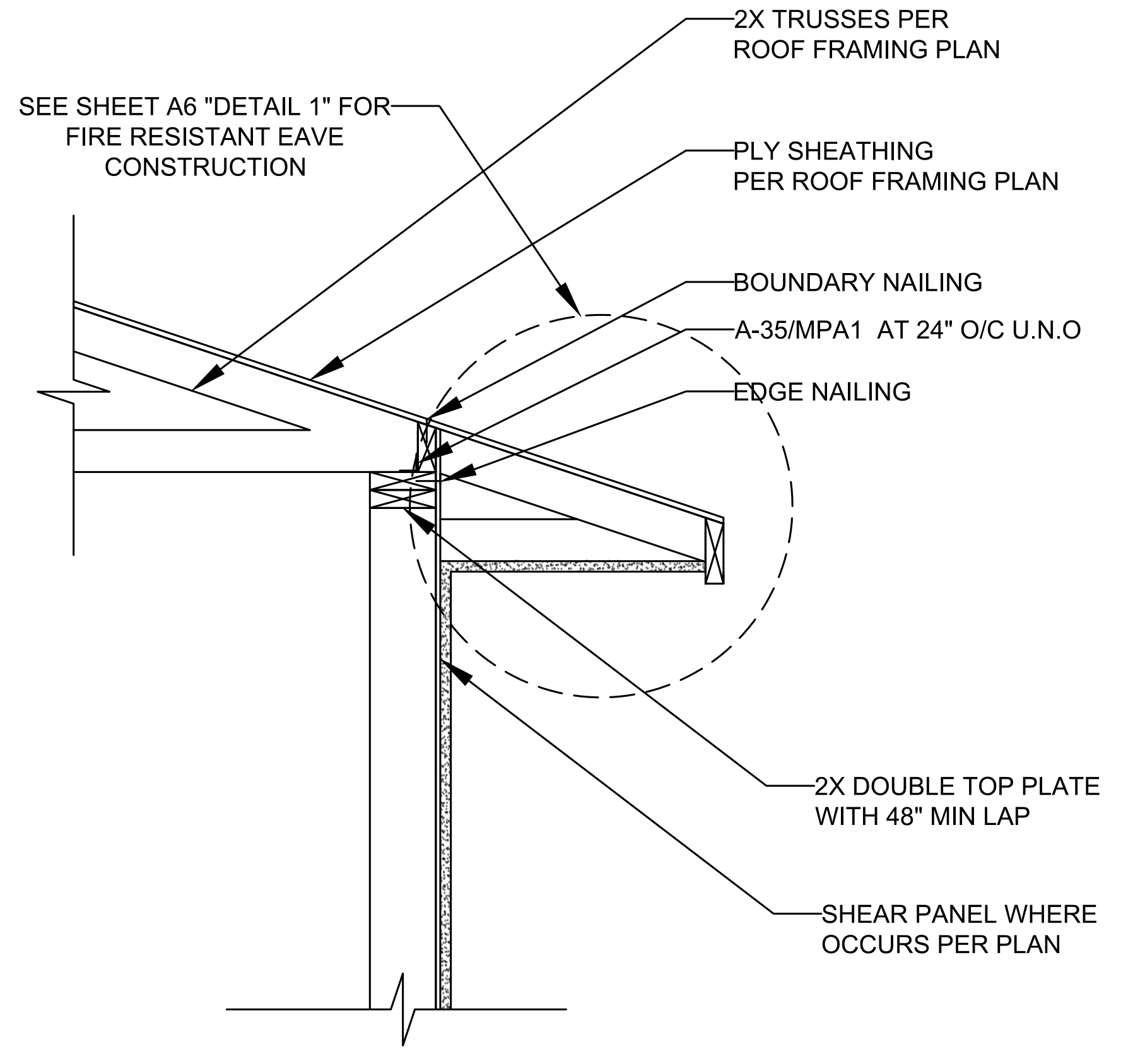
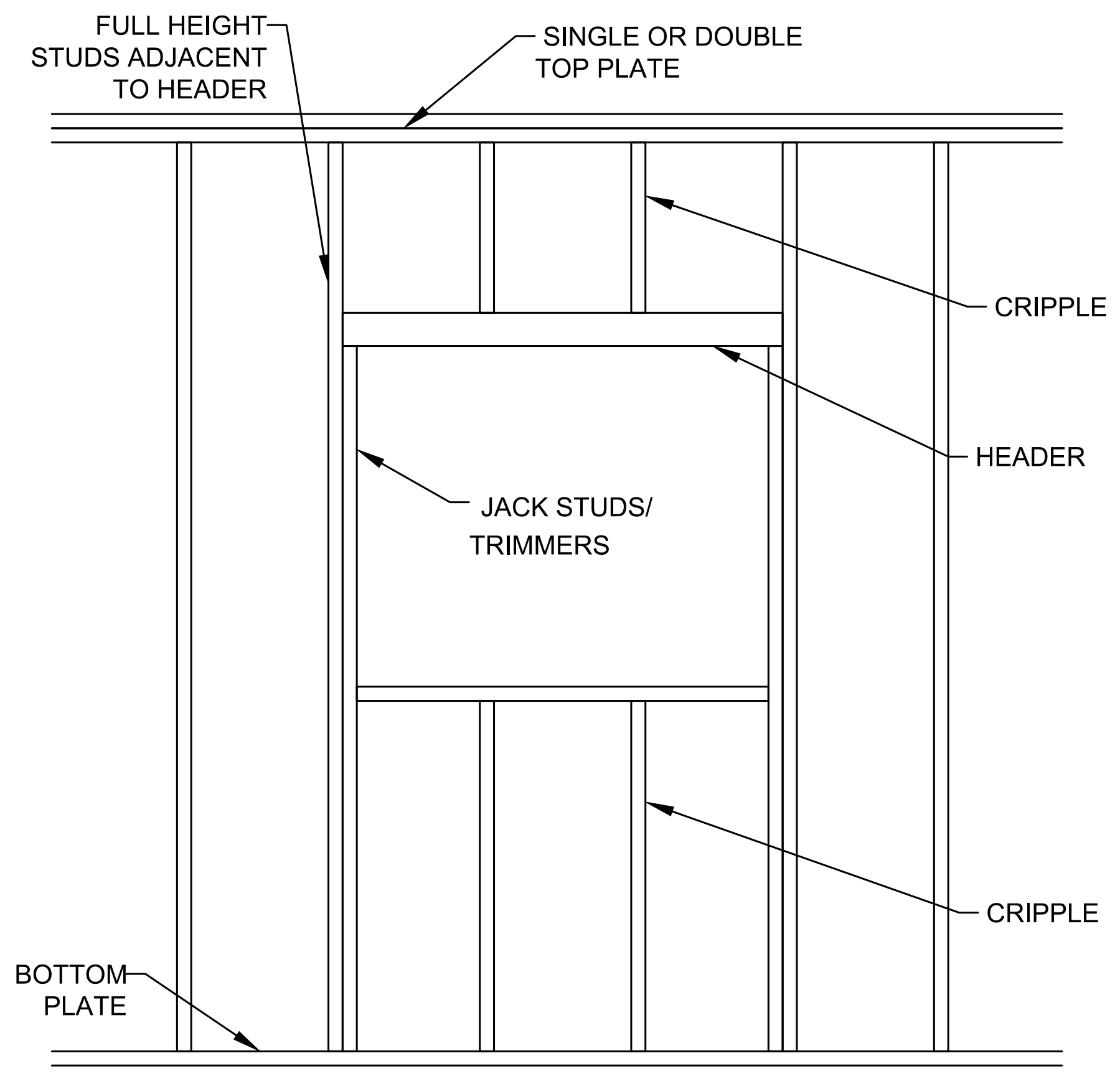
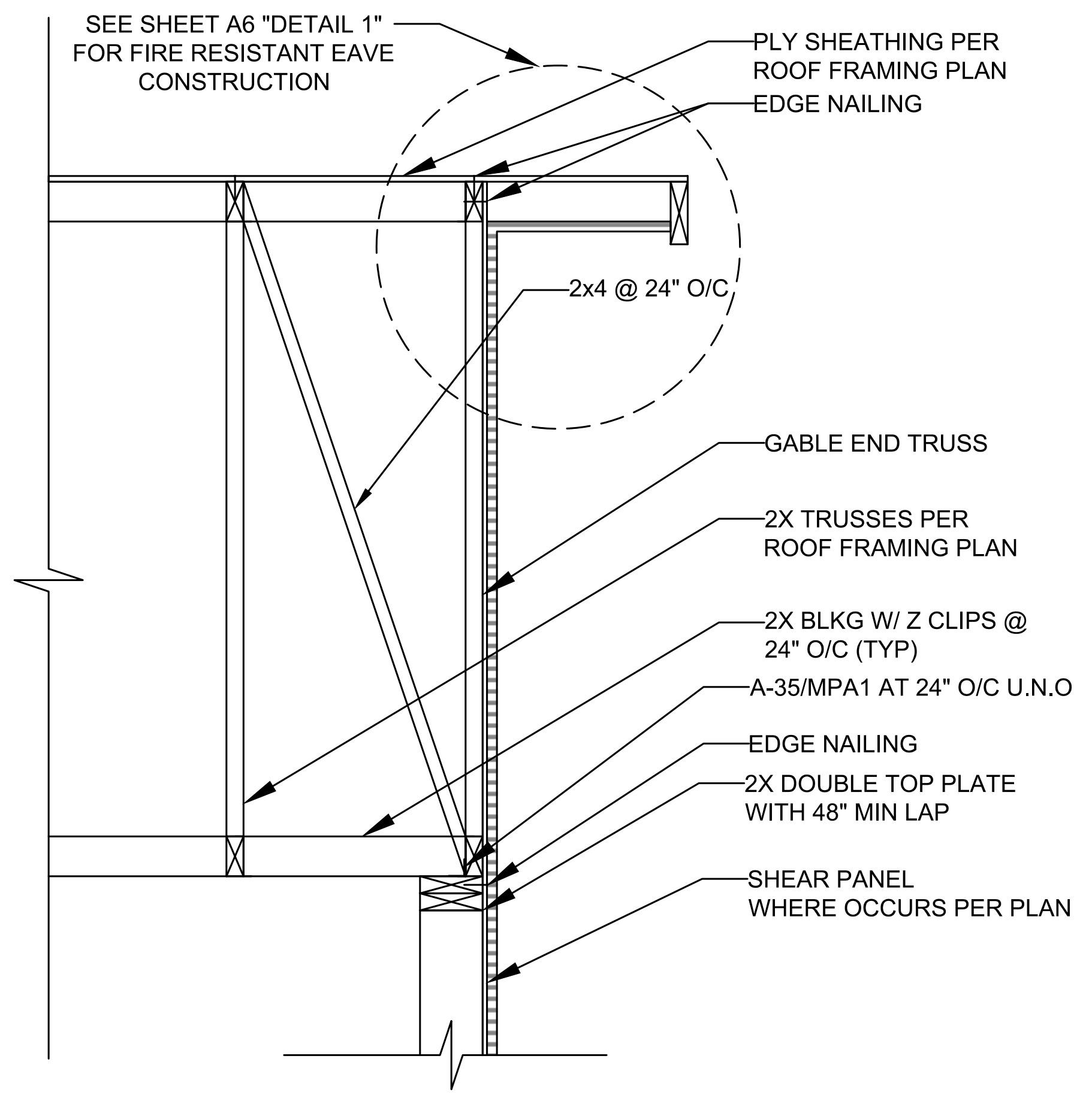
ROOF FRAMING PLAN
3/8" = 1'-0"


2X TRUSSES PER
ROOF FRAMING PLAN
SEE SHEET A6 "DETAIL 1" FOR
FIRE RESISTANT EAVE
CONSTRUCTION
PLY SHEATHING
PER ROOF FRAMING PLAN
BOUNDARY NAILING
A-35/MPA1 AT 24" O/C U.N.O
EDGE NAILING
2X DOUBLE TOP PLATE
WITH 48" MIN LAP
SHEAR PANEL WHERE
OCCURS PER PLAN

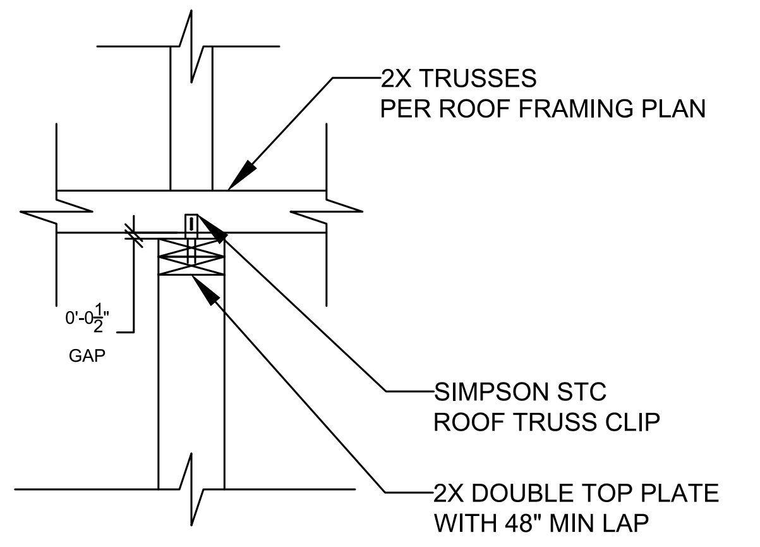
① EAVE (SHEAR TRANSFER)


FULL HEIGHT
STUDS ADJACENT
TO HEADER
SINGLE OR DOUBLE
TOP PLATE
CRIPPLE
HEADER
JACK STUDS/
TRIMMERS
CRIPPLE
BOTTOM
PLATE

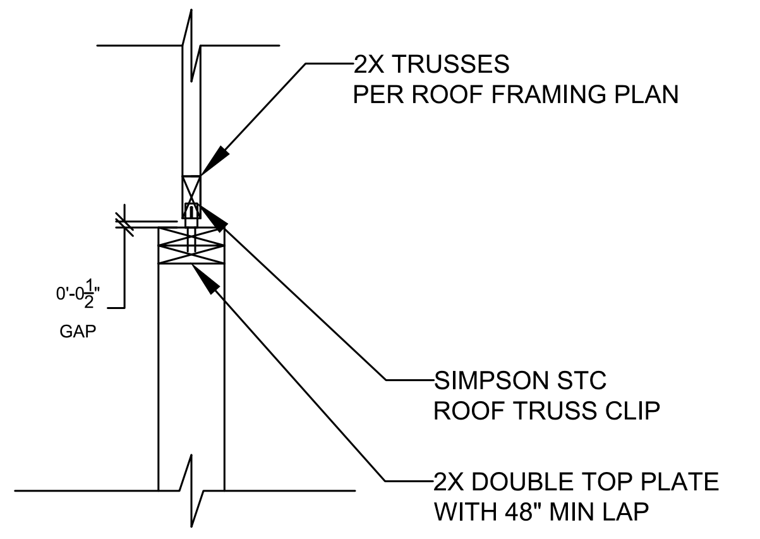
④ TYPICAL FRAMING AT OPENING


SEE SHEET A6 "DETAIL 1"
FOR FIRE RESISTANT EAVE
CONSTRUCTION
PLY SHEATHING PER
ROOF FRAMING PLAN
EDGE NAILING
2x4 @ 24" O/C
GABLE END TRUSS
2X TRUSSES PER
ROOF FRAMING PLAN
2X BLKG W/ Z CLIPS @
24" O/C (TYP)
A-35/MPA1 AT 24" O/C U.N.O
EDGE NAILING
2X DOUBLE TOP PLATE
WITH 48" MIN LAP
SHEAR PANEL
WHERE OCCURS PER PLAN

② GABLE-END (SHEAR TRANSFER)


2X TRUSSES
PER ROOF FRAMING PLAN
0'-0$\frac{1}{2}$"
GAP
SIMPSON STC
ROOF TRUSS CLIP
2X DOUBLE TOP PLATE
WITH 48" MIN LAP

TRUSS PERPENDICULAR TO FRAMING MEMBER


2X TRUSSES
PER ROOF FRAMING PLAN
0'-0$\frac{1}{2}$"
GAP
SIMPSON STC
ROOF TRUSS CLIP
2X DOUBLE TOP PLATE
WITH 48" MIN LAP

TRUSS PARALLEL TO FRAMING MEMBER

③ NON-BEARING WALL

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PROJECT TITLE

DETACHED
DWELLING UNIT
(PLAN D - 1000 SF)

SHEET TITLE

FRAMING PLAN
& DETAILS

SHEET NUMBER

S2

A. General

Applicable codes. All projects will comply with the following building codes and associated County of San Diego amendment.
2025 California Residential Code (CRC) and/or California Building Code (CBC)
2025 California Green Building Standards Code (CalGreen)
2025 California Electrical Code (CEC)
2025 California Mechanical Code (CMC)
2025 California Plumbing Code (CPC)
2025 California Fire Code (CFC)
2025 California Building Energy Efficiency Standards (CBEES)
2025 California Wildland-Urban Interface Code (CWUIC)

B. Electrical, Plumbing, and Mechanical

1. **Exterior lighting.** All projects shall comply with the County of San Diego lighting ordinance.
2. **GFCI outlets.** Ground Fault Circuit Interrupter (GFCI) outlets are required in bathrooms, at kitchen countertops, at laundry and wet bar sinks, in garages, in crawlspaces, in unfinished basements, and outdoors. (CEC 210.8)
3. **AFCI outlets.** Electrical circuits in bedrooms, living rooms, dining rooms, dens, closets, hallways, or similar rooms must be protected by Arc Fault Circuit Interrupters (AFCI). (CEC 210.12)
4. **Luminaire requirements.** Installed luminaires shall meet the efficacy and fixture requirements of CBEES 150.0(k).
5. **Smoke detectors in building remodels.** Smoke detectors are required in each existing sleeping room, outside each separate sleeping area in the immediate vicinity of sleeping rooms, and on each story of a dwelling including basements. Battery-operated detectors are acceptable in existing areas with no construction taking place and in alterations not resulting in removal of interior wall or ceiling finishes and without access via an attic, crawl space, or basement. (CRC R314.3)
6. **Carbon monoxide detectors in building remodels.** Carbon monoxide detectors are required outside each separate sleeping area in the immediate vicinity of sleeping rooms and on each story of a dwelling including basements. Battery-operated detectors are acceptable in existing areas with no construction taking place and in alterations not resulting in removal of interior wall or ceiling finishes and without access via an attic, crawl space, or basement. (CRC R315.3)
7. **Water heater seismic strapping.** Minimum two 3/4-inch-by-24-gauge straps required around water heaters, with 1/4-inch-by-3-inch lag bolts attached directly to framing. Straps shall be at points within upper third and lower third of water heater vertical dimension. Lower connection shall occur minimum 4 inches above controls. (CPC 507.2)
8. **Gas appliances in garages.** Water heaters and heating/cooling equipment capable of igniting flammable vapors shall be placed on minimum 18-inch-high platform unless listing report number provided showing ignition-resistant appliance. (CPC 507.13 and CMC 305.1)
9. **Impact protection of appliances.** Water heaters and heating/cooling equipment subject to vehicular impact shall be protected by bollards or an equivalent measure. (CPC 507.13.1 and CMC 305.11)
10. **Water closet clearance.** Minimum 30-inch-wide by 24-inch-deep clearance required at front of water closets. (CPC 402.5)
11. **Shower size.** Shower compartments shall have minimum area of 1024 square inches and be able to encompass a 30-inch-diameter circle. Shower doors shall have a minimum 22-inch unobstructed width. (CPC 408.5 and CPC 408.6)
12. **Fireplace appliances.** Fireplaces with gas appliances are required to have the flue damper permanently fixed in the open position and fireplaces with LPG appliances are to have no 'pit' or 'sump' configurations. (CMC 303.7.1)
13. **Chimney clearance.** Minimum 2-foot chimney clearance required above building within 10-foot horizontally of chimney. The chimney shall extend minimum 3 feet above highest point where chimney passes through roof. (CRC R1003.9)

C. Mechanical Ventilation and Indoor Air Quality (ASHRAE 62.2-2010)

1. **Transfer air.** Ventilation air shall be provided directly from the outdoors and not as transfer air from adjacent dwelling units or other spaces, such as garages, unconditioned crawlspaces, or unconditioned attics. (CBEES 150.0(o))
2. **Instructions and labeling.** Ventilation system controls shall be labeled, and the homeowner shall be provided with instructions on how to operate the system. (CBEES 150.0(o))
3. **Combustion and solid-fuel burning appliances.** Combustion appliances shall be properly vented and air systems shall be designed to prevent back drafting. (CBEES 150.0(o))
4. **Garages.** The wall and openings between occupiable spaces and the garage shall be sealed. HVAC systems that include air handlers or return ducts located in garages shall have total air leakage of no more than 6% of total fan flow when measured at 0.1 in. w.c. using California Title 24 or equivalents. (CBEES 150.0(o))
5. **Minimum filtration.** Mechanical systems supplying air to occupiable space through ductwork shall be provided with a filter having a minimum efficiency of MERV 6 or better. (CBEES 150.0(o))
6. **Air inlets.** Air inlets (not exhaust) shall be located away from known contaminants. (CBEES 150.0(o))
7. **Air moving equipment.** Air moving equipment used to meet either the whole-building ventilation requirement or the local ventilation exhaust requirement shall be rated in terms of airflow and sound. (CBEES 150.0(o))

   a. All continuously operating fans shall be rated at a maximum of 1.0 sone.

   b. Intermittently operated whole-building ventilation fans shall be rated at a maximum of 1.0 sone.

   c. Intermittently operated local exhaust fans shall be rated at maximum of 3.0 sone.

   d. Remotely located air-moving equipment (mounted outside of habitable spaces) need not meet sound requirements if at least 4 feet of ductwork between fan and intake grill.

D. Foundation and Underfloor

1. **Foundation reinforcement.** Continuous footings and stem walls shall be provided with a minimum two longitudinal No. 4 bars, one at the top and one at the bottom of the footing. (CRC R403.1.3.3)
2. **Shear wall foundation support.** Shear walls shall be supported by continuous foundations. (CRC 403.1.2)
3. **Concrete slabs-on-grade.** Slabs-on-grade shall be minimum 3-1/2-inches thick. (CRC R506.1)
4. **Vapor retarder.** A 10-mil polyethylene or approved vapor retarder with joints lapped minimum 6 inches shall be placed between a concrete slab-on-grade and the base course or subgrade. (CRC 506.2.3)
5. **Anchor bolts and sills.** Foundation plates or sills shall be bolted or anchored to the foundation or foundation wall per the following (CRC R403.1.6 and CRC R602.11.1):

   a. Minimum 1/2-inch-diameter steel bolts

   b. Bolts embedded at least 7 inches into concrete or masonry

   c. Bolts spaced maximum 6 feet on center

   d. Minimum two bolts per plate/sill piece with one bolt located maximum 12 inches and minimum 7 bolt diameters from each end of each sill plate/piece

   e. Minimum 3-inch by 3-inch by 0.299-inch steel plate washer between sill and nut on each bolt

6. Hold-**downs**. All hold-downs must be tied in place prior to foundation inspection.
7. **Protection of wood against decay.** Naturally durable or preservative-treated wood shall be provided in the following locations (CRC R317.1):

   a. All wood in contact with ground, embedded in concrete in direct contact with ground, or embedded in concrete exposed to weather

   b. Wood joists within 18 inches and wood girders within 12 inches of the exposed ground in crawl spaces shall be of naturally durable or preservative-treated wood

   c. Wood framing members that rest on concrete or masonry exterior foundation walls and are less than 8 inches from exposed earth shall be of naturally durable or preservative-treated wood

   d. Wood framing, sheathing, and siding on the exterior of the building and having clearance less than 6 inches from the exposed ground or less than 2 inches vertically from concrete steps, porch slabs, patio slabs, and similar horizontal surface exposed to weather

   e. Sills and sleepers on concrete or masonry slab in direct contact with ground unless separated from such slab by impervious moisture barrier

   f. Ends of wood girders entering masonry or concrete walls with clearances less than 1/2 inch on tops, sides, and ends

   g. Wood structural members supporting moisture-permeable floors or roofs exposed to weather, such as concrete or masonry slabs, unless separated from such floors or roofs by an impervious moisture barrier

   h. Wood furring strips or other wood framing members attached directly to interior of exterior concrete or masonry walls below grade except where vapor retarder applied between wall and furring strips or framing members

8. **Underfloor ventilation.** Underfloor areas shall have ventilation openings through foundation walls or exterior walls, with minimum net area of ventilation openings of 1 square foot for each 150 square feet of underfloor area. On such ventilating opening shall be within 3 feet of each corner of the building. (CRC R408.1)
9. **Underfloor access.** Underfloor areas shall be provided with a minimum 18-inch by 24-inch access opening. (CRC R408.4)

E. Wood Framing

1. **Fastener requirements.** The number, size, and spacing of fasteners connecting wood members/elements shall not be less than that set forth in CRC Table R602.3(1). (CRC R502.9, CRC R602.3, and CRC R802.2)
2. **Stud size, height, and spacing.** The size, height, and spacing of studs shall be in accordance with CRC Table R602.3(5). (CRC R602.3.1)

E. Wood Framing (Continued)

3. **Sill plate**. Studs shall have full bearing on nominal 2-inch thick or larger sill plate with width at least equal to stud width. (CRC R602.3.4)
4. **Bearing studs.** Where joists, trusses, or rafters are spaced more than 16 inches on center and the bearing studs below are spaced 24 inches on center, such members shall bear within 5 inches of the studs beneath. (CRC R602.3.3)
5. **Drilling and notching of studs.** Any stud in an exterior wall or bearing partition may be cut or notched to a depth not exceeding 25% of its width. Studs in nonbearing partitions may be notched to a depth not to exceed 40% of a single stud width. Any stud may be bored or drilled, provided the diameter of the resulting hole is no more than 60% of the stud width, the edge of the hole is no more than 5/8 inch to the edge of the stud, and the hole is not located in the same section as a cut or notch. Studs located in exterior wall or bearing partitions drilled over 40% and up to 60% shall also be doubled with no more than two successive studs bored. (CRC R602.6)
6. **Top plate.** Wood stud walls shall be capped with a double top plate installed to provide overlapping at corners and at intersections with other partitions. End joints in double top plates shall be offset at least 24 inches. Joints in plates need not occur over studs. Plates shall be minimum nominal 2 inches thick and have width at least equal to width of studs. (CRC R602.3.2)
7. **Top plate splices.** Top plate lap splices shall be face-nailed with minimum 8 16d nails on each side of splice. (CRC R602.10.8.1)
8. **Drilling and notching of top plate.** When piping or ductwork is placed in or partly in an exterior wall or interior load-bearing wall, necessitating cutting, drilling, or notching of the top plate by more than 50% of its width, a galvanized metal tie not less than 0.054-inch thick and 1-1/2-inches wide shall be fastened across and to the plate at each side of the opening with not less than 8 10d nails having a minimum length of 1-1/2 inches at each side or equivalent. The metal tie must extend minimum 6 inches past the opening. (CRC R602.6.1)
9. **Cripple walls.** Foundation cripple walls shall be framed of studs not less in size than the studding above. Cripple walls more than 4 feet in height shall have studs sized as required for an additional story. Cripple walls with stud height less than 14 inches shall be sheathed on at least one side with a wood structural panel fastened to both the top and bottom plates in accordance with Table R602.3(1), or the cripple walls shall be constructed of solid blocking. Cripple walls shall be supported on continuous foundations. (CRC R602.9)
10. **Wall bracing.** Buildings shall be braced in accordance with the methods allowed per CRC R602.10.2, CRC R602.10.4, and/or CRC R602.10.5.
11. **Braced wall line spacing.** Spacing between braced wall lines shall not exceed 20 feet or alternate provisions of CRC R602.10.1.3.
12. **Shear wall cumulative length.** The cumulative length of shear walls within each braced wall line shall meet the provisions of CRC Table R602.10.3(1) for wind loads and CRC Table R602.10.3(2) for seismic loads. (CRC R602.10.1.1)
13. **Shear wall spacing.** Shear walls shall be located not more than 25 feet on center. (CRC R602.10.2.2)
14. **Shear wall offset.** Shear walls may be offset out-of-plan not more than 4 feet from the designated braced wall line and not more than 8 feet from any other offset wall considered part of the same braced wall line. (CRC R602.10.1.2)
15. **Shear wall location.** Shear walls shall be located at the ends of each braced wall line or meet the alternate provisions of CRC R602.10.2.2.
16. **Individual shear wall length.** Shear walls shall meet minimum length requirements of CRC R602.10.6.5.1.
17. **Cripple wall bracing**. Cripple walls shall be braced per CRC R602.10.11.
18. **Shear wall and diaphragm nailing**. All shear walls, roof diaphragms, and floor diaphragms shall be nailed to supporting construction per CRC Table R602.3(1). (CRC R604.3)
19. **Shear wall joints.** All vertical joints in shear wall sheathing shall occur over, and be fastened to, common studs. Horizontal joints in shear walls shall occur over, and be fastened to, minimum 1-1/2-inch-thick blocking. (CRC R602.10.10)
20. **Framing over openings.** Headers, double joists, or trusses of adequate size to transfer loads to vertical members shall be provided over window and door openings in load-bearing walls and partitions. (CBC 2304.3.2)
21. **Joists under bearing partitions.** Joists under parallel bearing partitions shall be of adequate size to support the load. Double joists, sized to adequately support the load, that are separated to permit the installation of piping or vents shall be full-depth solid-blocked with minimum 2-inch nominal lumber spaced at maximum 4 feet on center. Bearing partitions perpendicular to joists shall not be offset from supporting girders, walls, or partitions more than the joist depth unless such joists are of sufficient size to carry the additional load. (CRC R502.4)
22. **Joists above or below shear walls.** Where joists are perpendicular to a shear wall above or below, a rim joist, band joist, or blocking shall be provided along the entire length of the shear wall. Where joists are parallel to a shear wall above or below, a rim joist, end joist, or other parallel framing shall be provided directly above and/or below the shear wall. Where a parallel framing member cannot be located directly above and/or below the shear wall, full-depth blocking at 16-inch spacing shall be provided between the parallel framing members to each side of the shear wall. (CRC R602.10.8)
23. **Floor member bearing.** The ends of each floor joist, beam, or girder shall have minimum 1-1/2 inches of bearing on wood or metal and minimum 3 inches of bearing on masonry or concrete except where supported on a 1-inch-by-4-inch ribbon strip and nailed to the adjoining stud or by the use of approved joist hangers. (CRC R502.6)
24. **Floor joist lap.** Floor joists framing opposite sides over a bearing support shall lap minimum 3 inches and shall be nailed together within minimum 3 10d face nails. A wood or metal splice with strength equal to or greater than that provided by the lap is permitted. (CRC R502.6.1)
25. **Floor joist-to-girder support.** Floor joists framing into the side of a wood girder shall be supported by approved framing anchors or on ledger strips minimum nominal 2 inches by 2 inches. (CRC R502.6.2)
26. **Floor joist lateral restraint.** Floor joists shall be supported laterally at ends and each intermediate support by minimum 2-inch full-depth blocking, by attachment to full-depth header, band joist, or rim joist, to an adjoining stud, or shall be otherwise provided with lateral support to prevent rotation. (CRC R502.7)
27. **Floor joist bridging.** Floor joists exceeding nominal 2 inches by 12 inches shall be supported laterally by solid blocking, diagonal bridging (wood or metal), or a continuous 1-inch-by-3-inch strip nailed across the bottom of joists perpendicular to joists at maximum 8-foot intervals. (CRC R502.7.1)
28. **Framing of floor openings.** Openings in floor framing shall be framed with a header and trimmer joists. When the header joist span does not exceed 4 feet, the header joist may be a single member the same size as the floor joist. Single trimmer joists may be used to carry a single header joist located within 3 feet of the trimmer joist bearing. When the header joist span exceeds 4 feet, the trimmer joists and header joist shall be doubled and of sufficient cross section to support the floor joists framing into the header. Approved hangers shall be used for the header-joist-to-trimmer-joist connections when the header joist span exceeds 6 feet. Tail joists over 12 feet long shall be supported at the header by framing anchors or on ledger strips minimum 2 inches by 2 inches. (CRC R502.10)
29. **Girders**. Girders for single-story construction or girders supporting loads from a single floor shall not be less than 4 inches by 6 inches for spans 6 feet or less, provided that girders are spaced not more than 8 feet on center. Other girders shall be designed to support the loads specified in the CBC. Girder end joints shall occur over supports. When a girder is spliced over a support, an adequate tie shall be provided. The ends of beams or girders supported on masonry or concrete shall not have less than 3 inches of bearing. (CBC 2308.7)
30. **Ridges, hips, and valleys**. Rafters shall be framed to a ridge board or to each other with a gusset plate as a tie. Ridge boards shall be minimum 1-inch nominal thickness and not less in depth than the cut end of the rafter. At all valley and hips, there shall be a valley or hip rafter not less than 2-inch nominal thickness and not less in depth than the cut end of the rafter. Hip and valley rafters shall be supported at the ridge by a brace to a bearing partition or be designed to carry and distribute the specific load at that point. Where the roof pitch is less than 3:12 slope (25% gradient), structural members that support rafters and ceilings joists, such as ridges, hips, and valleys, shall be designed as beams. (CRC R802.3)
31. **Ceiling joist and rafter connections.** Ceiling joists and rafters shall be nailed to each other per CRC Table R802.5.1(9), and the rafter shall be nailed to the wall top plate per CRC Table R602.3(1). Ceiling joists shall be continuous or securely joined per CRC Table R802.5.1(9) where they meet over interior partitions and are nailed to adjacent rafters to provide a continuous tie across the building when such joists are parallel to rafters. Where ceiling joists are not connected to the rafters at the wall top plate, joists connected higher in the attic shall be installed as rafter ties, or rafter ties shall be installed to provide a continuous tie. Where ceiling joists are not parallel to rafters, rafter ties shall be installed. Rafter ties shall be minimum 2 inches by 4 inches nominal, installed per CRC Table R802.5.1(9), or connections of equivalent capacities shall be provided. Where ceilings joists or rafter ties are not provided, the ridge formed by these rafters shall be supported by a wall or engineer-designed girder. (CRC R802.3.1)
32. **Ceiling joists lapped.** Ends of ceiling joists shall be lapped minimum 3 inches or butted overbearing partitions or beams and toenailed to the bearing element. Where ceiling joists provide resistance to rafter thrust, lapped joists shall be nailed together per CRC Table R602.3(1) and butted joists shall be tied together in a manner to resist such thrust. (CRC R802.3.2)
33. **Collar ties.** Collar ties or ridge straps to resist wind uplift shall be connected in the upper third of the attic space. Collar ties shall be a minimum 1 inch by 4 inches nominal and spaced at maximum 4 feet on center. (CRC R802.3.1)
34. **Purlins.** Purlins installed to reduce the span of rafters shall be sized not less than the required size of the rafters they support. Purlins shall be continuous and shall be supported by 2-inch-by-4-inch nominal braces installed to bearing walls at a minimum 45-degree slope from horizontal. The braces shall be spaced maximum 4 feet on center with a maximum 8-foot unbraced length. (CRC R802.5.1)
35. **Roof/ceiling member bearing.** The ends of each rafter or ceiling joist shall have not less than 1-1/2 inches of bearing on wood or metal and not less than 3 inches of bearing on masonry or concrete. (CRC R802.6)
36. **Roof/ceiling member lateral support.** Roof framing members and ceiling joists with a nominal depth-to-thickness ratio exceeding 5:1 shall be provided with lateral support at points of bearing to prevent rotation. (CRC R802.8)
37. **Roof/ceiling bridging.** Rafters and ceiling joists with a nominal depth-to-thickness ratio exceeding 6:1 shall be supported laterally by solid blocking, diagonal bridging (wood or metal), or a continuous 1-inch-by-3-inch wood strip nailed across the rafters or ceiling joists at maximum 8-foot intervals. (CRC R802.8.1)

E. Wood Framing (Continued)

38. **Framing of roof/ceiling openings.** Openings in roof and ceiling framing shall be framed with a header and trimmer joists. When the header joist span does not exceed 4 feet, the header joist may be a single member the same size as the ceiling joist or rafter. Single trimmer joists may be used to carry a single header joist located within 3 feet of the trimmer joist bearing. When the header joist span exceeds 4 feet, the trimmer joists and header joist shall be doubled and of sufficient cross section to support the ceiling joists or rafters framing into the header. Approved hangers shall be used for the header-joist-to-trimmer-joist connections when the header joist span exceeds 6 feet. Tail joists over 12 feet long shall be supported at the header by framing anchors or on ledger strips minimum 2 inches by 2 inches. (CRC R502.10)
39. **Roof framing above shear walls.** Rafters or roof trusses shall be connected to top plates of shear walls with blocking between the rafters or trusses. (CRC R602.10.8)
40. **Roof diaphragm under fill framing.** Roof plywood shall be continuous under California fill framing.
41. **Roof diaphragm at ridges.** Minimum 2-inch nominal blocking required for roof diaphragm nailing at ridges.
42. **Blocking of roof trusses.** Minimum 2-inch nominal blocking required between trusses at ridge lines and at points of bearing at exterior walls.
43. **Truss clearance.** Minimum 1/2-inch clearance required between top plates of interior non-bearing partitions and bottom chords of trusses.
44. **Drilling, cutting, and notching of roof/floor framing.** Notches in solid lumber joists, rafters, blocking, and beams shall not exceed one-sixth the member depth, shall be not longer than one-third the member depth, and shall not be located in the middle one-third of the span. Notches at member ends shall not exceed one-fourth the member depth. The tension side of members 4 inches or greater in nominal thickness shall not be notched except at member ends. The diameter of holes bored or cut into members shall not exceed one-third the member depth. Holes shall not be closer than 2 inches to the top or bottom of the member or to any other hole located in the member. Where the member is also notched, the hole shall not be closer than 2 inches to the notch. (CRC R502.8.1)
45. **Exterior landings, decks, balconies, and stairs.** Such elements shall be positively anchored to the primary structure to resist both vertical and lateral forces or shall be designed to be self-supporting. Attachment shall not be accomplished by use of toenails or nails subject to withdrawal. (CRC R311.3)
46. **Fireblocking.** Fireblocking shall be provided in the following locations (CRC R302.11 and CRC R1003.19):

   a. In concealed spaces of stud walls and partitions, including furred spaces, and parallel rows of studs or staggered studs, as follows:

      I. Vertically at the ceiling and floor levels

      II. Horizontally at intervals not exceeding 10 feet

   b. At all interconnections between concealed vertical and horizontal spaces such as occur at soffits, drop ceilings, and cove ceilings

   c. In concealed spaces between stair stringers at the top and bottom of the run

   d. At openings around vents, pipes, ducts, cables and wires at ceiling and floor level, with an approved material to resist the free passage of flame and products of combustion

   e. At chimneys and fireplaces per item E.49

   f. Cornices of a two-family dwelling at the line of dwelling-unit separation

47. **Fireblocking materials.** Except as otherwise specified in items E.48 and E.49, fireblocking shall consist of the following materials with the integrity maintained (CRC R302.11.1):

   a. Two-inch nominal lumber

   b. Two thicknesses of one-inch nominal lumber with broken lap joints

   c. One thickness of 23/32-inch wood structural panel with joints backed by 23/32-inch wood structural panel

   d. One thickness of 3/4-inch particleboard with joints backed by 3/4-inch particleboard

   e. 1/2-inch gypsum board

   f. 1/4-inch cement-based millboard

   g. Batts or blankets of mineral or glass fiber of other approved materials installed in such a manner as to be securely retained in place. Batts or blankets of mineral or glass fiber or other approved non-rigid materials shall be permitted for compliance with the 10-foot horizontal fireblocking in walls constructed using parallel rows of studs or staggered studs. Unfaced fiberglass batt insulation used as fireblocking shall fill the entire cross-section of the wall cavity to a minimum height of 16 inches measured vertically. When piping, conduit, or similar obstructions are encountered, the insulation shall be packed tightly around the obstruction. Loose-fill insulation material shall not be used as a fireblock unless specifically tested in the form and manner intended for use to demonstrate its ability to remain in place and to retard the spread of fire and hot gases.

48. **Fireblocking at openings around vents, pipes, ducts, cables, and wires at ceiling and floor level**. Such openings shall be fireblocked with an approved material to resist the free passage of flame and products of combustion. (CRC R302.11)
49. **Fireblocking of chimneys and fireplaces.** All spaces between chimneys and floors and ceilings through which chimneys pass shall be fireblocked with noncombustible material securely fastened in place. The fireblocking of spaces between chimneys and wood joists, beams, or headers shall be self-supporting or be placed on strips of metal or metal lath laid across the spaces between combustible material and the chimney. (CRC R1003.19)
50. **Draftstopping**. In combustible construction where there is usable space both above and below the concealed space of a floor/ceiling assembly, draftstops shall be installed so that the area of the concealed space does not exceed 1000 square feet. Draftstopping shall divide the concealed space into approximately equal areas. Where the assembly is enclosed by a floor membrane above and a ceiling membrane below, draftstopping shall be provided in floor/ceiling assemblies under the following circumstances (CRC R302.12):

   a. Ceiling is suspended under the floor framing

   b. Floor framing is constructed of truss-type open-web or perforated members

51. **Draftstopping materials.** Draftstopping shall not be less than 1/2-inch gypsum board, 3/8-inch wood structural panels, or other approved materials adequately supported. Draftstopping shall be installed parallel to the floor framing members unless otherwise approved by the building official. The integrity of draftstops shall be maintained. (CRC R302.12.1)
52. **Combustible insulation clearance.** Combustible insulation shall be separated minimum 3 inches from recessed luminaires, fan motors, and other heat-producing devices. (CRC R302.14)

F. General Material Specifications

1. **Lumber.** All joists, rafters, beams, and posts 2-inches to 4-inches thick shall be No. 2 grade Douglas Fir-Larch or better. All posts and beams 5 inches and thicker shall be No. 1 grade Douglas Fir-Larch or better. Studs not more than 8 feet long shall be stud-grade Douglas Fir-Larch or better when supporting not more than one floor, roof, and ceiling. Studs longer than 8 feet shall be No. 2 grade Douglas Fir-Larch or better.
2. **Concrete.** Concrete shall have a minimum compressive strength of 2,500 psi at 28 days and shall consist of 1 part cement, 3 parts sand, 4 parts 1-inch maximum size rock, and not more than 7-1/2 gallons of water per sack of cement. (CRC R402.2)
3. **Mortar.** Mortar used in construction of masonry walls, foundation walls, and retaining walls shall conform to ASTM C 270 and shall consist of 1 part portland cement, 2-1/4 to 3 parts sand, and 1/4 to 1/2 part hydrated lime. (CBC 2103.2)
4. **Grout.** Grout shall conform to ASTM C 476 and shall consist of 1 part portland cement, 1/10 part hydrated lime, 2-1/4 to 3 parts sand, and 1 to 2 parts gravel. Grout shall attain a minimum compressive strength of 2,000 psi at 28 days. (CBC 2103.3)
5. **Masonry.** Masonry units shall comply with ASTM C 90 for load-bearing concrete masonry units. (CBC 2103.1)
6. **Reinforcing steel.** Reinforcing steel used in construction of reinforced masonry or concrete structures shall be deformed and comply with ASTM A 615. (CBC 2103.4)
7. **Structural steel.** Steel used as structural shapes such as wide-flange sections, channels, plates, and angles shall comply with ASTM A36. Pipe columns shall comply with ASTM A53. Structural tubes shall comply with ASTM A500, Grade B.
8. **Fasteners for preservative-treated wood.** Fasteners for preservative-treated and fire-retardant-treated wood - including nuts and washers -- shall be of hot dipped zinc-coated galvanized steel, stainless steel, silicon bronze, or copper. (CRC R317.3.1)

   Exception: 1/2-inch diameter or greater steel bolts

   Exception: Fasteners other than nails and timber rivets may be of mechanically deposited zinc-coated steel with coating weights in accordance with ASTM B 695, Class 55 minimum

   Exception: Plain carbon steel fasteners acceptable in SBX/DOT and zinc borate preservative-treated wood in an interior, dry environment

9. **Fasteners for fire-retardant-treated wood.** Fasteners for fire-retardant-treated wood used in exterior applications or wet or damp locations shall be of hot dipped zinc-coated galvanized steel, stainless steel, silicon bronze, or copper. (CRC R317.3.3)

G. Roofing and Weatherproofing

1. **Roof covering.** All roof covering shall be installed per applicable requirements of CBC 1507. Roof coverings shall be at least Class A rated in accordance with ASTM E 108 or UL 790. (County Building Code 92.1.1505.1)
2. **Roof flashing.** Flashing shall be installed at wall and roof intersections, at gutters, wherever there is a change in roof slope or direction, and around roof openings. Where flashing is of metal, the metal shall be corrosion-resistant with a thickness of not less than 0.019 inch (No. 26 galvanized sheet). (CRC R903.2.1)
3. Crickets and saddles. A cricket or saddle shall be installed on the ridge side of any chimney or penetration more than 30 inches wide as measured perpendicular to the slope. Cricket or saddle covering shall be sheet metal or the same material as the roof covering. (CRC R903.2.2)

G. Roofing and Weatherproofing (Continued)

4. **Water-resistive barrier.** A minimum of one layer of No. 15 asphalt felt shall be attached to studs or sheathing of all exterior walls. Such felt or material shall be applied horizontally, with the upper layer lapped over the lower layer minimum 2 inches. Where joints occur, felt shall be lapped minimum 6 inches. The felt shall be continuous to the top of walls and terminated at penetrations and building appendages in a manner to maintain a weather-resistant exterior wall envelope. (CRC R703.2)
5. **Wall flashing.** Approved corrosion-resistant flashing shall be applied shingle fashion at the following locations to prevent entry of water into the wall cavity or penetration of water to the building structural framing components (CRC R703.8):

   a. Exterior door and window openings, extending to the surface of the exterior wall finish or to the water-resistive barrier for subsequent drainage

   b. At the intersection of chimneys or other masonry construction with frame or stucco walls, with projecting lips on both sides under stucco copings

   c. Under and at the ends of masonry, wood, or metal copings and sills

   d. Continuously above all projecting wood trim

   e. Where exterior porches, decks, or stairs attach to a wall or floor assembly of wood-frame construction

   f. At wall and roof intersections

   g. At built-in gutters

6. **Damp proofing.** Damp proofing materials for foundation walls enclosing usable space below grade shall be installed on the exterior surface of the wall and shall extend from the top of the footing to finished grade. (CRC R406.1)
7. **Weep screed.** A minimum 0.019-inch (No. 26 galvanized sheet gage), corrosion-resistant weep screed or plastic weep screed with a minimum vertical attachment flange of 3-1/2 inches shall be provided at or below the foundation plate line on exterior stud walls in accordance with ASTM C 92. The weep screed shall be placed a minimum 4 inches above the earth or 2 inches above paved areas and shall be of a type allowing trapped water to drain to the exterior of the building. (CRC R703.7.2.1)

H. Grading and soils

1. **Grading permit.** Grading permit required if volume of earth moved exceeds 200 cubic yards or if any cuts or fills exceed 8 feet in height/depth. (County Grading Ordinance 202)
2. **Compaction report.** Compaction report required for fill material 12 inches or more in depth. (CBC 1803.5.8)

I. Green Building Standards Code (CALGreen) Requirements

1. Applicability. CalGreen residential mandatory measures shall apply to every newly constructed building or structure and within any addition or alteration increasing a building's conditioned area, volume, or size. (CalGreen 101.3, CalGreen 301.1.1)

   Exception: All residential buildings undergoing permitted alterations, additions, or improvements shall replace noncompliant plumbing fixtures with water-conserving plumbing fixtures per CalGreen 301.1.1 and CalGreen 4.303.1

2. Water conserving plumbing fixtures and fittings. Plumbing fixtures and fittings shall comply with the following per CalGreen 4.303.1:

   a. Water closets: Maximum 1.28 gallons per flush

   b. Urinals: Maximum 0.5 gallons per flush

   c. Single showerheads: Maximum flow rate of 1.8 gallons per minute at 80 psi

   d. Multiple showerheads serving one shower: Maximum combined flow rate of 1.8 gallons per minute at 80 psi

   e. Lavatory faucets: Maximum flow rate of 1.2 gallons per minute at 60 psi, minimum flow rate of 0.8 gallons per minute at 20 psi

   f. Kitchen faucets: Maximum flow rate of 1.5 gallons per minute at 60 psi (County Green Building Code 97.1.4.303.1.4.4)

   Exception: Temporary increase allowed to maximum 2.2 gallons per minute at 60 psi if faucet defaults back to maximum 1.5 gallons per minute at 60 psi

   g. Appliances: At least one qualified ENERGY STAR dishwasher or clothes washer shall be installed in each dwelling unit. (County Green Building Code 97.1.4.303.3)

3. **Outdoor potable water uses in landscape areas.** Residential developments shall comply with local water efficient landscape ordinance or the current California Department of Water Resources Model Water Efficient Landscape Ordinance (MWELO), whichever is more stringent. (CalGreen 4.304.1)
4. **Joints and openings.** Openings in the building envelope separating conditioned space from unconditioned space needed to accommodate utility and other penetrations must be sealed in compliance with the California Energy Code. (CALGreen 4.406.1)

   Exception: Annular spaces around pipes, electric cables, conduits or other openings in plates at exterior walls shall be protected against the passage of rodents by closing such opening with cement mortar, concrete masonry or a similar method acceptable to the enforcing agency.

5. **Construction waste reduction, disposal, and recycling.** Recycle and/or salvage for reuse a minimum of 65 percent of the nonhazardous construction and demolition waste in accordance with either Section 4.408.2, 4.408.3, or 4.408.4, or meet a more stringent local construction and demolition waste management ordinance. (CalGreen 4.408.1)

   Exception: Excavated soil and land-clearing debris.

   Exception: Alternate waste reduction methods developed by working with local agencies if diversion or recycle facilities capable of compliance with this item do not exist or are not located reasonably close to the jobsite The County of San Diego, Department of Public Works, Construction & Demolition (C&D) Facilities Guide is online at:

   https://www.sandiegocounty.gov/content/dam/sdc/dpw/SOLID_WASTE_PLANNING_and_RECYCLING/UpdatedCDResources/CDFacility_QuickGuide.pdf

   Exception: The enforcing agency may make exceptions to the requirements of this section when isolated jobsites are located in areas beyond the haul boundaries of the diversion facility.

6. **Construction waste management plan.** A construction waste management plan in conformance with Items 1-5 shall be completed and available on the job site. The construction waste management plan shall be updated as necessary and shall be available during construction for examination by the enforcing agency. (CalGreen 4.408.2)

   a. Identify the construction and demolition waste materials to be diverted from disposal by recycling, reuse on the project or salvage for future use or sale.

   b. Specify if construction and demolition waste materials will be sorted on-site (source-separated) or bulk mixed (single stream).

   c. Identify diversion facilities where the construction and demolition waste materials will be taken.

   d. Identify construction methods employed to reduce the amount of construction and demolition waste generated.

   e. Specify that the amount of construction and demolition waste materials diverted shall be calculated by weight or volume, but not by both.

7. **Waste management company.** Utilize a waste management company, approved by the enforcing agency, which can provide verifiable documentation that the percentage of construction and demolition waste material diverted from the landfill complies with Section 4.408.1. (CalGreen 4.408.3)

   Note: The owner or contractor may make the determination if the construction and demolition waste materials will be diverted by a waste company.

8. **Waste stream reduction alternative [LR].** Projects that generate a total combined weight of construction and demolition waste disposed of in landfills, which do not exceed 3.4 pounds per square foot of the building area shall meet the 65 percent construction waste reduction requirement in Section 4.408.1. (CalGreen 4.408.4)

   4.408.4.1 Waste stream reduction alternative. Projects that generate a total combined weight of construction and demolition waste disposed of in landfills, which do not exceed 2 pounds per square foot of the building area shall meet the 65 percent construction waste reduction requirement in Section 4.408.1.

9. **Documentation.** Documentation shall be provided to the enforcing agency which demonstrates compliance with Section 4.408.2, Items 1-5, Section 4.408.3, or Section 4.408.4.
10. **Operation and maintenance manual.** Prior to final inspection, a manual, compact disc, web-based reference, or other acceptable media which includes all of the following shall be placed in the building (CALGreen 4.410.1):

    a. Directions to owner or occupant that manual shall remain with the building throughout the life cycle of the structure.

    b. Operation and maintenance instructions for the following:

       I. Equipment and appliances, including water-saving devices and systems, HVAC system, photovoltaic systems, water-heating systems and other major appliances and equipment.

       II. Roof and yard drainage, including gutters and downspouts.

       III. Space conditioning systems, including condensers and air filters.

       IV. Landscape irrigation systems.

       V. Water reuse systems.

    c. Information from local utility, water, and waste recovery providers on methods to further reduce resource consumption, including recycle programs and locations.

    d. Public transportation and/or carpool options available in the area.

    e. Educational material on the positive impacts of an interior relative humidity between 30-60 percent and what methods an occupant may use to maintain the relative humidity level in that range.

    f. Information about water-conserving landscape and irrigation design and controllers which conserve water.

    g. Instructions for maintaining gutters and downspouts and the importance of diverting water at least 5 feet away from the foundation.

I. (CalGreen) Requirements (Continued)

    h. Information on required routine maintenance measures, including, but not limited to, caulking, painting, grading around the building, etc.

    i. Information about state solar energy and incentive programs available.

    j. A copy of all special inspection verifications required by the enforcing agency or code.

    k. Information from the Department of Forestry and Fire Protection on maintenance of defensible space around residential structures.

    l. Information and/or drawings identifying the location of grab bar reinforcements.

11. **Covering of duct openings and protection of mechanical equipment during construction.** At the time of rough installation or during storage on the construction site and until final startup of the heating and cooling equipment, all duct and other related air distribution component openings shall be covered with tape, plastic, Sheetmetal or other methods acceptable to the enforcing agency to reduce the amount of dust or debris which may collect in the system. (CALGreen 4.504.1)
12. **Adhesives, sealants, caulks, paints, and coatings pollutant control.** Adhesives (including carpet adhesives), sealants, caulks, paints, and coatings shall comply with VOC limits per CALGreen 4.504.2. Verification of compliance shall be provided at the request of the enforcing agency. (CALGreen 4.504.2.1)
13. **Carpet systems.** All carpet installed in the building interior shall meet the testing and product requirements of one of the following (CALGreen 4.504.3):

    a. Carpet and Rug Institute's Green Label Plus Program (all carpet cushions must meet the requirements of this program).

    b. California Department of Public Health Standard Practice for the testing of VOCs (Specification 01350).

    c. NSF/ANSI 140 at the Gold level.

    d. Scientific Certifications Systems Indoor Advantage™ Gold.

14. **Resilient flooring systems.** At least 80 percent of the floor area receiving resilient flooring shall comply with one of or more of the following (CALGreen 4.504.4):

    a. VOC emission limits defined in the Collaborative for High Performance Schools (CHPS) High Performance Products Database

    b. Products compliant with CHPS criteria certified under the Greenguard Children & Schools program.

    c. Certification under the Resilient Floor Covering Institute (RFCI) FloorScore program.

    d. Meet the currently adopted version of California Department of Public Health, “Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers," (also known as Specification 01350)

15. **Composite wood products.** Hardwood plywood, particleboard and medium density fiberboard composite wood products used on the interior or exterior of the building shall meet the requirements for formaldehyde as specified in ARB's Air Toxics Control Measure for Composite Wood (17 CCR 93120 et seq.) by or before the dates specified in those sections, as shown in CalGreen Table 4.504.5. The following limits are in parts per million (CALGreen 4.504.5):

a. Hardwood plywood veneer core	0.05
b. Hardwood plywood composite core	0.05
c. Particle board	0.09
d. Medium-density fiberboard (MDF)	0.11
e. Thin MDF (5/16 inch or less)	0.13

16. **Moisture content of building materials.** Building materials with visible signs of water damage shall not be installed. Wall and floor framing shall not be enclosed when the framing members exceed 19 percent moisture content. Moisture content shall be verified in compliance with the following (CALGreen 4.505.3):

    a. Moisture content shall be determined with either a probe-type or contact-type moisture meter.

    b. Moisture readings shall be taken at a point 2 feet to 4 feet from the grade stamped end of each piece to be verified.

    c. At least three random moisture readings shall be performed on wall and floor framing with documentation acceptable to the enforcing agency provided at the time of approval to enclose the wall and floor framing.

    Insulation products which are visibly wet or have high moisture content shall be replaced or allowed to dry prior to enclosure in wall or floor cavities. Wet-applied insulation products shall follow the manufacturers' drying recommendations prior to enclosure.

17. **Bathrooms with a bathtub and/or shower** shall be mechanically ventilated per the following (CalGreen 4.506.1):

    a. Fans shall be ENERGY STAR compliant and ducted to terminate outside building

    b. Unless functioning as a component of a whole-house ventilation system, fans shall have humidity controls capable of adjustment - manually or automatically -- between a relative humidity range of 50% to 80%.

18. **Heating and air-conditioning system design.** Heating and air-conditioning systems shall be sized, designed, and have their equipment selected using the following methods (CALGreen 4.507.2):

    a. The heat loss and heat gain are established according to the currently adopted version of ANSI/ACCA 2 Manual J, ASHRAE handbooks, or other equivalent design software or methods.

    b. Duct systems are sized according to the currently adopted version of ANSI/ACCA 1 Manual D, ASHRAE handbooks, or other equivalent design software or methods.

    c. Select heating and cooling equipment according to the currently adopted version of ACCA 36-S Manual S or other equivalent design software or methods.

TABLE R602.3(1)
FASTENER SCHEDULE FOR STRUCTURAL MEMBERS

TABLE R602.3(1)
FASTENING SCHEDULE

ITEM	DESCRIPTION OF BUILDING ELEMENTS	NUMBER AND TYPE OF FASTENER[a, b, c]	SPACING AND LOCATION
Roof			
1	Blocking between ceiling joists, rafters or trusses to top plate or other framing below	4-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	Toe nail
	Blocking between rafters or truss not at the wall top plates, to rafter or truss	2-8d common ($2\frac{1}{2}$″ × 0.131″); or 2-3″ × 0.131″ nails	Each end toe nail
		2-16d common ($3\frac{1}{2}$″ × 0.162″); or 3-3″ × 0.131″ nails	End nail
	Flat blocking to truss and web filler	16d common ($3\frac{1}{2}$″ × 0.162″); or 3″ × 0.131″ nails	6″ o.c. face nail
2	Ceiling joists to top plate	4-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	Per joist, toe nail
3	Ceiling joist not attached to parallel rafter, laps over partitions [see Section R802.5.2 and Table R802.5.2(1)]	4-10d box (3″ × 0.128″); or 3-16d common ($3\frac{1}{2}$″ × 0.162″); or 4-3″ × 0.131″ nails	Face nail
4	Ceiling joist attached to parallel rafter (heel joint) [see Section R802.5.2 and Table R802.5.2(1)]	Table R802.5.2(1)	Face nail
5	Collar tie to rafter, face nail	4-10d box (3″ × 0.128″); or 3-10d common (3″ × 0.148″); or 4-3″ × 0.131″ nails	Face nail each rafter
6	Rafter or roof truss to plate	3-16d box ($3\frac{1}{2}$″ × 0.135″); or 3-10d common (3″ × 0.148″); or 4-10d box (3″ × 0.128″); or 4-3″ × 0.131″ nails	2 toe nails on one side and 1 toe nail on opposite side of each rafter or truss[i]
7	Roof rafters to ridge, valley or hip rafters or roof rafter to minimum 2″ ridge beam	4-16d box ($3\frac{1}{2}$″ × 0.135″); or 3-10d common (3″ × 0.148″); or 4-10d box (3″ × 0.128″); or 4-3″ × 0.131″ nails	Toe nail
		3-16d box ($3\frac{1}{2}$″ × 0.135″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	End nail
Wall			
8	Stud to stud (not at braced wall panels)	16d common ($3\frac{1}{2}$″ × 0.162″)	24″ o.c. face nail
		10d box (3″ × 0.128″); or 3″ × 0.131″ nails	16″ o.c. face nail
9	Stud to stud and abutting studs at intersecting wall corners (at braced wall panels)	16d box ($3\frac{1}{2}$″ × 0.135″); or 3″ × 0.131″ nails	12″ o.c. face nail
		16d common ($3\frac{1}{2}$″ × 0.162″)	16″ o.c. face nail
10	Built-up header (2″ to 2″ header with $\frac{1}{2}$″ spacer)	16d common ($3\frac{1}{2}$″ × 0.162″)	16″ o.c. each edge face nail
		16d box ($3\frac{1}{2}$″ × 0.135″)	12″ o.c. each edge face nail
11	Continuous header to stud $3\frac{1}{2}$″	5-8d box ($2\frac{1}{2}$″ × 0.113″); or 4-8d common ($2\frac{1}{2}$″ × 0.131″); or 4-10d box (3″ × 0.128″)	Toe nail
12	Adjacent full-height stud to end of header	4-16d box ($3\frac{1}{2}$″× 0.135″); or 3-16d common ($3\frac{1}{2}$″ × 0.162″); or 4-10d box (3″ × 0.128″); or 4-3″ × 0.131″ nails	End nail
13	Top plate to top plate	16d common ($3\frac{1}{2}$″ × 0.162″)	16″ o.c. face nail
		10d box (3″ × 0.128″); or 3″ × 0.131″ nails	12″ o.c. face nail
14	Double top plate splice	8-16d common ($3\frac{1}{2}$″ × 0.162″); or 12-16d box ($3\frac{1}{2}$″ × 0.135″); or 12-10d box (3″ × 0.128″); or 12-3″ × 0.131″ nails	Face nail on each side of end joint (minimum 24″ lap splice length each side of end joint)
15	Bottom plate to joist, rim joist, band joist or blocking (not at braced wall panels)	16d common ($3\frac{1}{2}$″ × 0.162″)	16″ o.c. face nail
		16d box ($3\frac{1}{2}$″ × 0.135″); or 3″ × 0.131″ nails	12″ o.c. face nail
Roof			
16	Bottom plate to joist, rim joist, band joist or blocking (at braced wall panel)	3-16d box ($3\frac{1}{2}$″ × 0.135″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″); or 4-3″ × 0.131″ nails	16″ o.c. face nail
17	Top or bottom plate to stud	4-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-16d box ($3\frac{1}{2}$″ × 0.135″); or 4-8d common ($2\frac{1}{2}$″ × 0.131″); or 4-10d box (3″ × 0.128″); or 4-3″ × 0.131″ nails	Toe nail
		3-16d box ($3\frac{1}{2}$″ × 0.135″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	End nail
18	Top plates, laps at corners and intersections	3-10d box (3″ × 0.128″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″); or 3-3″ × 0.131″ nails	Face nail
19	1″ brace to each stud and plate	3-8d box ($2\frac{1}{2}$″ × 0.113″); or 2-8d common ($2\frac{1}{2}$″ × 0.131″); or 2-10d box (3″ × 0.128″); or 2 staples $1\frac{3}{4}$″	Face nail
20	1″ × 6″ sheathing to each bearing	3-8d box ($2\frac{1}{2}$″ × 0.113″); or 2-8d common ($2\frac{1}{2}$″ × 0.131″); or 2-10d box (3″ × 0.128″); or 2 staples, 1″ crown, 16 ga., $1\frac{3}{4}$″ long	Face nail
21	1″ × 8″ and wider sheathing to each bearing	3-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 3 staples, 1″ crown, 16 ga., $1\frac{3}{4}$″ long	Face nail
		Wider than 1″ × 8″ 4-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 4 staples, 1″ crown, 16 ga., $1\frac{3}{4}$″ long	
Floor			
22	Joist to sill, top plate or girder	4-8d box ($2\frac{1}{2}$″ × 0.113″); or 3-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	Toe nail
23	Rim joist, band joist or blocking to sill or top plate (roof applications also)	8d box ($2\frac{1}{2}$″ × 0.113″)	4″ o.c. toe nail
		8d common ($2\frac{1}{2}$″ × 0.131″); or 10d box (3″ × 0.128″); or 3″ × 0.131″ nails	6″ o.c. toe nail
24	1″ × 6″ subfloor or less to each joist	3-8d box ($2\frac{1}{2}$″ × 0.113″); or 2-8d common ($2\frac{1}{2}$″ × 0.131″); or 3-10d box (3″ × 0.128″); or 2 staples, 1″ crown, 16 ga., $1\frac{3}{4}$″ long	Face nail
25	2″ subfloor to joist or girder	3-16d box ($3\frac{1}{2}$″ × 0.135″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″)	Blind and face nail
26	2″ planks (plank & beam—floor & roof)	3-16d box ($3\frac{1}{2}$″ × 0.135″); or 2-16d common ($3\frac{1}{2}$″ × 0.162″)	At each bearing, face nail
27	Band or rim joist to joist	3-16d common ($3\frac{1}{2}$″ × 0.162″); or 4-10 box (3″ × 0.128″); or 4-3″ × 0.131″ nails; or 4-3″ × 14 ga. staples, $\frac{7}{16}$″ crown	End nail
28	Built-up girders and beams, 2-inch lumber layers	20d common (4″ × 0.192″); or	Nail each layer as follows: 32″ o.c. at top and bottom and staggered.
		10d box (3″ × 0.128″); or 3″ × 0.131″ nails	24″ o.c. face nail at top and bottom staggered on opposite sides
		And: 2-20d common (4″ × 0.192″); or 3-10d box (3″ × 0.128″); or 3-3″ × 0.131″ nails	Face nail at ends and at each splice
29	Ledger strip supporting joists or rafters	4-16d box ($3\frac{1}{2}$″ × 0.135″); or 3-16d common ($3\frac{1}{2}$″ × 0.162″); or 4-10d box (3″ × 0.128″); or 4-3″ × 0.131″ nails	At each joist or rafter, face nail
30	Bridging or blocking to joist, rafter or truss	2-10d box (3″ × 0.128″); or 2-8d common ($2\frac{1}{2}$″ × 0.131″); or 2-3″ × 0.131″ nails	Each end, toe nail

ITEM	DESCRIPTION OF BUILDING ELEMENTS	NUMBER AND TYPE OF FASTENER[a, b, c]	SPACING OF FASTENERS	
			Edges[h] (inches)	Intermediate supports[c, e] (inches)
Wood structural panels, subfloor, roof and interior wall sheathing to framing and particleboard wall sheathing to framing [see Table R602.3(3) for wood structural panel exterior wall sheathing to wall framing]				
31	$\frac{3}{8}$″ – $\frac{1}{2}$″	6d common or deformed (2″ × 0.113″× 0.266″ head); or $2\frac{3}{8}$″ × 0.113″ × 0.266″ head nail (subfloor, wall)[i]	6	6[f]
		8d common ($2\frac{1}{2}$″ × 0.131″) nail (roof); or RSRS-01 ($2\frac{3}{8}$″ × 0.113″) nail (roof)[b]	6	6[f]
32	$\frac{19}{32}$″ – $\frac{3}{4}$″	8d common (2-$2\frac{1}{2}$″ × 0.131″) nail (subfloor, wall)	6	12
		8d common (21/2″ × 0.131″) nail (roof); or RSRS-01; ($2\frac{3}{8}$″ × 0.113″) nail (roof)[b]	6	6[f]
		Deformed $2\frac{3}{8}$″ × 0.113″ × 0.266″ head (wall or subfloor)	6	12
33	$\frac{7}{8}$″ – $1\frac{1}{4}$″	10d common (3″ × 0.148″) nail; or ($2\frac{1}{2}$″ × 0.131 × 0.281″ head) deformed nail	6	12
Other wall sheathing[g]				
34	$\frac{1}{2}$″ structural cellulosic fiberboard sheathing	$1\frac{1}{2}$″ × 0.120″ galvanized roofing nail,$\frac{7}{16}$″ head diameter; or $1\frac{1}{4}$″ long 16 ga. staple with $\frac{7}{16}$″ or 1″ crown	3	6
35	$\frac{25}{32}$″ structural cellulosic fiberboard sheathing	$1\frac{3}{4}$″ × 0.120″ galvanized roofing nail, $\frac{7}{16}$″ head diameter; or $1\frac{1}{4}$″ long 16 ga. staple with $\frac{7}{16}$″ or 1″ crown	3	6
36	$\frac{1}{2}$″ gypsum sheathing[d]	$1\frac{1}{2}$″ × 0.120″ galvanized roofing nail, $\frac{7}{16}$″ head diameter, or $1\frac{1}{4}$″long 16 ga.; staple galvanized, $1\frac{1}{2}$″ long; $\frac{7}{16}$″ or 1″ crown or $1\frac{1}{4}$″ screws, Type W or S	7	7
37	$\frac{5}{8}$″ gypsum sheathing[d]	$1\frac{3}{4}$″ × 0.120″ galvanized roofing nail, $\frac{7}{16}$″ head diameter, or $1\frac{1}{4}$″ long 16 ga.; staple galvanized, $1\frac{1}{2}$″ long; $\frac{7}{16}$″ or 1″crown or $1\frac{1}{4}$″ screws, Type W or S	7	7
Wood structural panels, combination subfloor underlayment to framing				
38	$\frac{3}{4}$″ and less	Deformed (2″× 0.113″) or Deformed (2″ × 0.120″) nail; or 8d common ($2\frac{1}{2}$″ × 0.131″) nail	6	12
39	$\frac{7}{8}$″ – 1″	8d common ($2\frac{1}{2}$″ × 0.131″) nail; or Deformed (2″ × 0.113″); or Deformed ($2\frac{1}{2}$″ × 0.120″) nail	6	12
40	$1\frac{1}{8}$″ – $1\frac{1}{4}$″	10d common (3″ × 0.148″) nail; or Deformed (2″ × 0.113″);or Deformed ($2\frac{1}{2}$″ × 0.120″) nail	6	12

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s; 1 ksi = 6.895 MPa.

a. Nails are smooth-common, box or deformed shanks except where otherwise stated. Nails used for framing and sheathing connections are carbon steel and shall have minimum average bending yield strengths as shown: 80 ksi for shank diameter of 0.192 inch (20d common nail), 90 ksi for shank diameters larger than 0.142 inch but not larger than 0.177 inch, and 100 ksi for shank diameters of 0.142 inch or less. Connections using nails and staples of other materials, such as stainless steel, shall be designed by accepted engineering practice or approved under Section R104.11.

b. RSRS-01 is a Roof Sheathing Ring Shank nail meeting the specifications in ASTM F1667.

c. Nails shall be spaced at not more than 6 inches on center at all supports where spans are 48 inches or greater.

d. Four-foot by 8-foot or 4-foot by 9-foot panels shall be applied vertically.

e. Spacing of fasteners not included in this table shall be based on Table R602.3(2).

f. For wood structural panel roof sheathing attached to gable end roof framing and to intermediate supports within 48 inches of roof edges and ridges, nails shall be spaced at 4 inches on center where the ultimate design wind speed is greater than 130 mph in Exposure B or greater than 110 mph in Exposure C.

g. Gypsum sheathing shall conform to ASTM C1396 and shall be installed in accordance with ASTM C1280 or GA 253. Fiberboard sheathing shall conform to ASTM C208.

h. Spacing of fasteners on floor sheathing panel edges applies to panel edges supported by framing members and required blocking and at floor perimeters only. Spacing of fasteners on roof sheathing panel edges applies to panel edges supported by framing members and required blocking. Blocking of roof or floor sheathing panel edges perpendicular to the framing members need not be provided except as required by other provisions of this code. Floor perimeter shall be supported by framing members or solid blocking.

i. Where a rafter is fastened to an adjacent parallel ceiling joist in accordance with this schedule, provide two toe nails on one side of the rafter and toe nails from the ceiling joist to top plate in accordance with this schedule. The toe nail on the opposite side of the rafter shall not be required.

THERE ARE MINIMUM REQUIREMENTS AND SHALL NOT SUPERSEDE MORE RESTRICTIVE SPECIFICATIONS ON THE PLANS OR AS REQUIRED BY APPLICABLE CODE

BUILDING RECORD ID:

OWNER OR CONTRACTOR SIGNATURE:

County of San Diego, Planning & Development Services
MINIMUM CONSTRUCTION SPECIFICATIONS
BUILDING DIVISION

Sheet Number

CS-1

2025 Single-Family Residential Mandatory Requirements Summary

NOTE: Single-family residential buildings subject to the Energy Code must comply with all applicable mandatory measures, regardless of the compliance approach used. Review the respective sections for more information.

Space Conditioning, Water Heating, and Plumbing System:

§ 110.0-110.3:	**Certification.** Heating, ventilation, and air conditioning (HVAC) equipment, water heaters, and all other regulated appliances must be certified by the manufacturer to the California Energy Commission. *
§ 110.2(a):	**HVAC Efficiency.** Equipment must meet the applicable efficiency requirements in Tables 110.2-A through -L. *
§ 110.2(b):	**Controls for Heat Pumps with Supplementary Heaters.** Heat pumps with supplementary heaters must have control requirements as specified in § 150.0(h)7 and § 150.0(i)2. *
§ 110.2(c):	**Thermostats.** All heating or cooling systems not controlled by a central energy management control system (EMCS) must have a setback thermostat. *
§ 110.3(c)3:	**Insulation.** Unfired service water heater storage tanks and solar water-heating backup tanks must have adequate insulation, or tank surface heat loss rating.
§ 110.3(c)6:	**Isolation Valves.** Instantaneous water heaters with an input rating greater than 6.8 kBtu per hour (2 kW) must have isolation valves with hose bibbs or other fittings on both cold and hot water lines to allow for flushing the water heater when the valves are closed.
§110.3(c)7:	**Backup Heat and Ventilation.** Air-source heat pump water heaters must have backup heat (internal or external) if inlet air is unconditioned, unless compressor cut-off temperature is below Heating Winter Median of Extremes from Reference Appendix JA2. Heat pump water heater installation space and/or communicating space(s) must meet minimum volume, ducting, and/or grille net free area by kBtu/h of compressor capacity.
§ 110.5:	**Pilot Lights.** Continuously burning pilot lights are prohibited for natural gas: fan-type central furnaces; household cooking appliances (except appliances without electrical supply voltage connection with pilot lights that consume less than 150 Btu/h); pool and spa heaters. *
§ 150.0(h)1:	**Building Cooling and Heating Loads.** Heating and/or cooling loads are calculated in accordance with the ASHRAE Handbook, Equipment Volume, Applications Volume, and Fundamentals Volume; the SMACNA Residential Comfort System Installation Standards Manual; or the ACCA Manual J using design conditions specified in § 150.0(h)2. *
§ 150.0(h)3A:	**Clearances.** Air conditioner and heat pump outdoor condensing units must have a clearance of at least five feet from any dryer outlet or vent.
§ 150.0(h)3B:	**Liquid Line Drier.** Air conditioners and heat pump systems must be equipped with liquid line filter driers if required, as specified by the manufacturer's instructions.
§ 150.0(h)5:	**System Selection.** Equipment sizing and selection must meet the cooling and heating loads of § 150.0(h)1 & 2, and systems must be sized per ACCA Manual S-2023 with no minimum cooling capacity. Furnace heating capacity must meet ACCA Manual S. Heat pump heating capacity must meet minimum CBC requirements without including supplementary heat with no limit on maximum heating capacity.
§ 150.0(h)6:	**Defrost.** Installer-adjustable defrost delay timers must be set to greater than or equal to 90 minutes. CF2R certification required. *
§ 150.0(h)7:	**Supplementary Heating Control Configuration:** Heat pumps with supplementary heating must have controls to lock supplementary heating above outside air temperature no greater than 35°F, allowed during defrost or emergency operation. CF2R certification required. *
§ 150.0(h)8:	**Sizing of Electric Resistance Supplementary Heat:** When heat pumps have electric resistance heat, the capacity of electric resistance heat must not exceed the heat pump nominal cooling capacity (at 95°F ambient conditions) multiplied by 2.7 kW per ton, rounded up to the closest kW.
§ 150.0(h)9:	**Capacity Variation with Third-party Thermostats:** For variable or multi-speed systems, the space conditioning system and thermostat must be capable of responding to heating and cooling loads by modulating system compressor speed and must meet thermostat requirements of § 150.0(i)2. CF2R certification required.
§ 150.0(i):	**Thermostat.** All heating or cooling systems including heat pumps which are not controlled by energy management control system (EMCS) must have setback thermostat. Additional requirements for thermostats that control heat pumps with supplemental heating include thermostat must display outdoor air temperature, must lock out supplementary heat when outdoor air temperature is above 35°F, and must notify when supplemental heat is in use.*
§ 150.0(j)1:	**Water Piping, Solar Water-heating System Piping, and Space Conditioning System Line Insulation.** All domestic hot water piping must be insulated as specified in § 609.12 of the California Plumbing Code. *
§ 150.0(j)2:	**Insulation Protection.** Piping insulation must be protected from damage, including from sunlight, moisture, equipment maintenance, and wind, as required by §120.3(b). Insulation exposed to weather must be water retardant and protected from UV light (no adhesive tapes). Insulation covering chilled water piping and refrigerant suction piping located outside the conditioned space must include, or be protected by, a Class I or II vapor retarder. Pipe insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve.
§ 150.0(n)1:	**Gas or Propane Water Heating Systems.** Systems using gas or propane water heaters to serve individual dwelling units must designate a space at least 2.5' x 2.5' x 7' suitable for the future installation of a heat pump water heater, and meet electrical and plumbing requirements, based on the distance between this designated space and the water heater location; and a condensate drain no more than 2" higher than the base of the water heater
§ 150.0(n)2:	**Solar Water Heating Systems.** Solar water heating systems and collectors must be certified and rated by the Solar Rating and Certification Corporation (SRCC), the International Association of Plumbing and Mechanical Officials, Research and Testing (IAPMO R&T), or by a listing agency that is approved by the Executive Director.

Ducts and Fans

§110.8(d)3:	**Ducts.** Insulation installed on an existing space-conditioning duct must comply with § 604.0 of the California Mechanical Code (CMC). If a contractor installs the insulation, the contractor must certify to the customer, in writing, that the insulation meets this requirement.
§ 150.0(m)1:	**CMC Compliance.** All air-distribution system ducts and plenums must meet CMC §§ 601.0-605.0 and ANSI/SMACNA-006-2006 HVAC Duct Construction Standards Metal and Flexible 3rd Edition. Portions of supply- and return-air ducts and plenums must be insulated to R-6.0 or higher. Ducts located entirely in conditioned space as confirmed via field verification and diagnostic testing (RA3.1.4.3.8) do not require insulation; in dwelling units with attics, the ducts must be below the ceiling separating occupiable space from the attic. Connections of metal ducts and inner core of flexible ducts must be mechanically fastened. Openings must be sealed with mastic, tape, or other duct-closure system that meets applicable UL requirements; or aerosol sealant that meets UL 723. The combination of mastic and either mesh or tape must be used to seal openings greater than ¼", if mastic or tape is used. Building cavities, air handler support platforms, and plenums designed or constructed with materials other than sealed sheet metal, duct board or flexible duct must not be used to convey conditioned air. Building cavities and support platforms may contain ducts; ducts installed in these spaces must not be compressed. *
§ 150.0(m)2:	**Factory-Fabricated Duct Systems** must comply with applicable requirements of UL 181 for duct construction, connections, and closures; joints and seams of duct systems and their components must not be sealed with cloth back rubber adhesive duct tapes unless such tape is used in combination with mastic and draw bands.
§ 150.0(m)3:	**Field-Fabricated Duct Systems** must comply with applicable requirements for: pressure-sensitive tapes, mastics, sealants, and other requirements specified for duct construction.
§ 150.0(m)7:	**Backdraft Damper.** Fan systems that exchange air between conditioned space and outdoors must have backdraft or automatic dampers.
§ 150.0(m)8:	**Gravity Ventilation Dampers.** Gravity ventilating systems serving conditioned space must have either automatic or readily accessible, manually operated dampers in all openings to the outside, except combustion inlet and outlet air openings and elevator shaft vents.
§ 150.0(m)9:	**Protection of Insulation.** Insulation must be protected from damage due to sunlight, moisture, equipment maintenance, and wind. Insulation exposed to weather must be suitable for outdoor service (e.g., protected by aluminum, sheet metal, painted canvas, or plastic cover). Cellular foam insulation must be protected as above or painted with a water retardant and solar radiation-resistant coating.
§ 150.0(m)10:	**Porous Inner Core Flex Duct.** Flex ducts with porous inner cores must have a non-porous layer or air barrier between the inner core and outer vapor barrier.
§ 150.0(m)11:	**Duct System Sealing and Leakage Test.** When space conditioning systems use forced air duct systems to supply conditioned air to an occupiable space, the ducts must be sealed and duct leakage tested, as confirmed through field verification and diagnostic testing, in accordance with Reference Residential Appendix RA3.1.
§ 150.0(m)12:	**Air Filtration.** Space conditioning systems with ducts exceeding 10 feet and the supply side of ventilation systems must have MERV 13 or equivalent filters. Filters for space conditioning systems must have a two inch depth, or can be one inch if sized per Equation 150.0-A. Clean-filter pressure drop and labeling must meet the requirements in §150.0(m)12. Filters must be accessible for regular service. Filter racks or grilles must use gaskets, sealing, or other means to close gaps around the inserted filters and prevents air from bypassing the filter. *
§ 150.0(m)13:	**Space Conditioning System Airflow Rate and Fan Efficacy.** Space conditioning systems that use ducts to supply cooling must have a hole for the placement of a static pressure probe, or a permanently installed static pressure probe in the supply plenum. Airflow must be ≥ 350 CFM per ton of nominal cooling capacity, and an air-handling unit fan efficacy ≤ 0.45 watts per CFM for gas furnace air handlers and ≤ 0.58 watts per CFM for all others. Small duct high velocity systems must provide an airflow ≥ 250 CFM per ton of nominal cooling capacity, and an air-handling unit fan efficacy ≤ 0.62 watts per CFM. Field verification testing is required in accordance with Reference Residential Appendix RA3.3. *

Ventilation and Indoor Air Quality:

§ 150.0(o)1:	**Requirements for Ventilation and Indoor Air Quality**. All dwelling units shall meet the requirements of ASHRAE Standard 62.2, Ventilation and Acceptable Indoor Air Quality in Residential Buildings subject to the amendments specified in §150.0(o)1. *
§ 150.0(o)1B:	**Central Fan Integrated (CFI) Ventilation Systems.** Continuous operation of CFI air handlers used in CFI systems is not a permissible method to provide whole-dwelling unit ventilation airflow required per §150.0(o)1C. A motorized damper(s) shall be installed on the ventilation duct(s) that prevents all airflow through the space conditioning duct system when the damper(s) is closed and controlled per §150.0(o)1Biii&iv. CFI ventilation systems must have controls that track outdoor air ventilation run time, and either open or close the motorized damper(s) for compliance with §150.0(o)1C.
§ 150.0(o)1C:	**Whole-Dwelling Unit Mechanical Ventilation for Single-Family dwellings.** Single-family dwellings shall have mechanical ventilation specified in §150.0(o)1Ci and 150.0(o)1Civ.Single family detached dwelling units and attached dwelling units not sharing ceilings or floors with other dwelling units, occupiable spaces, public garages, or commercial spaces may reduce their required mechanical ventilation rates in accordance with §150.0(o)1Cii-iii.* Vertically attached dwelling units shall not reduce their minimum ventilation rates in accordance with §150.0(o)1Cii-iii.
§ 150.0(o)1G:	**Local Mechanical Exhaust.** Kitchens and bathrooms shall have local mechanical exhaust; nonenclosed kitchens shall have demand-controlled exhaust system meeting requirements of §150.0(o)1Giii, enclosed kitchens and bathrooms shall either use demand-controlled as specified in §150.0(o)1Giii or continuous exhaust as specified in §150.0(o)1Giv. Airflow shall be measured by the installer per §150.0(o)1Gv, and rated for sound per §150.0(o)1Gvi. *
§ 150.0(o)1H-J:	**Airflow Measurement, Sound Ratings, Label of Whole-Dwelling Unit Ventilation Systems.** The airflow required per §150.0(o)1C shall be measured by using a flow hood, flow grid, or other airflow measuring device at the fan's inlet or outlet terminals/grilles per Reference Residential Appendix RA3.7. Whole-Dwelling unit ventilation systems shall be rated for sound per ASHRAE 62.2 §7.3 at no less than the minimum airflow rate required by §150.0(o)1C. Manual on-off switches must be labeled per §150.0(o)1J.
§ 150.0(o)2:	**Field Verification and Diagnostic Testing.** Whole-Dwelling Unit ventilation airflow, vented range hood airflow and sound rating, and HRV and ERV fan efficacy shall be verified in accordance with Reference Residential Appendix RA3.7. Vented range hoods shall be verified per Reference Residential Appendix RA3.7.4.3 to confirm if it is rated by HVI or AHAM to comply with the airflow rates and sound requirements per §150.0(o)1G

Electric and Battery Energy Storage Ready:

§ 150.0(s):	**Battery Energy Storage System (BESS) Ready.** All single-family residences that include one or two dwelling units, which a load serving entity provides with electrical service for the dwelling unit that is greater than 125 amps, shall meet at least one of the following: Either BESS-ready interconnection equipment with backed up capacity of 60 amps or more and four or more BESS supplied branch circuits in §150.0(s)2, *or* a dedicated raceway from the main service to a subpanel that supplies the branch circuits in § 150.0(s)2; at least four branch circuits must be identified and have their source collocated at a single panelboard suitable to be supplied by the BESS, with one circuit supplying the refrigerator, one lighting circuit near the primary exit, and one circuit supplying a sleeping room receptacle outlet; main panelboard must have a minimum busbar rating of 225 amps; sufficient space must be reserved to allow future installation of a system isolation equipment/transfer switch within 3 feet of the main panelboard, with raceways installed between the panelboard and the switch location to allow the connection of backup power source. *
§ 150.0(t):	**Heat Pump Space Heater Ready.** Systems using gas or propane furnaces to serve individual dwelling units must include: A dedicated unobstructed 240V branch circuit wiring installed within 3' of the furnace with circuit conductors rated at least 30 amps with the blank cover identified as "240V ready;" and a reserved main electrical service panel space to allow for the installation of a double pole circuit breaker permanently marked as "For Future 240V use."
§ 150.0(u):	**Electric Cooktop Ready.** Systems using gas or propane cooktop to serve individual dwelling units must include: A dedicated unobstructed 240V branch circuit wiring installed within 3' of the cooktop with circuit conductors rated at least 50 amps with the blank cover identified as "240V ready;" and a reserved main electrical service panel space to allow for the installation of a double pole circuit breaker permanently marked as "For Future 240V use."
§ 150.0(v):	**Electric Clothes Dryer Ready.** Clothes dryer locations with gas or propane plumbing to serve individual dwelling units must include: A dedicated unobstructed 240V branch circuit wiring installed within 3' of the dryer location with circuit conductors rated at least 30 amps with the blank cover identified as "240V ready;" and a reserved main electrical service panel space to allow for the installation of a double pole circuit breaker permanently marked as "For Future 240V use."

Lighting:

§ 110.9:	**Lighting Controls and Components.** All lighting control devices and systems, light sources, and luminaires must meet the applicable requirements of §110.9. *
§ 150.0(k)1A:	**Luminaire Efficacy.** All installed luminaires and light sources must meet the requirements of Reference Joint Appendix JA8, except lighting integral to exhaust fans, kitchen range hoods, bath vanity mirrors, and garage door openers; navigation lighting less than 5 watts; lighting internal to drawers, cabinets, and linen closets with an efficacy of at least 45 lumens per watt; and ceiling fan kits subject to DOE's Appliance and Equipment Standards. *
§ 150.0(k)1C:	**Recessed Downlight Luminaires in Ceilings.** Luminaires recessed into ceilings must not contain screw-based sockets, be airtight with air leakage rate < 2.0 cfm at 75 Pascals, and leak paths be sealed with a gasket or caulk. California Electrical Code Article 410.116 must also be met. *
§ 150.0(k)1D:	**Light Sources in Enclosed or Recessed Luminaires.** Lamps and other separable light sources installed in enclosed or recessed luminaires must meet the JA8 elevated temperature requirements, including marking requirements.
§ 150.0(k)1E:	**Blank Electrical Boxes.** The number of electrical boxes that are more than five feet above the finished floor and do not contain a luminaire or other device shall be no more than the number of bedrooms. These boxes must be served by a dimmer, vacancy sensor control, low voltage wiring, or fan speed control.
§ 150.0(k)2A:	**Indoor Lighting Controls.** Lighting must have readily accessible wall-mounted controls that allow the lighting to be manually turned on and off.
§ 150.0(k)2D:	**Controls Permitted.** An energy management control system (EMCS) may be used to comply with dimming, occupancy, and control requirements if it provides the functionality of the specified control per §110.9 and the physical controls specified in §150.0(k)2A. Controls must not bypass a dimmer, occupant sensor, or vacancy sensor function installed to comply with §150.0(k)2.
§ 150.0(k)2E:	**Automatic Shutoff Controls.** In bathrooms, garages, laundry rooms, utility rooms and walk-in closets, at least one installed luminaire must be controlled by an occupancy or vacancy sensor providing automatic-off functionality. Lighting inside drawers and cabinets with opaque fronts or doors must have controls that turn the light off when the drawer or door is closed.
§ 150.0(k)2F:	**Dimmers.** Lighting in habitable spaces (e.g., living rooms, dining rooms, kitchens, bedrooms) must have readily accessible wall-mounted dimming controls that allow the lighting to be manually adjusted up and down, except lighting integral to kitchen range hoods and bathroom exhaust fans, navigation lighting less than 5 watts, lighting internal to drawers, cabinetry with opaque fronts or doors and controlled by automatic-off controls, luminaires in a circuit with controlled lighting power less than 20 watts or controlled by an occupancy or vacancy sensor. Forward phase cut dimmers controlling LED light sources in these spaces must comply with NEMA SSL 7A. Lighting integrated with ceiling fan may be controlled via a remote control.
§ 150.0(k)2G:	**Independent Controls.** Lighting integral to exhaust fans shall be controlled independently from the fans. Lighting under cabinets or shelves, lighting in display cabinets, and switched outlets must be controlled separately from ceiling-installed lighting.
§ 150.0(k)3:	**Residential Outdoor Lighting.** Outdoor lighting permanently mounted to a single-family residential building or other buildings on the same lot must meet luminaire efficacy requirements of §150.0(k)1A, have a manual on/off switch and either a photocell and motion sensor, a photocell and automatic time switch, or an astronomical time clock. Controls that override to ON are not allowed, unless that override automatically returns the automatic control to its normal operation within 6 hours. An EMCS or other controls that provides the specified control functionality may be used but must meet all applicable requirements.
§ 150.0(k)4:	**Internally Illuminated Address Signs** must either comply with §140.8 or consume no more than 5 watts of power.
§ 150.0(k)5:	**Residential Garages for Eight or More Vehicles.** Lighting for residential parking garages for eight or more vehicles must comply with the applicable requirements for nonresidential garages in §§ 110.9, 130.0, 130.1, 130.4, 140.6, and 141.0.

Building Envelope:

§ 110.6(a)1:	**Air Leakage.** Manufactured fenestration, exterior doors, and exterior pet doors must limit air leakage to 0.3 CFM per square foot or less when tested per NFRC-400, ASTM E283, or AAMA/WDMA/CSA 101/I.S.2/A440-2011. *
§110.6(a)2-4:	**U-factor, Solar heat gain coefficient, Visible transmittance.** The fenestration product and exterior door's U-factor and solar heat gain coefficient (SHGC) shall be rated in accordance with NFRC 100 and NFRC 200, or use the applicable defaults in Table 110.6-A and TABLE 110.6-B. The visible transmittance (VT) shall be rated in accordance with NFRC 200 or ASTM E972, for tubular daylighting devices VT shall be rated using NFRC 203. *
§ 110.6(a)5:	**Labeling.** Fenestration products and exterior doors must have a label meeting the requirements of §10-111(a).
§ 110.6(b):	**Field-fabricated Exterior Doors and Fenestration** products must use U-factors from Table 110.6-A and solar heat gain coefficient (SHGC) values from Table 110.6-B. Must be caulked and/or weather-stripped. *
§ 110.7:	**Air Leakage.** All joints, penetrations, and other openings in the building envelope that are potential sources of air leakage must be caulked, gasketed, or weather-stripped.
§ 110.8(a):	**Insulation Certification by Manufacturers.** Insulation must be certified by the Department of Consumer Affairs, Bureau of Household Goods and Services (BHGS).
§ 110.8(g):	**Insulation Requirements for Heated Slab Floors.** Heated slab floors must be insulated per the requirements of §110.8(g).
§ 110.8(i):	**Roofing Products Solar Reflectance and Thermal Emittance.** The thermal emittance and aged solar reflectance values of the roofing material must meet the requirements of § 110.8(i) and be labeled per §10-113 when the installation of a cool roof is specified on the CF1R.
§ 110.8(j):	**Radiant Barrier.** When required, radiant barriers must have an emittance of 0.05 or less and be certified to the Department of Consumer Affairs.
§ 150.0(a):	**Roof Deck, Ceiling, and Rafter Roof Insulation.** Roof decks in newly constructed attics above conditioned space in climate zones 4 and 8-16 area-weighted average U-factor not exceeding U-0.184. Ceiling and rafter roofs minimum R-22 insulation in wood-frame ceiling; or area-weighted average U-factor must not exceed 0.043. Rafter roof alterations minimum R-19 or area-weighted average U-factor of 0.054 or less. Attic access doors must have permanently attached insulation using adhesive or mechanical fasteners. The attic access must be gasketed to prevent air leakage. Insulation must be installed in direct contact with a roof or ceiling that is sealed to limit infiltration and exfiltration as specified in §110.7, including but not limited to placing insulation either above or below the roof deck or on top of a drywall ceiling. *
§ 150.0(b):	**Loose-fill Insulation.** Loose fill insulation must meet the manufacturer's required density for the labeled R-value.
§ 150.0(c):	**Wall Insulation.** Minimum R-15 insulation in 2x4 inch wood framing wall or have a U-factor of 0.095 or less, or R-21 in 2x6 inch wood framing or have a U-factor of 0.069 or less. Opaque non-framed assemblies must have an overall assembly U-factor not exceeding 0.102. Masonry walls must meet Tables 150.1-A. *
§ 150.0(d):	**Raised-floor Insulation.** Minimum R-19 insulation in wood framed raised-floor or 0.037 maximum U-factor. *
§ 150.0(f):	**Slab Edge Insulation.** Slab edge insulation must meet the following: water absorption rate no greater than 0.3 percent for insulation material alone without facings; water vapor permeance no greater than 2.0 perm per inch; protected from physical damage and UV light deterioration; and, when installed as part of a heated slab floor, meet the requirements of §110.8(g).
§ 150.0(g)1:	**Vapor Retarder.** In climate zones 1 through 16, the earth floor of unvented crawl space must be covered with a Class I or Class II vapor retarder. This requirement also applies to controlled ventilation crawl space for buildings complying with the exception to §150.0(d).
§ 150.0(g)2:	**Vapor Retarder.** In climate zones 14 and 16, a Class I or Class II vapor retarder must be installed on the conditioned space side of all insulation in all exterior walls, vented attics, and unvented attics with air-permeable insulation.
§ 150.0(q):	**Fenestration Products.** Fenestration, including skylights, separating conditioned space from unconditioned space or outdoors must have a maximum area-weighted average U-factor of 0.40. *

Solar Readiness:

§ 110.10(a)1:	**Single-family Residences.** Single-family residences located in subdivisions with 10 or more single-family residences and where the application for a tentative subdivision map for the residences has been deemed complete and approved by the enforcement agency, which do not have a photovoltaic system installed, must comply with the requirements of §110.10(b)-(e).
§110.10(b)1A:	**Minimum Solar Zone Area.** The solar zone must comply with access, pathway, smoke ventilation, and spacing requirements per Title 24, Part 9, or other parts of Title 24, or any requirements adopted by a local jurisdiction. The solar zone must be located on the roof or overhang of the building and have a total area no less than 250 square feet. The solar zone total area must be comprised of areas with no dimension less than 5 feet and no less than 80 square feet each for buildings with roof areas up to 10,000 square feet; or no less than 160 square feet each for buildings with roof areas greater than 10,000 square feet. *
§ 110.10(b)2:	**Azimuth.** All sections of the solar zone located on steep-sloped roofs must have an azimuth between 90-300° of true north.
§ 110.10(b)3A:	**Shading.** No obstructions in solar zone, including but not limited to: vents, chimneys, architectural features, and roof mounted equipment. *
§ 110.10(b)3B:	**Shading.** Any obstruction located on the roof or any other part of the building that projects above a solar zone must be located at least twice the horizontal distance of the height difference between the highest point of the obstruction and the horizontal projection of the nearest point of the solar zone, measured in the vertical plane. *
§ 110.10(b)4:	**Structural Design Loads on Construction Documents.** For areas of the roof designated as a solar zone, the structural design loads for roof dead load and roof live load must be clearly indicated on the construction documents.
§ 110.10(c):	**Interconnection Pathways.** The construction documents must indicate: a location reserved for inverters and metering equipment and a pathway reserved for routing of conduit from the solar zone to the point of interconnection with the electrical service; and for single-family residences and central water-heating systems, a pathway reserved for routing plumbing from the solar zone to the water-heating system.
§ 110.10(d):	**Documentation.** A copy of the construction documents or a comparable document indicating the information from §110.10(b)-(c) must be provided to the occupant.
§ 110.10(e)1:	**Main Electrical Service Panel.** The main electrical service panel must have a minimum busbar rating of 200 amps.
§ 110.10(e)2:	**Main Electrical Service Panel.** The main electrical service panel must have a reserved space to allow for the installation of a double pole circuit breaker for a future solar electric installation. The reserved space must be permanently marked as "For Future Solar Electric."

Fireplaces, Decorative Gas Appliances, and Gas Log:

§ 110.5(e):	**Pilot Light.** Continuously burning pilot lights are not allowed for indoor and outdoor fireplaces.
§ 150.0(e)1:	**Closable Doors.** Masonry or factory-built fireplaces must have a closable metal or glass door covering the entire opening of the firebox.
§ 150.0(e)2:	**Combustion Intake.** Masonry or factory-built fireplaces must have a combustion outside air intake, which is at least six square inches in area and is equipped with a readily accessible, operable, and tight-fitting damper or combustion-air control device. *
§ 150.0(e)3:	**Flue Damper.** Masonry or factory-built fireplaces must have a flue damper with a readily accessible control. *

Pool and Spa Systems and Equipment:

§ 110.4(a):	**Certification by Manufacturers.** Pool/spa heating equipment must be certified as compliant with Appliance Efficiency Regulations; have an externally mounted, readily accessible on-off switch that allows shutting off the heater without adjusting the thermostat; permanent, easily readable weatherproof plate or card with energy efficiency rating and operating instructions.
§ 110.4(b)1:	**Heating Equipment.** Pool/spa heating equipment must meet applicable standards, by fuel type.
§ 110.4(b)2:	**Piping.** At least 18 inches of horizontal or vertical pipe installed between the filter and the heater; or dedicated suction and return lines; or built-in or built-up connections installed to allow for the future addition of solar heating equipment.
§ 110.4(b)3:	**Covers.** Outdoor pools or spas with electric or gas heater must have a cover.
§ 110.4(b)4:	**Directional Inlets and Time Switches for Pools.** Pools must have directional inlets that adequately mix the pool water, and a permanently installed time switch that will allow all pumps to be set or programmed to run only during off-peak electric demand periods.
§ 110.4(c):	**Heat Source Sizing.** Pool/spa heating systems must be solar heating system with collector area of at least 60% of pool/spa surface area; heat pump sized per JA16.3; combination of solar and heat pump; or use 60% or more on-site renewable/recoverable energy. *
§ 110.4(d):	**Heat Pump Pool Heaters with Supplementary Heating.** Controls for such systems must limit supplementary heating when the heating load can be met by the heat pump alone.
§ 110.5:	**Pilot Light.** Natural gas pool and spa heaters must not have a continuously burning pilot light.
§ 150.0(p):	**Pool Systems and Equipment Installation.** Single-family pool systems or equipment must meet the specified requirements for pump sizing, flow rate, piping, filters, and valves.

*Exceptions may apply.

COUNTY OF SAN DIEGO
PLANNING & DEVELOPMENT SERVICES
BUILDING DIVISION

SINGLE-FAMILY RESIDENTIAL MANDATORY REQUIREMENTS SUMMARY

SHEET NUMBER

MR